AI-enabled Process and Design for Microelectronic Packaging

by

Peilun YAO

A Thesis Submitted to

The Hong Kong University of Science and Technology

in Partial Fulfillment of the Requirements for

the Degree of Doctor of Philosophy

in Mechanical Engineering

December 2024, Hong Kong

Authorization

I hereby declare that I am the sole author of the thesis.

I authorize the Hong Kong University of Science and Technology to lend this thesis to other institutions or individuals for the purpose of scholarly research.

I further authorize the Hong Kong University of Science and Technology to reproduce the thesis by photocopying or by other means, in total or part, at the request of other institutions or individuals for the purpose of scholarly research.

Signature redacted

Peilun YAO

December 2024

AI-enabled Process and Design for Microelectronic Packaging

by

Peilun YAO

This is to certify that I have examined the above PhD thesis
and have found that it is complete and satisfactory in all respects,
and that any and all revisions required by
the thesis examination committee has been made.

Signature redacted

Prof. Jinglei YANG, Thesis Supervisor

Signature redacted

Prof. Jingshen WU, Thesis Supervisors

Signature redacted

Prof. Fumin ZHANG, Acting Head of Department

Department of Mechanical and Aerospace Engineering

December 2024

Acknowledgements

Firstly, I would like to express my heartfelt gratitude to my thesis supervisors, Prof. Jingshen Wu and Prof. Jinglei Yang. Prof Wu's open-minded thinking and highly logical problem-solving approach provided novel feedback and support for my thesis research, and discussing research issues with him regularly helped me find the right research direction. I am fortunate and happy to conduct my thesis research under his guidance. At the same time, I would also like to thank Prof. Yang. His enthusiastic work style and rigorous work attitude have deeply influenced me, encouraged me to conduct deeper exploration in the research field, and will have a remarkable influence on my future research career.

I extend my thanks to the teachers and friends in the MCPF of HKUST and the funding sources, Nexperia, that facilitated my research. Thanks to the technical and project support they provided, I was able to conduct and deepen my research work.

Many thanks are also extended to my group members and friends, Dr. Haibin Chen, Dr. Yonglin Zhang, Dr. Zunyu Guan, Mr. Zhenteng Wu, and Miss. Qianwen Xu. I would like to express my sincerest gratitude to them for their technical support in my research.

Last but not least, I would like to thank my parents for providing many years of love, support, understanding, and encouragement. Without them giving me strength, I would not have been able to overcome the difficulties throughout over the years.

TABLE OF CONTENTS

LIST OF ABBREVIATIONS AND SYMBOLS

AI	Artificial Intelligence
BO	Bayesian Optimization
CEEQ	Accumulated Creep Strain
CNN	Convolutional Neural Network
CTE	Coefficient of Thermal Expansion
CZM	Cohesive Zone Model
DOE	Design of Methods
EMC	Epoxy Molding Compound
FEM	Finite Element Method
FOPLP	Fan-Out Panel-Level Package
GP	Gaussian Process
HF	High-Fidelity
LF	Low-Fidelity
MAPE	Mean Absolute Percentage Error
MCMC	Markov Chain Monte Carlo
MF	Multi-Fidelity
ML	Machine Learning
MO	Multi-Objective
MSE	Mean Square Error
NN	Neural Network
NPCNN	Nested Physics-Constrained Neural Network
PEEQ	Accumulated Plastic Strain
RNN	Recurrent Neural Network
TCT	Thermal Cycling Test
WLF	Williams-Landel-Ferry
α_T	WLF shift factor

ε_{acc}	Accumulated creep strain per cycle
η	Material related Mix-mode coefficient in CZM
$\rho(x)$	Radius of curvature
$\rho_{x,y}$	Pearson correlation coefficient
σ_{ij}, σ_{kk}	Deviatoric stress and dilatational stress in 2.1.1
σ_{top}	Die stress along the top line in multilayer shells
Φ(z), ϕ(z)	Cumulative distribution function and density function in 3.4
Ψ	Model mixity
$\omega(x)$	Deflection
D_{β}	Dundurs mismatch parameter
E	Young's modulus
E_i	Normalized Prony Series
G	Energy release rate
K	Stress intensity factor
G, K	Shear modulus and Bulk modulus
G^c	Critical fracture energy in CZM
G_n^c, G_n	Critical and transient fracture energy in normal direction
G_s^c, G_s	Critical and transient fracture energy in in-plane direction
G_t^c, G_t	Critical and transient fracture energy in out-of-plane direction
$k(x,x^{'})$	Covariance function for GP in 1.3.1
$m(x)$	Mean function for GP in 1.3.1
N_f	Fatigue lifetime
Tg	Glass transition temperature
t_n, t_n^c	Transient and peak stress in the normal direction
t_s ,t_s^c	Transient and peak stress in the in-plane direction
t_t, t_t^c	Transient and peak stress in the out-of-plane direction

LIST OF FIGURES

LIST OF TABLES

AI-enabled Process and Design for Microelectronic Packaging

by Peilun YAO

Department of Mechanical and Aerospace Engineering

The Hong Kong University of Science and Technology

ABSTRACT

With the challenge of Moore's law, system-on-chip (SoC) devices that focus on miniaturization and system-in-packaging (SiP) devices that focus on diversification are proposed. Applications such as artificial intelligence, high-performance computing, electric vehicles, mobile devices, and intelligent cities continuously drive the semiconductor industry to build systems with higher computational power. In the context of the vigorous application of these high-power devices, electronic packaging technologies face challenges under higher temperatures, higher-power working conditions, and more complex structures. The traditional design method, namely the DOE method, meets its limitations in the current situation due to its sensitivity to design parameters and the exponential growth of calculation time with the increase of design parameters. To meet the design requirements of the advanced packaging industry in the new era, an AI-enabled method for the microelectronic packaging industry is proposed.

This research focuses on optimizing and selecting epoxy molding compound (EMC) in AI-enabled IC packaging design. In the first part of the work, an automatic EMC material selection framework is proposed for IC packaging design, and the framework contains two sub-models. The first prediction model is developed to identify EMC constitutive parameters through the

flexural modulus at several temperature points, and the second optimization model with a multi-fidelity and multi-objective approach is constructed and applied in different reliability tests to help select appropriate EMC.

The second part develops a theory-guided multi-fidelity neural network model for identifying the cohesive zone model parameters under mix-mode failure to identify and analyze the EMC-lead frame interface behavior. Firstly, the proposed model shows acceptable performance and has been validated by testing results.

In summary, this work proposed an AI-enabled IC packaging design framework focused on EMC selection; the proposed framework can characterize the EMC constitutive parameter and EMC-lead frame interface parameter and select the suitable EMC under certain conditions effectively and accurately.

CHAPTER 1

INTRODUCTION

1.1 Background

The semiconductor industry has seen rapid advancement over the last few decades as the number of transistors in a functional chip increased based on Moore’s Law. As Figure 1.1 shows, to meet the challenges of Moore's Law, system-on-chip (SoC) devices focusing on miniaturization and system-in-packaging (Sip) devices focusing on diversification are proposed. Applications such as artificial intelligence, high-performance computing, electric vehicles, mobile devices, and smart cities continuously drive the semiconductor industry to build systems with higher computational power.

With the fast development of AI technology, chip power consumption has met a new challenge in fulfilling computing power, such as LLMs [2]. Figure 1.2 shows the road map of semiconductor packaging. As the I/O pitch becomes smaller and the I/O density becomes more extensive, the package design becomes more and more complex, pushing a higher pressure on reliability design. Building high-performance and highly integrated systems inevitably increases design complexity, which introduces significant challenges to deriving a modeling framework to rely on during the design cycle.

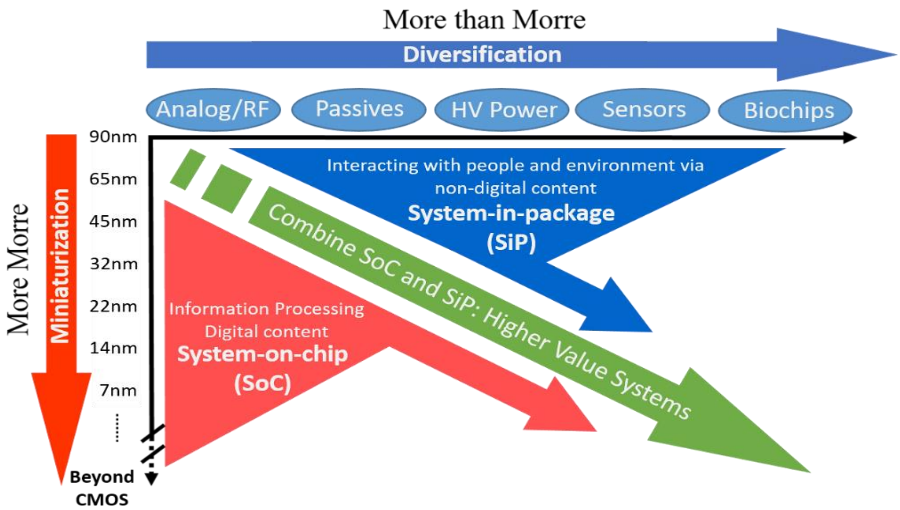

Figure 1.1 The concepts of More Moore and More than Moore [1].

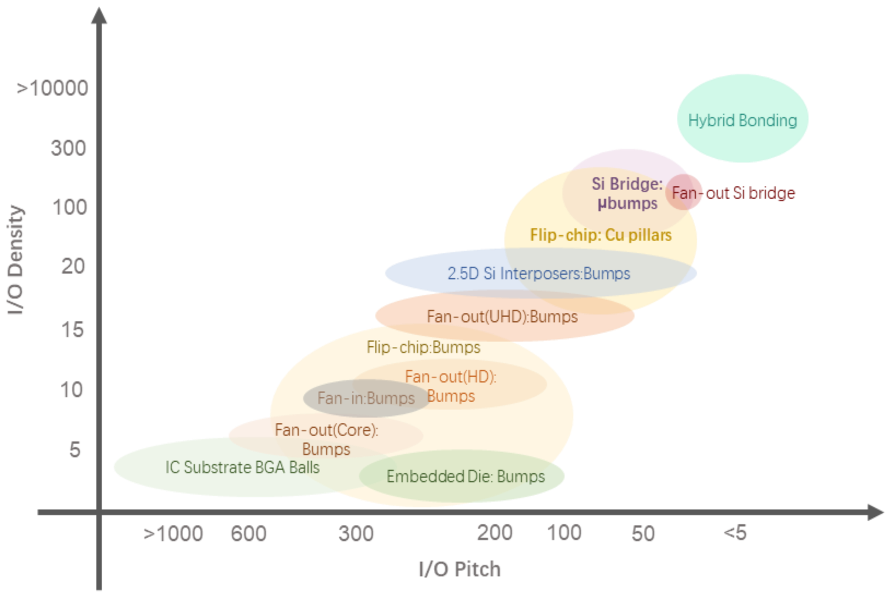

Figure 1.2 Advanced Packaging Technology Roadmap based on I/O pitch versus I/O density [3].

Current modeling frameworks fall into two categories, namely accurate but slow models and fast but inaccurate models. The former framework includes high-fidelity simulations such as FEM or CFD [4-6], and the latter framework is usually an analytical method that uses assumptions to simplify the original problem. However, using complex computer simulations in the design loop comes at the cost of increased simulation times and significantly delays the design closure. Using analytical models addresses the speed of simulations by introducing approximations or assumptions to the modeling framework but can lead to inaccurate design analysis. Such inaccurate modeling frameworks are one of the significant causes of design re-spins due to mistuned design parameters.

An alternative framework is to utilize data-driven models derived from machine learning (ML) techniques. Such models can have the same level of accuracy as simulation tools and be as fast as analytical models, potentially getting the best of both worlds. Here, the accurate modeling framework generates a finite amount of training data. This data is then used to train a learning-based model that can accurately predict the outcome of the simulation without running it. The derived ML model can then be used in an ample design space to analyze the effect of different design choices. Meanwhile, the ML model can be used to optimize the design to maximize the performance and analyze the impact of process variations on the performance. Considering the current semiconductor design industry, complex structures and multi-coupled loading conditions have introduced many variable parameters and uncertain factors, making traditional design methods increasingly unsuitable for the current situation. Although there are still some challenges in machine learning-based design methods, including data efficiency and accuracy, they are still a very competitive method.

1.2 Literature for Semiconductor Packaging Design

The traditional method for microelectronic packaging design is the Design of Experiments (DOE) method [7-8]. As shown in Figure 1.3, DOE is a systematic methodology determining the relationship between factors affecting a process and its output. This powerful statistical approach allows researchers and engineers to efficiently investigate multiple variables rather than testing one factor simultaneously. By carefully selecting and combining different input parameters at various levels, DOE helps identify the main effects of individual factors and their interactions. Standard designs include factorial, fractional factorial, and response surface methodologies. The technique finds widespread applications in manufacturing, product development, process optimization, and quality improvement initiatives, offering significant cost reduction and process understanding advantages while minimizing the number of required experiments.

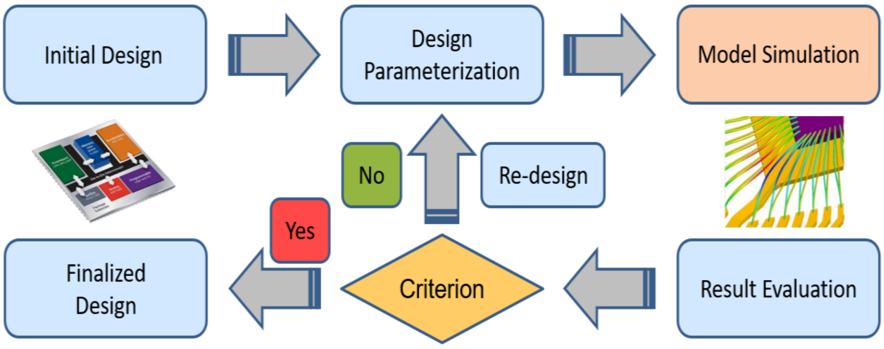

Figure 1.3 The typical workflow of the DOE design method.

DOE has been widely used in engineering and research for IC packaging design and optimization. Chaing [9] compared three different algorithms for solder ball geometry prediction and designed a DOE matrix for testing and validation; the results show that all three methods can have an acceptable result in solder ball geometry prediction in different conditions. Gao [10] uses the Taguchi method and variance analysis to study factors affecting the durability of TSVs and solder joints, and the 2D finite element models are constructed for evaluation. Results demonstrate that TSV diameter and underfill stiffness are the most critical variables for TSVs and solder joints. Qi [11] did the work to optimize SiP design for size reduction and cost efficiency by proposing a 100 DOE matrix that considers two types of bumped silicon wafers, three different substrate designs, and four different MUF and FU materials. Sun [12] used the DOE method to analyze the POP warpage behavior and found that the EMC thickness, die thickness, and substrate thickness are the most essential factors in POP warpage control. Xu [13] proposed a simulation-based DOE method for multi-objective optimization in packaging design; the modulus, CTE, PWB thickness, and solder height are selected as inputs for optimization, and the optimized design shows a higher solder joint reliability under the thermal cycling test. Wu [14] analyzes the warpage issue in FOWLP manufacturing through a combination of simulation and experimental methods, identifies key factors affecting warpage, and proposes optimization design schemes.

The researchers' work illustrates that the DOE method is effective for IC packaging. However, as the packaging structure becomes more and more complex, the drawbacks of the DOE method gradually appear, including:

1. Effectiveness drops with the increasing complexity.
2. Requirements for variable correlation
3. Time consuming
4. Discretized design space

These limits make the DOE method inappropriate for current microelectronic packaging design. Due to the increase in computing power, modern computers are already sufficient to deal with complex problems. Therefore, the AI-enabled design method has been re-proposed and has shown strong applicability in microelectronic packaging design.

AI-enabled design methodology [15-16] in the context of microelectronic packaging design most commonly refers to replacing an accurate but slow simulation framework with a learned model that is very fast to evaluate while providing the same level of accuracy as the actual simulation framework. $X = \{x_1, x_2 ... x_D\}$ represent the design parameters of interest as a D-dimensional vector, and y is the output of interest. The simulation framework can then be interpreted as a non-linear "black box" function that inputs the design parameters and provides the output of interest. The AI-enabled design method aims to build a machine learning model, commonly called a surrogate model, with some parameters for training, learning, and replacing the simulation model. Once the correct parameters are found, they can replace the actual simulation framework.

To derive such a model, training data consisting of T samples is first collected using the simulation framework. The training data is then used to determine the model parameters that minimize the error between the model and training data. After the training, the model is verified on an independent set of validation samples to assess its predictive accuracy. If the validation accuracy is high enough for a given application, the model derivation process is said to be completed. The derived model can then be used to infer the output corresponding to any input vector, eliminating the simulation framework from the design loop as in Figure 1.4.

In other words, one can use the derived model instead of the actual simulation framework to perform rapid and accurate parametric analysis to explore the design space, quantify the effect of process variations, and perform design optimization.

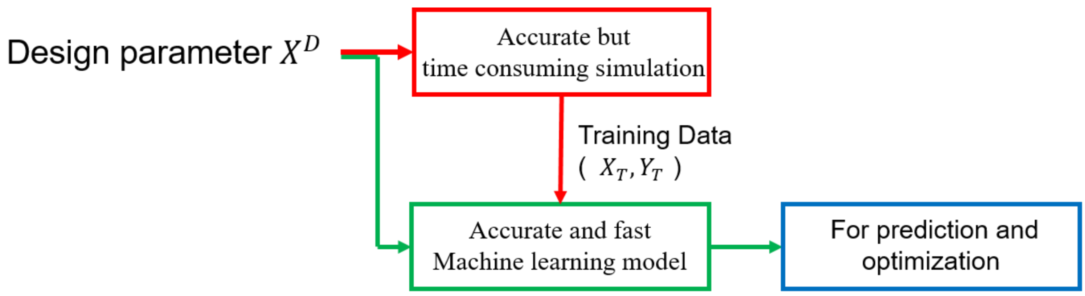

Figure 1.4 Fundamentals of AI-enabled design method.

In recent years, as artificial intelligence (AI) technology develops, various machine learning methods have been widely applied in different research domains [17-20] and the semiconductor packaging field. Commonly used machine learning models such as neural networks (NN), Gaussian Process (GP), random forest, and further deep learning models such as Convolutional neural network (CNN) and recurrent neural network (RNN) have shown an Unfathomable capability of dealing with complicated data and can solve complex, nonlinear and multivariate relationships. In the IC packaging field, an AI-assisted design approach has been applied in many areas of concern, including warpage control, thermal management, layout design, and interface delamination [21-25]. Selvanayagam [26] implements a Bayesian inference framework with MCMC to learn the in-plane and out-of-plane CTE in the warpage control area. A neural network model constructed with a single-layer neural network can replace high-fidelity FEA models for package warpages. The results show a good accuracy ($R^2 > 0.95$) and significant speed up of more than 3 orders of magnitude. Ahmed [27] proposes an ensemble machine learning model based on a random forest and gradient-boosted regression tree to implement the warpage prediction under different barrel temperatures, holding temperatures, holding time, and molding temperature; results show that the mean absolute percentage errors of random forest and gradient boosted regression tree model are 3.25% and 9.37%, respectively, and showing a potential application for the warpage prediction of injection molded part. In the stress degradation area, Prisacaru [28] uses Gated-RNN to provide promising prognostic results on stress distribution inside of encapsulated standard packages in different use cases; the results show a certain NN model can accurately predict the behavior of a single chip; however, the forecasting could not be reliably extrapolated for the behavior of a second chip under stress due to lack of adequate data. In the thermal management area, Lai [29] developed an ML model based on the least squares support vector regression (LSSVR) with Genetic Algorithms (GA) (LSSVR-GA) to predict the resistance (R), inductance (L), and capacitance (C) to capture the nonlinear data patterns and input-output electrical relationships of analog circuits, the results show an acceptable performance in predicting RLC value. Park [30] uses Bayesian Optimization (BO) and focuses on the temperature distribution on the die and the resultant skew on the clock distribution network (CDN) to improve the 3D packaging design thermal performance; the result demonstrates that BO shows a good performance in iteration speed compared with other optimization algorithms.

In the drop evaluation area, Mao [31] developed an RNN to learn and predict the solder stress, strain, energy density, and PCB warpage deformation of BGA under impact conditions; the proposed model shows a satisfactory performance (R^2>0.95). In the interface behavior area, Su [32] constructed an ANN model to learn the force-displacement curve under the shear test to predict the Mode II cohesive parameters; the results show a quite higher R^2 score (>0.999) for both shear traction stress and displacement. Freed [33] developed an ML model based on GPR; the model uses two steps to learn the stress and energy, respectively. The results are acceptable, although there are too many assumptions in the model's construction and training, which reduces the model's reliability in application. Wei [34] proposes a DNN model to learn the traction and separation laws of the interface. To avoid the limitations of a data-driven approach, the energy relationship and fracture angle are considered and added as an extra loss function for training, and the trained model shows pretty good performance and appeals to a new point-to-point constitutive modeling concept for interface mechanics.

1.3 Literature Review for Machine Learning Algorithms

1.3.1 Gaussian process

Gaussian process regression (GPR) is a machine learning regression algorithm that has been developed in recent years. As a promising technique, it has a strict statistical learning theory and has attracted widespread attention. Due to the advantages in uncertainty predictions, GPR is adaptable to complex problems such as high dimensionality, small samples, and nonlinearity. Meanwhile, GPR has strong generalization ability [35]. Unlike neural networks that use multiple hidden layers to handle nonlinear, complex classification and regression problems, GPR uses kernel tricks similar to SVR or SVM, making it easier to understand and implement in practice. As a concept of probability statistics, the Gaussian process (GP) is a kind of Stochastic process. It calculates the posterior degradation estimated by constraining the prior distribution.

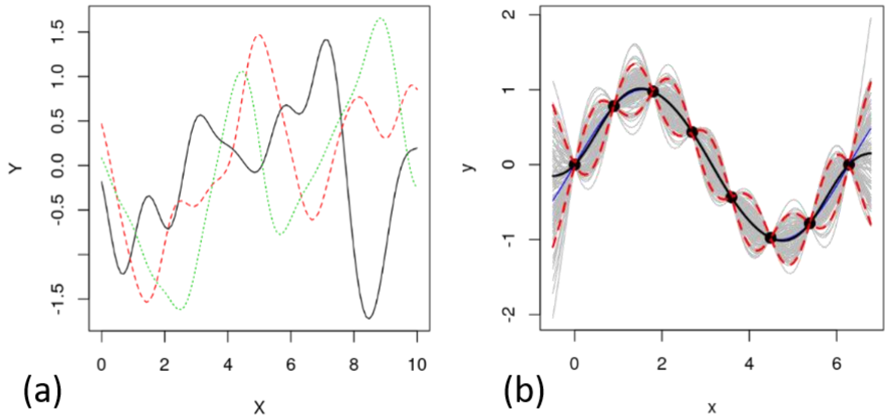

Figure 1.5 Demonstration of sampling functions from a GP using a few observations. (a) Samples from the GP prior without conditioning on the observed data. (b) A few samples from the GP posterior obtained by conditioning on a few observations..

Similar to Gaussian distribution, GP is determined by its mean and covariance functions. According to the definition of GP, assuming $f(x_i)$ obey multivariate Gaussian distribution, GP can be defined as:

$$f(x) \sim N\left(m(x), k(x, x^{'})\right) \tag{1.1}$$

where $m(x)$ and $k(x, x^{'})$ represent mean function and covariance function, respectively [36] and $x,\ x^{'} \in X$ are random variables.

Since the collected data often contains noise, the Gaussian process regression model becomes:

$$y = f(x_i) + \varepsilon \tag{1.2}$$

where $\varepsilon \sim N(0, \sigma_n^2)$. According to Bayesian principle, For the observed data $D = \{x_i, y_i\}_{i=n+1}^{n+n}$, GP can get the corresponding posterior knowledge as shown in Figure 1.5, and a joint Gaussian prior distribution of training output vector y and testing output vector y^*can be established as follows:

$$\begin{bmatrix} y \\ y^* \end{bmatrix} \sim N\left(0, \begin{bmatrix} K(X,X)+\sigma_n^2 I & K(X,x^*) \\ K(x^*,X) & K(x^*,x^*) \end{bmatrix}\right) \tag{1.3}$$

$$K(X,X)=\begin{bmatrix} k(x_1,x_1) & k(x_1,x_2) & \cdots & k(x_1,x_n) \\ k(x_2,x_1) & k(x_2,x_2) & \cdots & k(x_2,x_n) \\ \vdots & \vdots & \ddots & \vdots \\ k(x_n,x_1) & k(x_n,x_2) & \cdots & k(x_n,x_n) \end{bmatrix} \tag{1.4}$$

$$K(X,x^*)=\left[k(x^*,x_1),k(x^*,x_2)...k(x_*,x_n)\right] \tag{1.5}$$

$$K(x^*,x^*)=k(x^*,x^*) \tag{1.6}$$

where $K(X,X)$ is symmetric positive definite covariance matrix for training data X, $K(X,x^*)$ is covariance matrix for training data X and testing data x^* and $K(x^*,x^*)$ is covariance matrix for testing data x^*.

Thus, according to the conditional distribution form of multivariate Gaussian distribution, we can obtain the posterior probability formula y^* as follows:

$$y^* \middle| X,y,X^* \sim N(m(x^*),\mathrm{cov}(y^*)) \tag{1.7}$$

$$m(x^*)=K(x^*,X)(K(X,X)+\sigma_n^2 I)^{-1}y \tag{1.8}$$

$$\begin{aligned}\mathrm{cov}(y^*)=K(x^*,x^*)-&\\ &K(x^*,X)(K(X,X)+\sigma_n^2 I)^{-1}K(X,x^*)\end{aligned} \tag{1.9}$$

Since the covariance matrix in GPR needs to meet Mercer conditions, Equation (1.8) can be written as follows:

$$m(x^*)=\sum_i^n \alpha_i k(x^i,x^*) \tag{1.10}$$

where $k(x^i,x^*)$ is called kernel function.

In GPR, the kernel function maps the data of the non-linear relationships to a high-dimensional feature space for linear regression and constructs the hypothesis function by defining the relationship among the data points. Therefore, the selection of the covariance kernel function is very important for Gaussian process modeling.

Kernels are the similarity between two values of a function evaluated on each object, and the patterns of generalization in GP depend heavily on the choice of kernel function. Different kernels can impose widely varying modeling assumptions, such as smoothness, linearity, or periodicity. Capturing appropriate kernel structures can be crucial for interpretability and extrapolation. Commonly used basic kernel functions include squared exponential (SE) kernel function, rational quadratic (RQ) kernel function, and Matern class kernel function [37]. The specific definition is given as follows:

$$k_{SE}(x_i, x_j) = \sigma_f^2 \exp\left(-\frac{(x_i, x_j)^2}{2l^2}\right) \tag{1.11}$$

$$k_{RQ}(x_i, x_j) = \sigma_f^2 \left(1 + \frac{(x_i, x_j)^2}{2\alpha l^2}\right)^{-\alpha} \tag{1.12}$$

$$k_M(x_i, x_j) = \sigma_f^2 \left[\left(1 + \frac{\sqrt{3}}{l}(x_i, x_j)\right) \exp\left(-\frac{\sqrt{3}}{l}(x_i, x_j)\right)\right] \tag{1.13}$$

Each kernel has several parameters that specify the precise shape of the covariance function. These are sometimes referred to as hyper-parameters since they can be viewed as specifying a distribution over function parameters instead of being parameters that specify a function directly. Most kernel functions have at least 2 parameters, namely ℓ and σ; ℓ specifies the width of the kernel and, thereby, the smoothness of the functions in the model, and σ specifies the amplitude of the kernel function. Figure 1.6 shows the GP prior to different basic kernel functions. SE is a smooth, infinitely differentiable kernel function, and RQ is an alternative to SE that is less time-consuming but more sensitive. Matern is a moderately smooth kernel function.

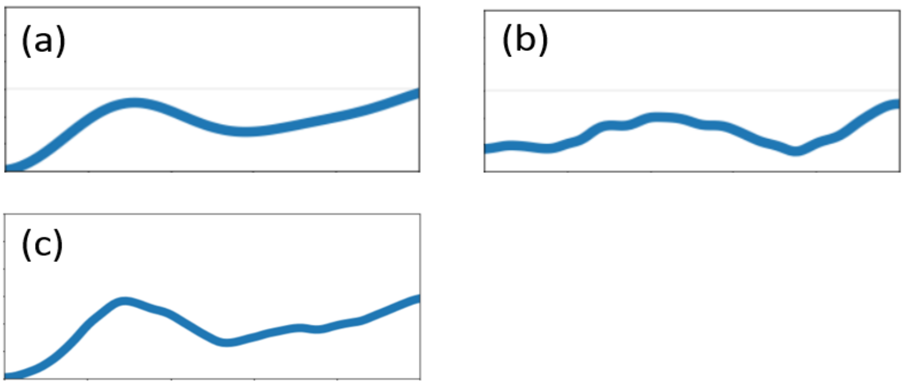

Figure 1.6 Shape and smoothness of Sample from GP prior distribution using different basic kernel functions: (a) SE (b) RQ and (c) Matern 5/2.

Under some conditions, a single kernel function may not be able to accurately express the relationship among variables. In view of the shortcomings of a basic kernel function, the composite kernel function may perform better in certain fields. In various composite methods, the additive kernel function is widely used.

An additive function can be expressed as $f(x) = f_a(x) + f_b(x)$. Additivity is a useful modeling assumption in a wide variety of contexts, especially if it allows us to make strong assumptions about the individual components which make up the sum. Restricting the flexibility of component functions often aids in building interpretable models, and sometimes enables extrapolation in high dimensions [38].

It is easy to encode additivity into GP models. Suppose functions f_a, f_b are drawn independently from GP priors:

$$f_a \sim GP(\mu_a, k_a) \tag{1.14}$$
$$f_b \sim GP(\mu_b, k_b)$$

Then the distribution of the sum of those functions is simply another GP:

$$f_a + f_b \sim GP(\mu_a + \mu_b, k_a + k_b) \tag{1.15}$$

Kernels k_a and k_b can be of different types, allowing us to model the data as a sum of independent functions, each possibly representing a different type of structure. Any number of components can be summed this way, and Figure 1.7 shows the GP prior to different addictive kernel functions.

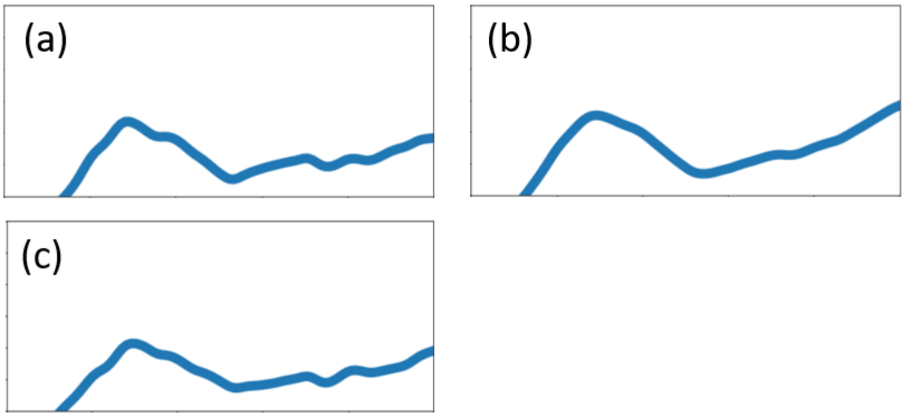

Figure 1.7 Shape and smoothness of Sample from GP prior distribution using different additive kernel functions: (a) SE+Matern (b) SE+RQ and (c) RQ+Matern.

There are several reasons why GP algorithms are so frequently used in regression.

- GP is a probabilistic model, where, when a kernel function and some observations are given, the predictive posterior distribution can be computed exactly in closed form. This can help researchers obtain the Confidence interval to judge the applicability of the results.
- GP uses kernel tricks to fit non-linear functions. Since the kernel function is simple and designable, we can express a wide range of modeling assumptions through the choice of kernel functions.
- GP has fewer hyperparameters than other similar algorithms, such as Neural networks; this makes GP less overfitting and easy to estimate.

There are several issues that limit the use of GP.

- The posterior distribution needs to compute the inverse matrix, which will take O(N3) time; this causes a slow inference when features are large. However, this problem can be addressed by approximate inference schemes [39] [40] [41].
- The predictive distribution of a standard GP model is Gaussian. We may sometimes use non-Gaussian predictive likelihoods, so using non-Gaussian likelihoods requires approximate inference [42] [43] [44].
- The kernel function will highly affect the performance of GP. Though Kernel parameters can be set automatically by maximizing the marginal likelihood, until recently, human experts were required to choose the parametric form of the kernel [45].

1.3.2 Neural network

One of the most well-known and commonly used machine learning models is neural networks (NN). The goal is to mimic the neuron-based decision-making system of the human brain. Figure 1.8 (a) shows the building block of an NN is a neuron, a neuron takes a vector of inputs$X = (x_1, \ldots x_n)$, constructs their weighted sum, $WX = (w_1x_1 \ldots w_nx_n)$ and adds a bias to generate $WX + b$, where b is similar to an intercept term in conventional linear regression. This is then passed through a non-linear activation function to get $\sigma(WX + b)$, and the purpose of this activation function is to introduce non-linearity to the overall framework.

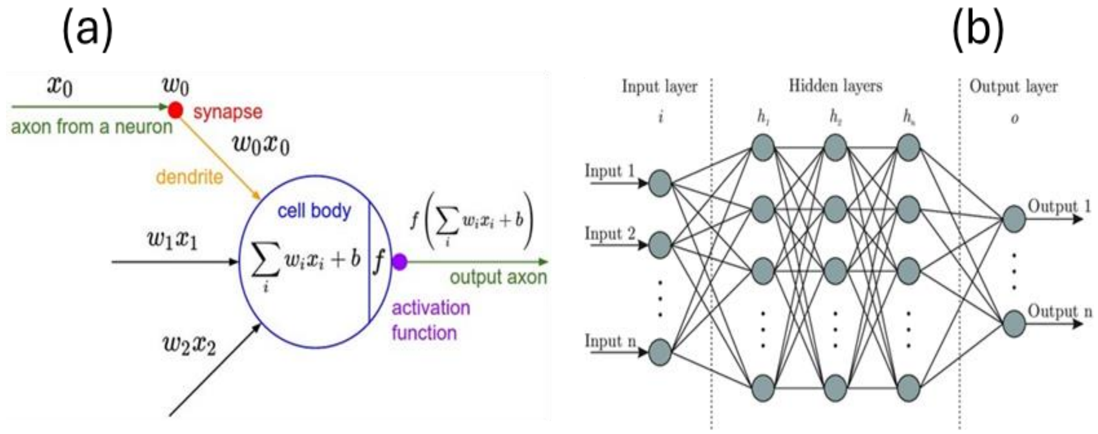

Figure 1.8 (a) Fundamental of a single neuron in multi-input neural network to obtain output through activation function (b) Typical structure of feed forward neural network include multi-input, multi-outputs, multi-layer and multi-nots.

A NN connects several of these neurons, as shown in Figure 1.8 (b). A typical architecture consists of an input layer, multiple hidden layers, and an output layer. The input layer consists of the variables we want to map to an output. The purpose of the hidden layers is to capture non-obvious interactions between the overall input-output relationship. We use multiple neurons in each hidden layer and connect each one to all the other neurons in the subsequent layers, where each connection describes a different interaction pattern. As the number of hidden layers increases to capture more complex patterns in data, the NN becomes a Deep Neural Network (DNN). The output of the output layer, i.e. NN generated output, is then compared with the actual output to calculate error. This error is then minimized by adjusting the weights in each layer through their gradients, and this process is known as training the NN. It is important to note that the type of activation function, number of neurons per layer, and number of layers in the NN, collectively referred to as the NN's hyperparameters, can vary based on the data sets. These hyperparameters should be optimized for each modeling task to minimize the validation error. Later chapters will introduce hyperparameter optimization further.

In semiconductor packaging, neural networks are widely used for many applications, including warpage prediction [46], solder fatigue analysis [47], and multi-physics analysis in 3D IC devices [48].

1.3.3 Bayesian optimization

Commonly, machine learning-based optimization starts from training a surrogate model. Once the model is complete, the fast and accurate surrogate model can combine with various optimization algorithms to find the design parameters that minimize or maximize the objective function. However, the prerequisite of this optimization framework is an accuracy surrogate model, which can have a very similar solution to the high-fidelity simulation model. Furthermore, the dataset we collect to build the model spans the entire design space X^D. Since we do not know where the optimum point lies in the design space, we waste a significant portion of our computational budget to collect training data where f(x) is potentially sub-optimal. A promising method to overcome these challenges is Bayesian Optimization (BO).

Bayesian optimization is a sample-efficient optimization algorithm that suits the optimization of expensive, black-box systems. By “black box,” we mean that the objective function does not have a closed-form representation, does not provide function derivatives, and only allows point-wise evaluation. Figure 1.9 shows the workflow of BO. Different from other optimization algorithms that perform optimization by evaluating a function multiple times, Bayesian optimization focuses on sample efficiency and uses a model-based approach with an adaptive sampling strategy to minimize the number of function evaluations.

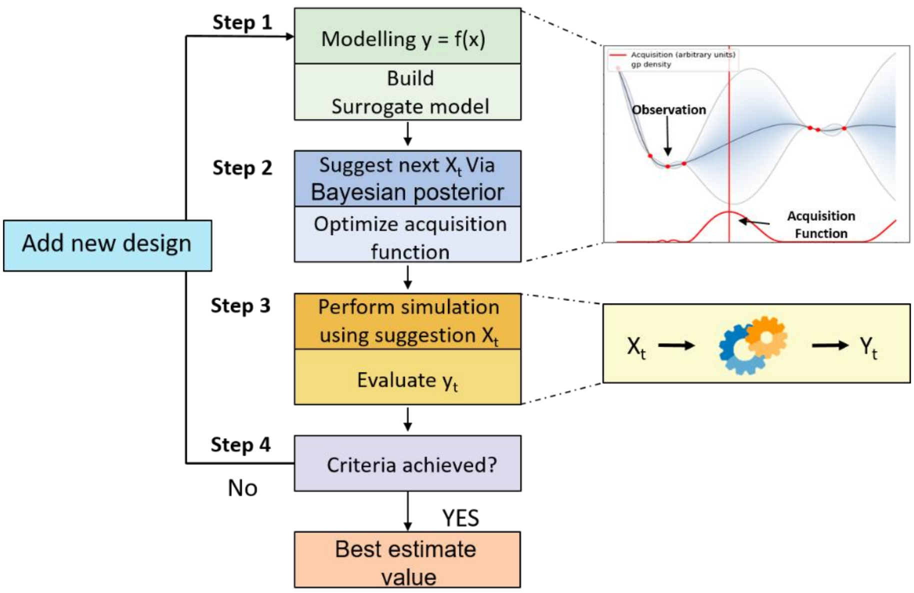

Figure 1.9 Workflow of Bayesian optimization.

Bayesian optimization incorporates two main ideas: a probabilistic model and a sampling strategy. There are several probabilistic models used as surrogate models in Bayesian optimization, including Bayesian Neural network [49], random forest [50], and Parzen estimators [51]. However, these methods have a relatively poor capability of predicting the posterior, which will affect the sampling efficiency in optimization. Thus, GPs, as a probabilistic model, have many theoretical and practical advantages that align with the purposes of BO. From the theoretical point of view, Artificial neural networks with one hidden layer converge to a GP when the number of hidden units tends to infinity [52]; as GP is a non-parametric model, a DNN model with multiple hidden layers can be equivalent to GP. So recently, Bayesian optimization with DNN has been proposed, which has a competitive ability in predicting posterior [53]. Using the GP as a surrogate model, an acquisition function is constructed as a sampling strategy to represent the most promising setting for the next experiment. Acquisition functions are mainly derived from the $\mu(x)$ and $\sigma(x)$ of the GP model. The acquisition function allows a balance between exploitation (sampling where the objective mean μ is high) and exploration (sampling where the uncertainty σ is high), and its global maximum value is used as the following experimental setting. Some commonly used acquisition functions are EI [54], PI [55], and UCB [56], which will have a detailed explanation in a later section.

Bayesian optimization still has some unresolved problems in the following aspects:

- High dimensional optimization. Traditional Bayesian optimization shows a good performance in low dimensions, but performance degrades above about 15-20 dimensions. Several methods have been proposed to make BO work in a higher dimension [57] [58] [59] [60], but these methods reduce the convergence efficiency and increase the complexity of iteration.
- Multi-objective optimization. Several multi-objective optimization methods have been proposed for BO [61] [62]. However, most methods are affected by high dimensions, and the computational cost increases linearly with the number of targets.

1.3.4 Differential evolutionary algorithm

Differential evolution (DE) [63] [64] [65] [66] [67] is arguably one of the most potent stochastic real-parameter optimization algorithms in current use. It operates through similar computational steps as employed by a standard evolutionary algorithm (EA). DE is a simple fundamental parameter optimization algorithm. It works through a simple cycle of stages presented in Figure 1.10. The first step is population initialization. DE searches for a global optimum point in a D-dimensional real parameter space R^D. It begins with a randomly initiated population of D-dimensional real-valued parameter vectors, which can be described as follows:

$$\vec{X}_i = \left[x_{1,i}, x_{2,i}, x_{3,i} \dots x_{D,i} \right] \tag{1.16}$$

The next step is mutation. Biologically, “mutation” means a sudden change in the gene characteristics of a chromosome, In DE literature, a parent vector from the current generation is called a target vector, and a mutant vector obtained through the differential mutation operation is known as a donor vector, and finally an offspring formed by recombining the donor with the target vector is called a trial vector. In one of the simplest forms of DE-mutation, to create the donor vector for each i th target vector from the current population, three other distinct parameter vectors, say $\vec{X}_{r_1^i}$, $\vec{X}_{r_2^i}$ and $\vec{X}_{r_3^i}$ are sampled randomly from the current population.

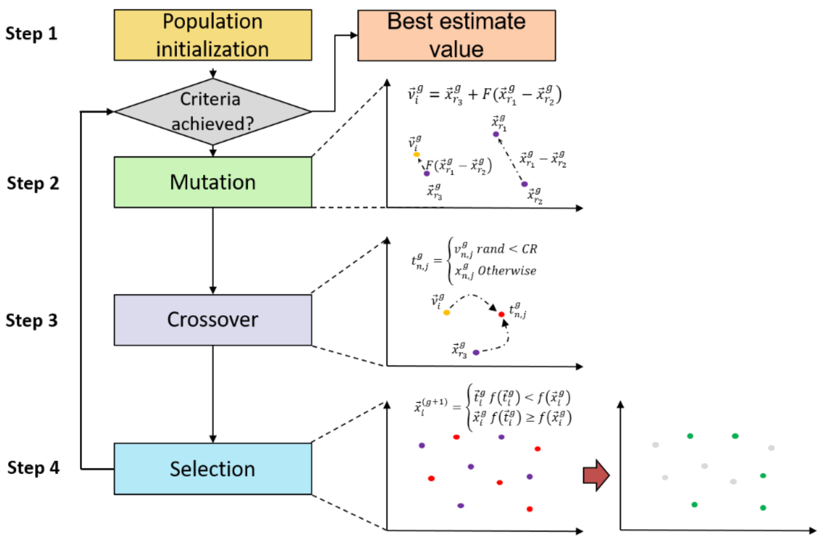

Figure 1.10 Workflow of Differential evolution

The indices r_1^i, r_2^i, and r_3^i are mutually exclusive integers randomly chosen from the range [1, NP], which are also different from the base vector index i. These indices are randomly generated once for each mutant vector. Now the difference of any two of these three vectors is scaled by a scalar number F and the scaled difference is added to the third one whence we obtain the donor vector $\vec{V_i}$. We can express the process as:

$$\vec{V_i} = \vec{X}_{r_1^i} + F \cdot (\vec{X}_{r_2^i} - \vec{X}_{r_3^i}) \tag{1.17}$$

The third step is crossover. Which aims to enhance the potential diversity of the population. The DE algorithm commonly uses two kinds of crossover methods—exponential and binomial [68]. In exponential crossover, an integer n is randomly selected among the numbers [1, D]. This integer acts as a starting point in the target vector, from where the crossover or exchange of components with the donor vector starts. We also choose another integer L from the interval [1, D]. L denotes the number of components the basic vector contributes to the target vector. After choosing n and L the trial vector is obtained as:

$$t_{j,i}^G = \begin{cases} v_{j,i}^G & j = \langle n \rangle_D, \langle n+1 \rangle_D, \langle n+L-1 \rangle_D \\ x_{j,i}^G & j = all\ other \end{cases} \tag{1.18}$$

For binomial crossover, each D variable will randomly generate a number between 0 and 1 and compare whether it is less than or equal to the Cr value. In this case, the number of parameters inherited from the donor has a binomial distribution. The scheme may be outlined as:

$$t_{j,i}^G = \begin{cases} v_{j,i}^G & f\left(rand_{i,j}[0,1] \le Crorj = j_{rand}\right) \\ x_{j,i}^G & Otherwise \end{cases} \tag{1.19}$$

Cr is called the crossover rate and appears as a control parameter of DE, just like F. The last step is selection. To keep the population size constant over subsequent generations, the next step of the algorithm calls for selection to determine whether the target or the trial vector survives to the next generation, i.e., at $G = G + 1$. The selection operation is described as:

$$\vec{X}_i^{g+1} = \begin{cases} \vec{t_i^g} & if \quad f(\vec{t_i^g}) < f(\vec{x_i^g}) \\ \vec{x_i^g} & if \quad f(\vec{t_i^g}) \ge f(\vec{x_i^g}) \end{cases} \tag{1.20}$$

Where $f(\vec{X})$ is the objective function to be minimized.

Therefore, if the new trial vector yields an equal or lower objective function value, it replaces the corresponding target vector in the next generation; otherwise, the target is retained in the population. Hence, the population either gets better or remains the same regarding fitness status.

There are several reasons why DE algorithms are so frequently used in optimization.

- Compared with most other EA algorithms, the implementation of DE is much simpler.
- As indicated by the recent studies on DE [69] [70] [71], despite its simplicity, DE exhibits much better performance in comparison with several other complex optimization algorithms and so on of current interest in a wide variety of problems, including unimodal, multimodal, separable, non-separable and so on.
- The number of control parameters in DE is small, and the simple adaptation rules F and Cr have been designed to improve the performance of the algorithm to a large extent without causing any severe computational burden [72] [73] [74].
- The space complexity of DE is low. Compared with some optimization methods [75], it is difficult to expand to higher dimensions due to the storage, update, and inversion operations of the square matrix with the same variable size and the number of variables. DE is good at dealing with large-scale and expensive optimization problems.

Also, DE has some limitations that need to be resolved.

- DE faces huge difficulties in a non-linear separable function [76] [77]. In such a function, DE must mainly rely on its differential mutation process. This mutation strategy lacks sufficient selection pressure when specifying the target and donor vector. Therefore, its performance on such functions is limited.
- In some cases, DE sometimes has a limited ability to move its population large distances across the search space, which may lead to the optimization falling into a local optimal.
- Many objective optimization problems usually deal with more than three objective functions. Many traditional MOEAs that use Pareto optimality as a ranking indicator may not perform well on many objective functions [78] [70].

1.4 Research Objective

From the previous review, the ML technique has been widely applied in semiconductor design and optimization. However, most of the research is still in its infancy, with scattered research directions and many boundary conditions and assumptions. At the same time, most of the research focuses on design and optimization under a single working condition, which affects the versatility of the model.

As a very important component of semiconductor packaging, the thermal and mechanical properties of materials will significantly affect the heat dissipation and mechanical damage resistance of the packaging structure. As shown in Figure 1.11, several steps may cause failure when examining the materials and processes of IC packaging. These include a mechanical die crack in the die pick-and-place process, a thermal crack of epoxy molding compound (EMC) and die under the thermal loading process, and EMC-lead frame interface delamination under the thermal loading process. The EMC is essential in IC packaging design among those failure modes. As a polymer-based composite material, the different combinations of epoxy resin matrix, fillers, and curing agent affect the mechanical properties of EMC, resulting in a different mechanical behavior in the following aspect.

1. Glass transition temperature (Tg)
2. Thermal Transport coefficient (TC)
3. Coefficient of thermal expansion (CTE)
4. Interface adhesion
5. Viscosity

Such different mechanical properties make the EMC [79-81] the most complex and designable material in IC packaging design and selecting the EMC with proper material properties and interface properties can effectively reduce the risk of packaging structure failure. However, as a complex material dependent on time and temperature, EMC has a lot of parameters that affect its mechanical properties and interface properties. Therefore, it is essential to establish a framework to determine the appropriate material parameters of EMC effectively and accurately.

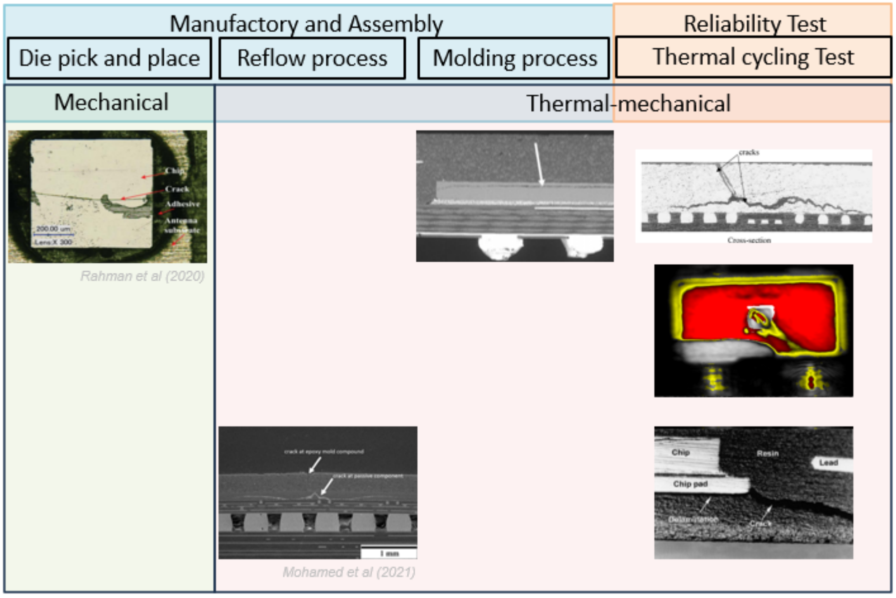

Figure 1.11 The failure modes and corresponding failure materials under different packaging processes, first columns show the failure mode of die, middle columns show that of lead frame and last columns for the EMC.

As introduced and reviewed before, the challenge of Moore's law has made the IC device more miniaturized and diversified. The microelectronic system design is a complex unknown function, and the complexity of IC devices makes the traditional DOE method inappropriate for current IC design. In contrast, due to the development of computing power, optimization work based on machine learning has made achievements in many fields, including the semiconductor industry. Many algorithms have been developed in prediction and optimization and have shown a powerful ability to fit and search. But there are still some challenges in applying machine learning-based design into microelectronic systems, which include:

1. The data is usually generated by high-fidelity simulation results, and this is a time-consuming process. Making compelling predictions with insufficient data or limited high-fidelity data becomes a problem.
2. The current IC design contains multiple variables and objectives, mix-mode input, various constraints, and high dimensions. Maintaining good prediction optimization accuracy under complex conditions is a problem.

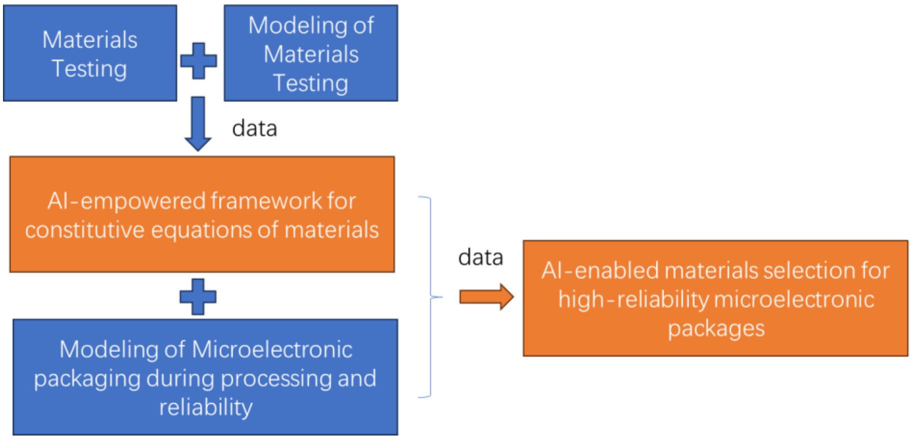

Figure 1.12 Workflow of the AI-enabled process and design framework.

Therefore, the core proposition of this research is to establish a novel machine learning optimization framework (Figure 1.12) that can deal with the structure and process design in microelectronic systems. Details of the research objectives are listed below:

1. To apply and evaluate AI models and algorithms in the application in the process and structure design for microelectronic devices.
2. To develop a novel multi-objectives global optimization framework to specifically address challenges related to the optimization of microelectronic packaging systems.

CHAPTER 2

AI-ENABLED MATERIAL CHARACTERIZATION MODEL FOR EPOXY MOLDING COMPOUND

This chapter establishes an AI-enabled model for EMC material property characterization. Figure 2.1 shows the workflow of the AI-enabled EMC material selection framework. The framework consists of two sub-frameworks: the left one is for the determination of EMC material properties, and the right one is for EMC material property optimization under specific conditions which will be fully illustrated in the next chapter.

About 60 different EMCs are prepared and implemented for the first sub-framework on the left for the DMA and TMA testing. The data collection and processing are done to separate the data to the modulus at different temperatures as input data and EMC mechanical properties as output data. After that, a physics-constrained NN model is established to train, validate, and test the data. The well-trained model can effectively and accurately determine EMC's material properties by inputting the modulus at some temperature points. The following EMC material properties determined in the previous framework will be delivered to the right framework in the figure to perform the optimization process.

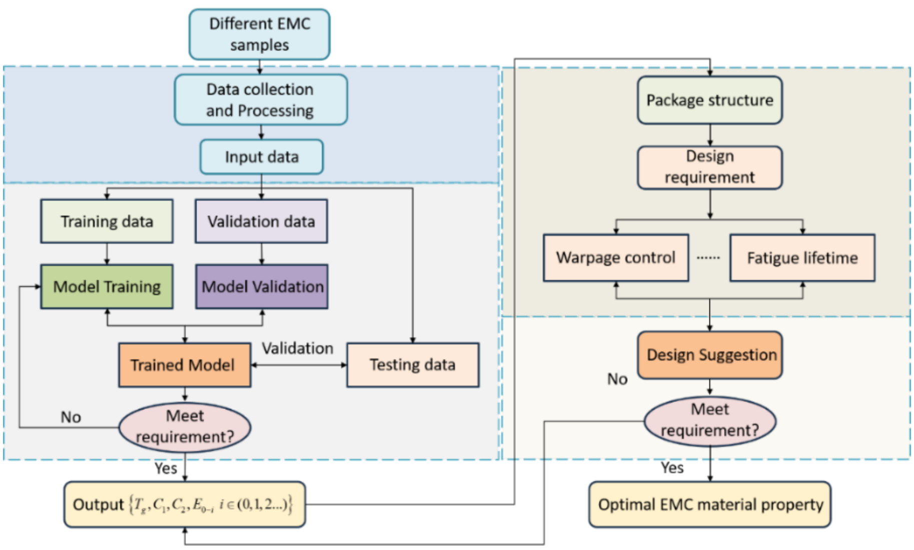

Figure 2.1 Workflow of the AI-enabled EMC material selection framework.

2.1 NPCNN for EMC Constitutive Parameters Determination

2.1.1 Constitutive equation for viscoelastic material

As a time- and temperature-dependent material, the EMC shows viscoelastic behavior [82-83] under different time domains and temperature profiles. It can be described via a linear viscoelastic constitutive equation.

$$\sigma_{\mathrm{ij}}(t) = 2\int_{-\infty}^{t} G(t-\tau)\frac{d\varepsilon_{ij}(\tau)}{d\tau}d\tau \tag{2.1}$$

$$\sigma_{kk}(t) = 3\int_{-\infty}^{t} K(t-\tau)\frac{d\varepsilon_{kk}(\tau)}{d\tau}d\tau \tag{2.2}$$

$$\sigma_{kk} = 3K\varepsilon_{kk} \quad \sigma_{ij} = 2G\varepsilon_{ij} \tag{2.3}$$

$$\sigma_{ij} = \int_{-\infty}^{t} 2G(t-\tau)\frac{d\varepsilon_{ij}(\tau)}{d\tau} + \delta_{ij}\int_{-\infty}^{t}\left[K(t-\tau) - \frac{2}{3}G(t-\tau)\right]\frac{d\varepsilon_{kk}(\tau)}{d\tau}d\tau \tag{2.4}$$

The σij and σkk are deviatoric stress and dilatational stress, respectively. Equation 2-4 is the general form for the viscoelastic constitutive equation, and G and K are still unknown. Assuming viscoelastic can be regarded as a combination of elastic and viscous behavior, the generalized Maxwell model is proposed to describe G and K. Figure 2.2 illustrates the form of the generalized Maxwell model. The spring represents the elastic behavior, and the damper represents the viscous behavior. The series connection of a set of springs and dampers in the figure is called the Maxwell model. It is used to describe stress relaxation behavior. The parallel connection of a set of springs and dampers is called the Kelvin model, which describes creep behavior. The generalized Maxwell model connects multiple sets of series Maxwell models in parallel to describe the time-varying mechanical behavior of materials under complex working conditions.

The Prony series [84], as equation 2.5, presents the viscoelasticity of the polymer-based EMC material. Where E(t) denotes the elastic modulus at relaxing time t, E0 is the instantaneous modulus, $\tau_i$are the relaxation times with coefficients of Ei, and N is the series number.

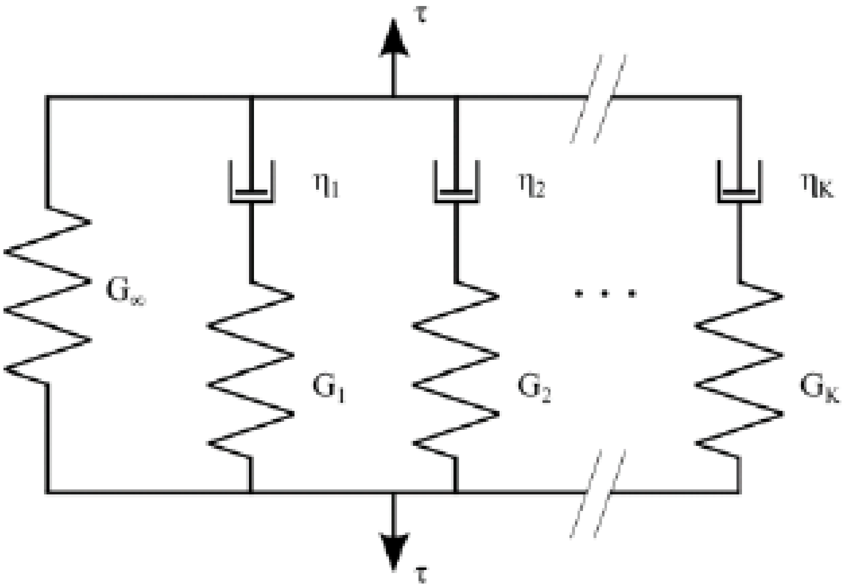

Figure 2.2 Schematic diagram of the generalized Maxwell model [84].

$$E(t) = E_0 - \sum_{i=1}^{N} E_i \exp\left(-\frac{t}{\tau_i}\right) \tag{2.5}$$

The viscoelastic constitutive equation determines the time-dependent behavior, and the Williams-Landel-Ferry (WLF) equation [85] is employed to determine the time-temperature superposition for the viscoelastic behavior of the composite. The function is described as

$$\log(a_T) = \frac{-C_1(T - T_r)}{C_2 + (T - T_r)} \tag{2.6}$$

Where α_T is the WLF shift factor, T is the temperature, Tr is the reference temperature selected to develop the compliance master curve, and C_1, C_2 are constants fitted by experimental data.

Combining the shift function and viscoelastic constitutive relation above, we can derive the core parameters describing the viscoelastic constitutive relation of EMC, namely Tg, C1, C2, E0, and Ei. Accurately and effectively determining these parameters and performing analysis and optimization under different working conditions can help select the best EMC for packaging design.

$$
\begin{cases}
\log(\alpha_T) = \dfrac{-C_1(T-T_r)}{C_2+(T-T_r)} \\
E(t,T) = E_0\left[1-\sum_{i=1}^{n} e_i\left(1-\exp(-\dfrac{t}{\tau_i\alpha_T})\right)\right]
\end{cases}
\tag{2.7}
$$

$$
\Rightarrow \{T_g, C_1, C_2, E_{0\sim i}(i \in (0,1,2...))\}
$$

2.1.2 Design method for EMC material characterization

The traditional method uses DMA and TMA testing to identify the EMC constitutive parameters in the previous section, and Figure 2.3 shows the workflow. The EMC is first to do the DMA and TMA test and collect the testing data including the storage modulus, loss modulus, etc. The second step is to identify the glass transition point through a typical storage modulus curve, which is the drop point on the curve. The WLF shift function is then to be fit after the identification of Tg. The third step is to collect the storage modulus curve and loss modulus curve under different temperatures and make a movement to generate the master curve. This master curve represents the time-dependent mechanical behavior of the material, and the pony series is then used for a data fitting to identify the viscoelastic constitutive parameters.

After these three steps, the viscoelastic parameters of EMC can be obtained, and the next analysis task can be carried out. The testing, data collection, and manual data fitting processes are complex and time-consuming. To solve this problem, this work develops a nested physics-constrained neural network (NPCNN) model to optimize the process flow and improve work efficiency, and the model framework is shown in Figure 2.4.

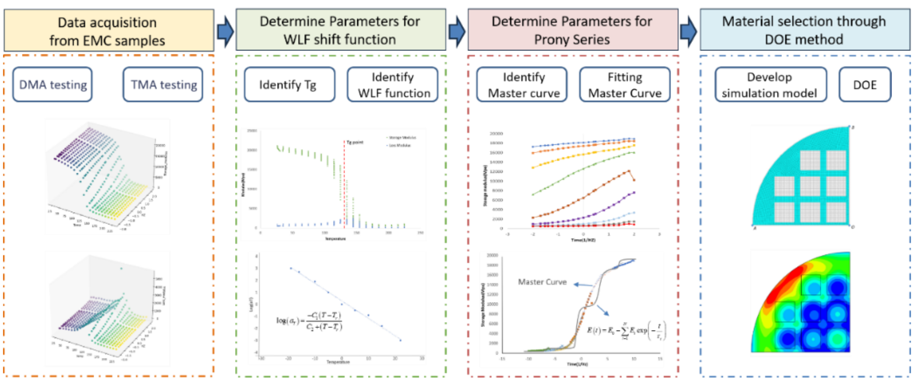

Figure 2.3 Flow chart for the traditional EMC constitutive parameter identification, four steps to generate, preprocess, manual fitting and application.

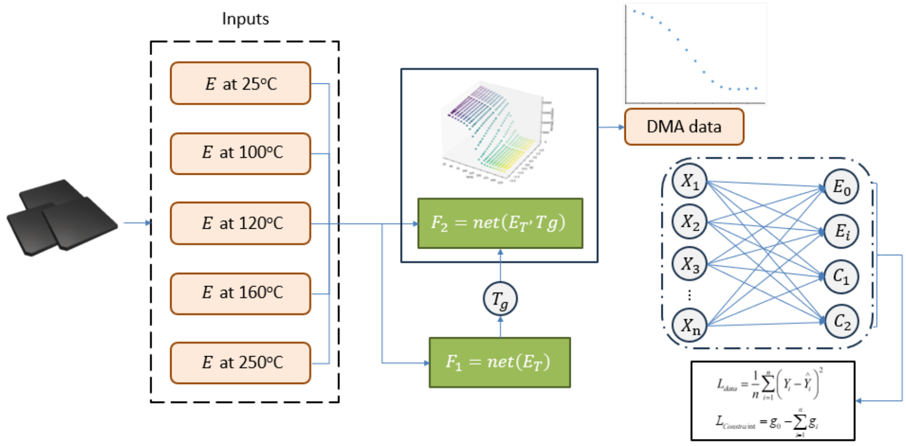

Figure 2.4 Proposed NPCNN framework for the EMC constitutive parameter identification.

This work collected and implemented 60 different EMCs with different modulus, CTE, and Tg for DMA and TMA testing. The storage modulus at the temperature ranges from 25 degrees to 250 degrees, and the Tg, CTE, and elastic modulus at several temperature points (seen in Figure 2.4) are extracted for the dataset.

The proposed NPCNN model includes a nested structure consisting of three sub-models, including F1 for Tg parameters prediction, F2 for storage modulus data prediction, and a PCNN model for the Prony series parameters prediction. In network F1, elastic modulus under different temperature points (25,100,120,160,250 degrees) are set as input variables to predict Tg. The predicted Tg will subsequently combine the elastic modulus as inputs for F2, and the network F2 calculates the prediction value of storage modulus at different temperatures. Since the Tg determines the form of the master curve and thus has an indirect effect on the constitutive parameters, the model employs the predicted Tg as an extra input to expand the content of prior knowledge for the model to learn the beneath relationship. The well-trained F1 and F2 models then generate the storage modulus versus frequency as input factors to predict WLF parameters and the Prony series. The definition of the Prony series is to characterize the modulus drop during time and temperature. The factor parameter E0 in the equation represents the instantaneous modulus, which means the modulus at t equals zero, and the rest of the parameters Ei represent the decrease in modulus over time, and the equation should obey the rules that the modulus should not drop below zero. Therefore, the loss of Prony series parameters is added to the loss function of PCNN to ensure the prediction value is under physics constraint. The loss function for the PCNN is shown below.

$$L_{data} = \frac{1}{n}\sum_{i=1}^{n}\left(Y_i - \hat{Y_i}\right)^2 \tag{2.8}$$

$$L_{Constraint} = g_0 - \sum_{i=1}^{n} g_i \tag{2.9}$$

$$L_{total} = L_{data} + L_{Constraint} \tag{2.10}$$

2.1.3 Model performance

The R squared, is used for model comparison in this work to reflects the degree of dispersion, and the specific definition is given as follows

$$R^2 = 1 - \frac{SS_{res}}{SS_{tot}} \tag{2.11}$$

$$SS_{res} = \sum_i (y_i - f_i)^2 \tag{2.12}$$

$$SS_{tot} = \sum_i (y_i - \bar{y})^2 \tag{2.13}$$

NN models have many hyperparameters and low correlation, so an optimization algorithm is used for NN hyperparameter selection. In this study, learning rate, batch size, epochs, weight decay, net layers, and net notes are considered hyperparameters to optimize. The objective function is the MSE loss of validation data. Table 2.1 lists the hyperparameter optimization results of NN.

Table 2.1 Hyperparameter optimization of NN models

Model	Epochs	Batches	Learning rate	Net layer	Net not	Weight decay
Range	50~800	16/32/64	(1e-5,1e-2)	1/2/3/4	3~20	(1e-5,1e-2)
F1	227	32	0.0012	3	9	5.123e-4
F2	412	32	0.001	3	8	0
PCNN	358	16	0.001	3	12	0

Comparing the runtime with the well-trained model and traditional manual fitting process in Figure 2.5, the proposed model's runtime is 13.2s, while that for the manual fitting process is about 5 hours. This demonstrates that the proposed model is both accurate and efficient in identifying the EMC constitutive parameters.

For the model without nested structure, optimal networks include 4 and 3 hidden layers, 7 and 9 nodes, and a learning rate of 0.002 and 0.001 for F2 and PCNN, respectively. Figure 2.6 shows the loss of training and testing. On the other hand, for the model with a nested structure, the optimal networks include 3, 3, and 3 hidden layers, 9, 8, and 12 nodes, and a learning rate of 0.0012, 0.001, and 0.001 for F1, F2, and PCNN, respectively. Figure 2.7 shows the training loss and testing loss of the proposed model. The performance of both models is within acceptable limits, and the R squared for most of the EMC constitutive parameters is above 0.95, showing good prediction accuracy. Comparing the nested structure model with the non-nested structure model, nearly all the constitutive parameters show a higher prediction accuracy. The R square for C1 is 0.9585 for the proposed model compared with 0.9425 for the non-nested model. The R square for C2 is 0.9765 for the proposed model compared with 0.9435 for the non-nested model. The R square for Ei is 0.9274 for the proposed model compared with 0.8646 for the non-nested model. The results demonstrate that adding expert physical relationships and experience as prior knowledge to the new network configuration generated during network creation does have higher accuracy when targeting specific problems.

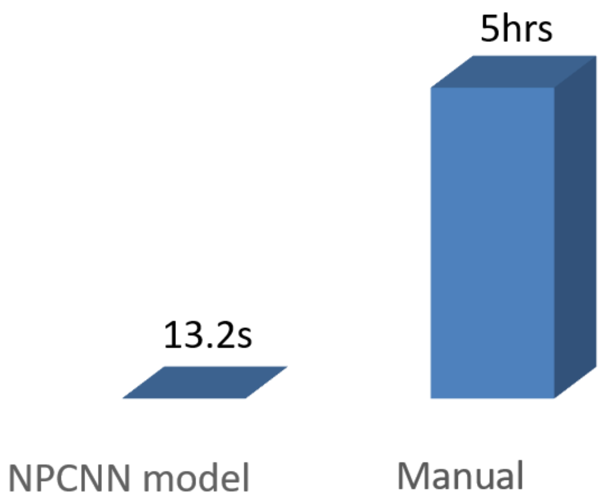

Figure 2.5 Comparison of Runtime for NPCNN model and Manual fitting process.

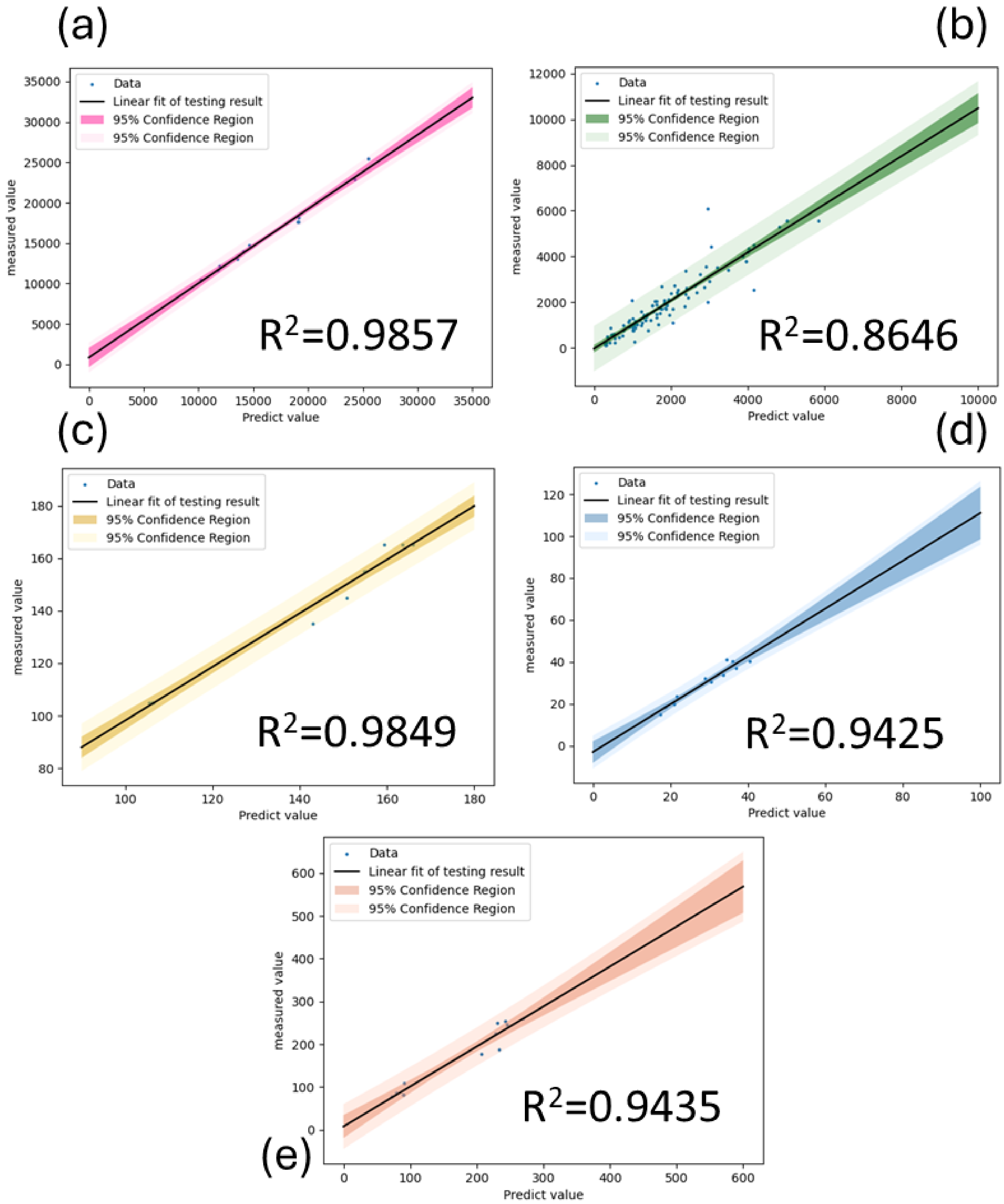

Figure 2.6 Illustration of the R-squared error and data distribution of EMC constitutive parameters obtained by FNN model for validation sets, Shaded areas represent 95% confidence bounds (a) E0 (b) Ei (c) RT (d) C1 (e) C2.

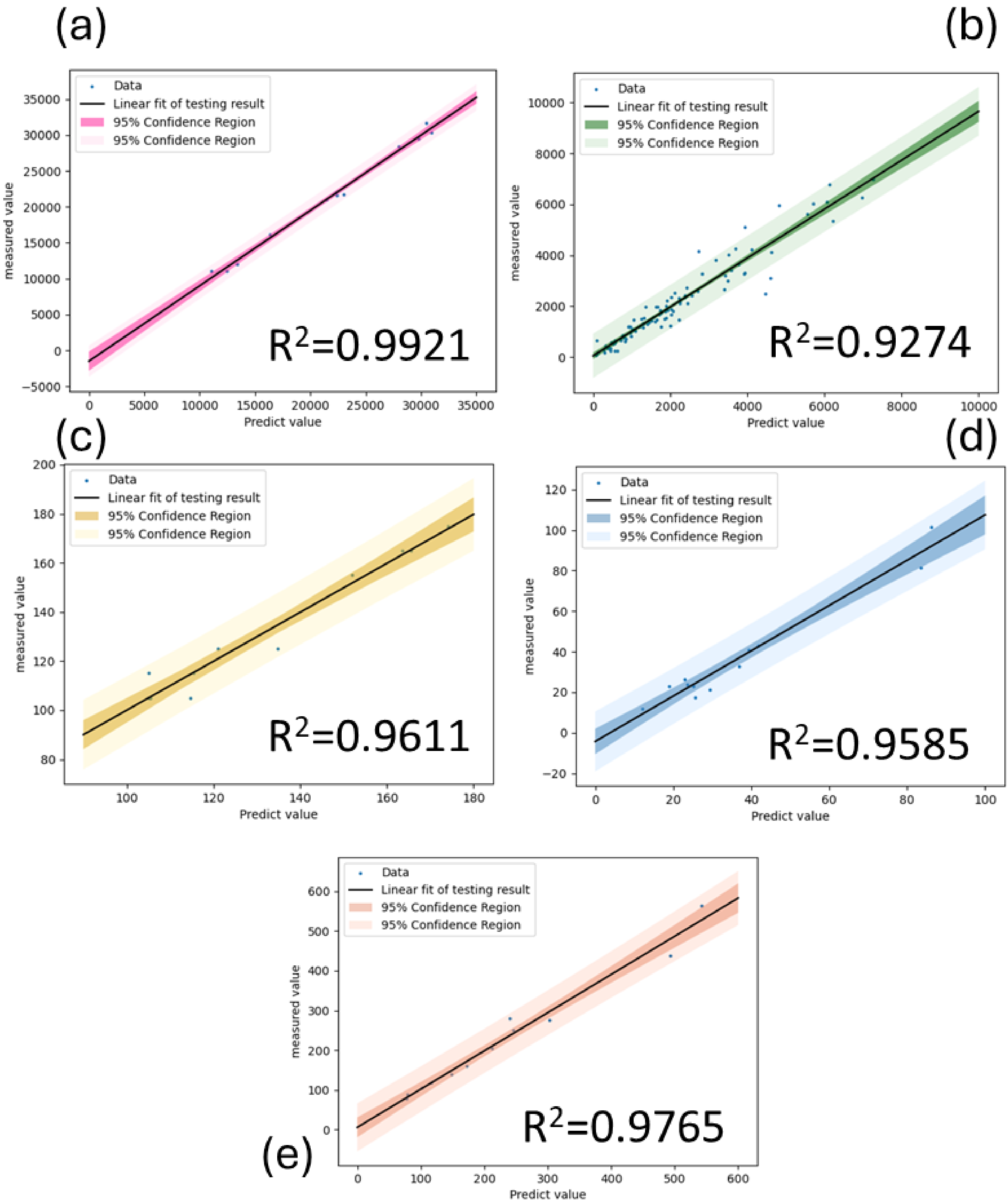

Figure 2.7 Illustration of the R-squared error and data distribution of EMC constitutive parameters obtained by NPCNN model for validation sets, Shaded areas represent 95% confidence bounds (a) E0 (b) Ei (c) RT (d) C1 (e) C2.

2.2 Chapter Summary

In this chapter, we first review and derive the core parameters that characterize the viscoelastic constitutive relation of EMC, and then an NPCNN model is constructed to learn the correct parameters of EMC viscoelastic constitutive parameters from the modulus at different temperature points. The proposed model shows good prediction accuracy and fast prediction time. The output EMC viscoelastic constitutive parameters can be used for simulation analysis to help design and select appropriate EMC under different working conditions.

CHAPTER 3

MULTI-FIDELITY APPROACH FOR AUTOMATIC EMC SELECTION UNDER THERMAL FATIGUE LOADING

Solder joint fatigue crack is one of the critical failure modes in IC packaging, particularly for power packages under a thermal cycling test (TCT). Design optimization and material selection were widely reported to achieve a high solder joint reliability performance. As an encapsulant, EMC is vital in package reliability. Its major parameters include modulus, CTE and Tg. For EMC selection, the traditional DOE approach based on reliability test and/or FE modeling is constrained by time and cost, limiting the possibility of getting the best combinations. The development of AI technology enables package design and material selection automatically, which can enhance the efficiency and robustness of devices.

This section developed an AI-enabled framework for automatic EMC selection for power packages to achieve high solder joint reliability performance under TCT. The model consists of a FE model, a multi-fidelity Bayesian optimization (MFBO), and an artificial neural network (ANN). A FE model was first constructed to characterize the accumulated creep strain in the solder joint under TCT. In the MFBO model, the properties of EMC, including Prony series, WLF shift function, and CTE, were employed as input, and accumulated creep strain per cycle (CEEQ) in the solder joint was used as output. The optimized Prony series and WLF shift function corresponding to the lowest creep strain in the solder joint was then forwarded to the trained ANN model and translated into modulus at different temperatures and Tg.

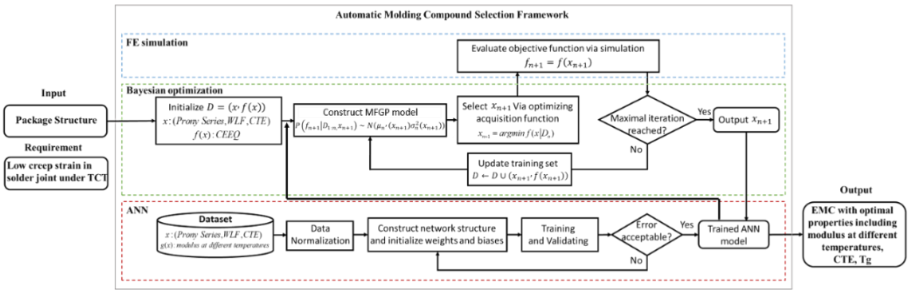

Figure 3.1 AI-enabled framework for automatic EMC selection under thermal fatigue loading.

3.1 Introduction of Solder Fatigue Failure

Solder crack [86-88], as shown in Figure 3.2, is a standard failure mode in package reliability tests, such as the thermal cycling test (TCT). With the requirement of high-power devices, more challenges are pointed to solder thermal and stress management.

As shown in Figure 3.3, the solder crack is mainly due to the CTE mismatch [89] among different components. The package structure will deform periodically during the temperature cycle. Due to the viscoelastic behavior of the EMC, the EMC will show a different modulus during the temperature cycle, and the rapid change of CTE above and below Tg increases the risk of thermal fatigue damage. Considering that EMC is not only the most significant factor affecting fatigue damage during thermal cycling but also a designable polymer composite material, it is necessary to optimize the mechanical behavior to select the best EMC to reduce the risk of package failure.

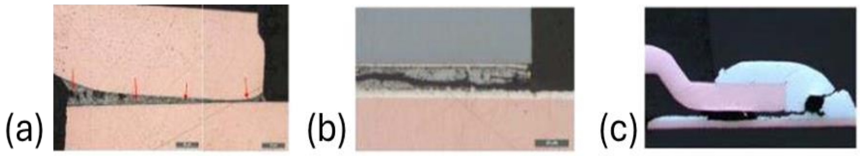

Figure 3.2 Typical fracture failure in packaging under thermal fatigue loading (a) Clip solder crack (b) Die attach crack (c) PCB solder crack.

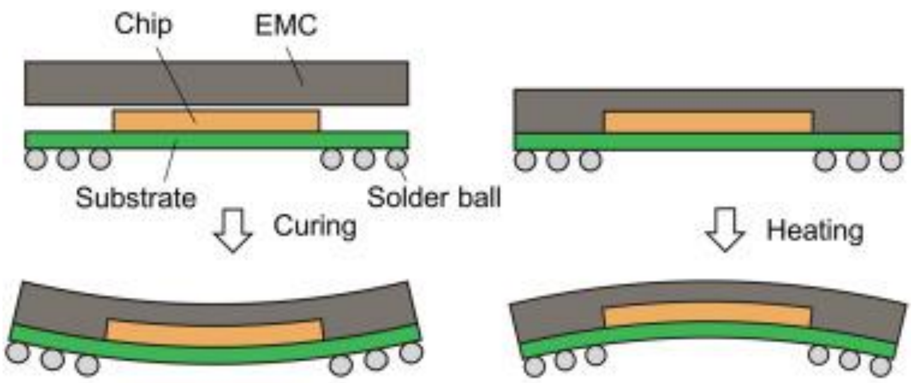

Figure 3.3 Solder cracks during thermal cycling testing due to CTE mismatch.

3.2 FE Model Development

The proposed framework has three parts: the FE simulation, the Bayesian optimization, and the ANN model. For the first part, an FE model was developed as a data generator to generate high-precision data. The previous section determined the core constitutive parameter for EMC, which can be used as the input variable for further simulation and optimization. This section develops an FE model to simulate and analyze thermal fatigue life under different EMC constitutive parameters.

A specific power package design under TCT loading is investigated as the benchmark for the proposed automatic molding compound selection framework. A full FE model includes the components of the EMC, PCB board, solder joint, lead frame, die-attach solder, die, clip, and clip solder, as shown in Figure 3.4. The package goes through varying temperatures during the TCT process, as illustrated in Figure 3.5. The material properties with time and temperature-dependent features are considered to ensure the high accuracy of the simulation results. In this model, the PCB board and die are assumed to be elastic and time-independent; the lead frame and clip are set as an elastoplastic model; die-attach solder and solder joint are considered Visco-plastic models. For molding compounds, the Prony series, WLF shift function, and thermal coefficient are considered to characterize the viscoelastic behaviors and are used as input variables for subsequent optimizations.

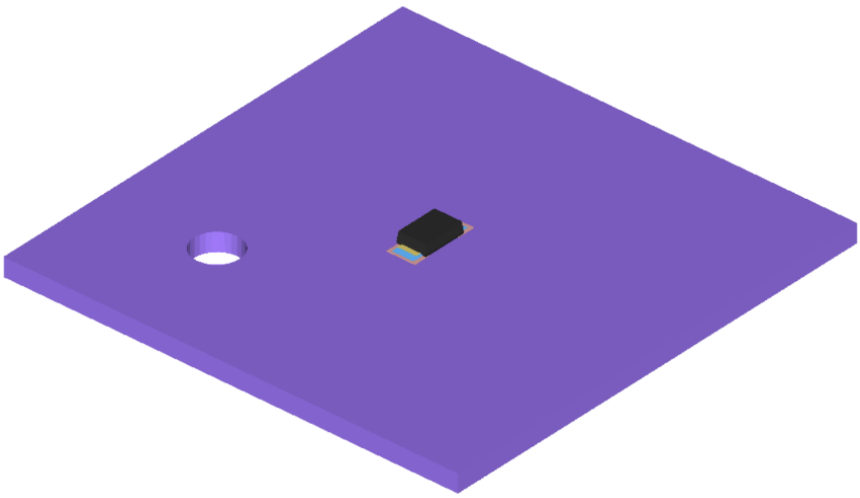

Figure 3.4 3D view of typical packaging architecture under consideration.

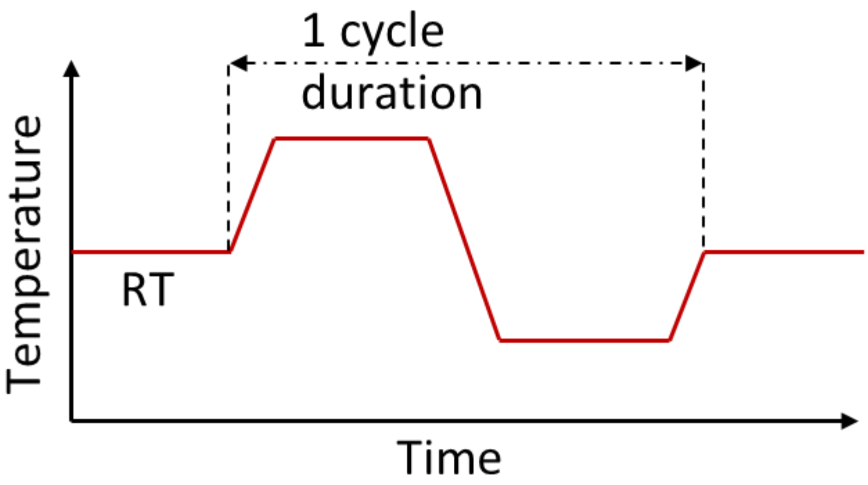

Figure 3.5 Thermal cycling testing profile.

The input variables for the FE model are the EMC constitutive parameters and the package structure, and the CEEQ of the solder joints is extracted as output and considered the optimized target. There is a correlation between CEEQ and the structural fatigue life of the solder joint [90]. The following equation can describe the relationship:

$$N_f = \left(C^{'}\varepsilon_{acc}\right)^{-1} \tag{3.1}$$

Where N_f represents the number of cycles to failure, ε_{acc} is accumulated creep strain per cycle and $C^{'}$ denotes the inverse of creep ductility.

As the FE simulation is regarded as the data generator for further optimization, it is necessary to ensure the accuracy of simulation results. The finite element method discretizes the analysis structure into a certain number of elements, connects the multi-segment elements through simple equations, and uses them to estimate the complex equations on a larger area. Therefore, if the number of division elements is huge and reasonable, the result is consistent with the actual situation. However, the large number of units will bring high-fidelity results while significantly increasing the simulation time, so finding a trade-off between accuracy and calculation time is necessary. Thus, mesh sensitivity must be considered to make the simulation result credible and establish an accurate quantitative numerical model. Figure 3.6 shows the runtime versus accuracy. As the mesh density increases, the accuracy of the finite element simulation will gradually approach a fixed value, that is, the analytical solution. In contrast, the calculation time increases exponentially with the mesh density. Figure 3.7 shows the mesh sensitivity analysis under the TCT process. Mesh sizes of 0.01mm, 0.02mm, 0.03mm, 0.04mm, and 0.05mm are considered in the simulation, and the calculation time and accuracy are compared. Assuming that the stress result and calculation time are 100% when the grid size is 0.01mm, the simulation result demonstrates that when the mesh size is 0.03mm, the model can obtain a pretty good accuracy (98%), and the runtime is only 4% compared with the model with mesh size 0.01mm. Therefore, this study uses a model with a mesh size of 0.03mm as a data generator.

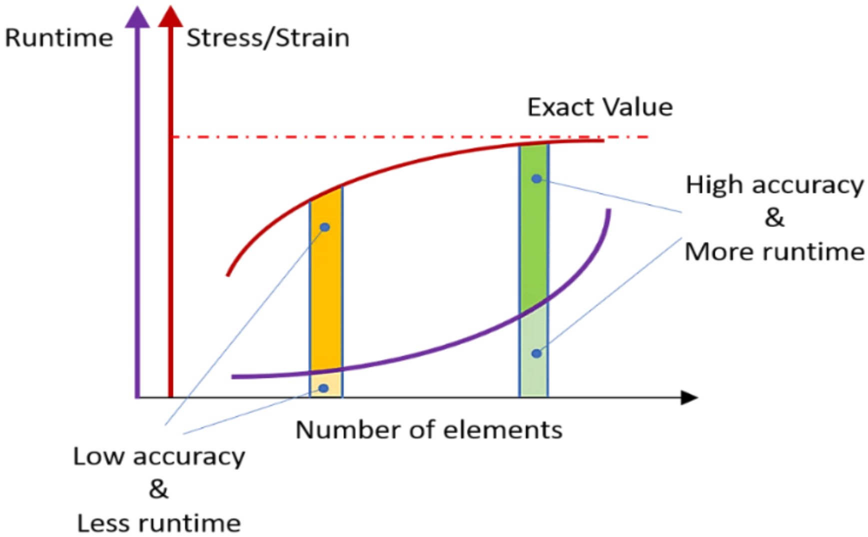

Figure 3.6 Illustration of the effect of mesh size on FE model accuracy and efficiency.

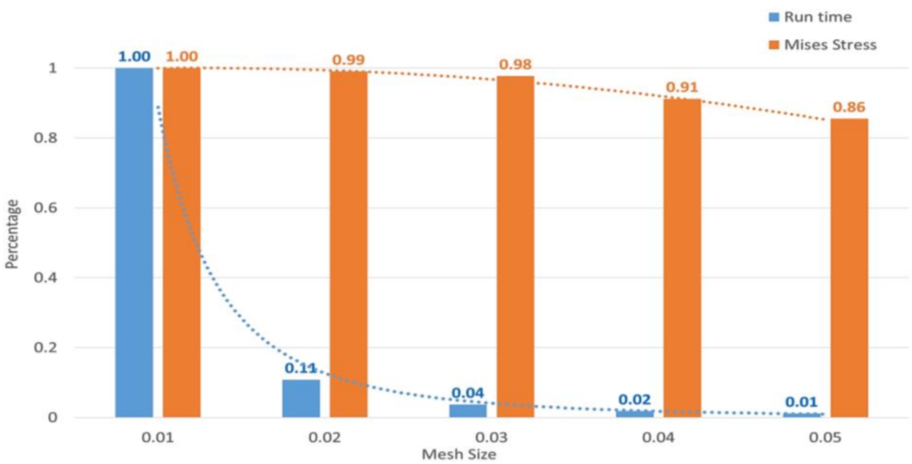

Figure 3.7 Comparison of normalized runtime and stress results under different mesh sizes.

3.3 Multi-Fidelity Approach

The FE models must be accurate enough to generate high-fidelity (HF) data as a data generator. Methods, including reducing mesh size and selecting high-order mesh types, are generally used to achieve this. However, these methods will lead to longer computation time and lower optimization efficiency. Therefore, MF technology is used to increase efficiency.

The MF method combines different surrogate models, which are built on scarce but accurate datasets (HF datasets) and large, approximate datasets (LF datasets) to improve prediction accuracy [91]. In general, a sparse mesh model is constructed to generate a low-accuracy but fast-evaluating model as a low-fidelity model and a finer mesh model to create a high-accuracy but slow-evaluating model as an HF model. However, this method is not suitable for models with components having a large size range and with viscoelastic or viscoplastic materials. This is because large mesh sizes can lead to difficulties in convergence, which will affect the accuracy of the results and the computational costs.

$$PEEQ = \int_0^t \sqrt{{}^2/_3\,[(PE_{11})^2 + (PE_{22})^2 + (PE_{33})^2 + 2(PE_{12})^2 + 2(PE_{23})^2 + 2(PE_{31})^2]}\,dt \tag{3.2}$$

$$CEEQ = \int_0^t \sqrt{{}^2/_3\,[(CE_{11})^2 + (CE_{22})^2 + (CE_{33})^2 + 2(CE_{12})^2 + 2(CE_{23})^2 + 2(CE_{31})^2]}\,dt \tag{3.3}$$

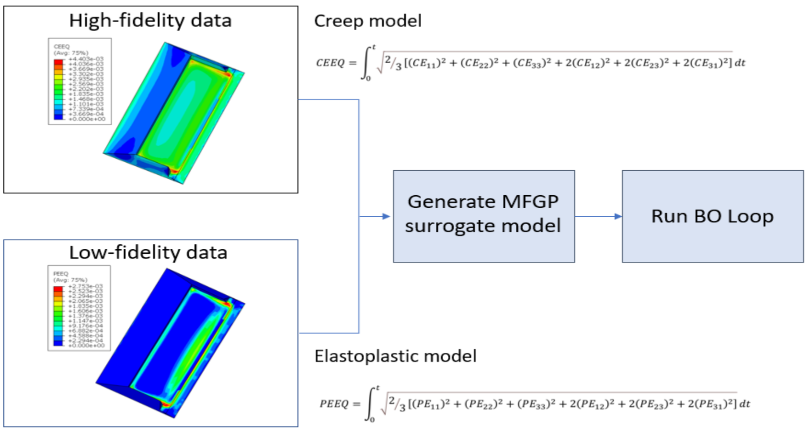

Figure 3.8 Workflow of applied multi-fidelity approach with Creep model for high-fidelity data and elastoplastic model for low-fidelity data.

A specially designed MF approach is developed in this work. The workflow is illustrated in Figure 3.8. An elastoplastic model is a low-fidelity model that replaces the EMC's viscoelastic behavior and the solder joint's viscoplastic behavior. The material property of the EMC in the low-fidelity model is changed from the Prony series, WLF shift function parameters to the elastic modulus under different temperatures. The material property of the solder in the low-fidelity model is changed from viscoplastic behavior to elastoplastic behavior, and the accumulated equivalent plastic strain (PEEQ) is put in, as a result, to replace the CEEQ in the high-fidelity model. An MF surrogate model is constructed for further optimization by combining a low-fidelity and high-fidelity model.

Figure 3.9 shows the calculation time for the creep model and elastoplastic model. The runtime for the creep model is 6618 seconds, while that for plastic is only 1066 seconds. This is because the plastic material is time-independent, and therefore, there is no need to consider the impact of the specific duration of the loading, unloading, and insulation stages on the simulation results. Equations 3.2 and 3.3 show the relationship between CEEQ and PEEQ, and Figure 3.10 shows the solder strain distribution for the creep and plastic models. Although the results of the elastic-plastic model are not accurate, it can be seen that the results of the elastic-plastic model have the same trend as the creep model.

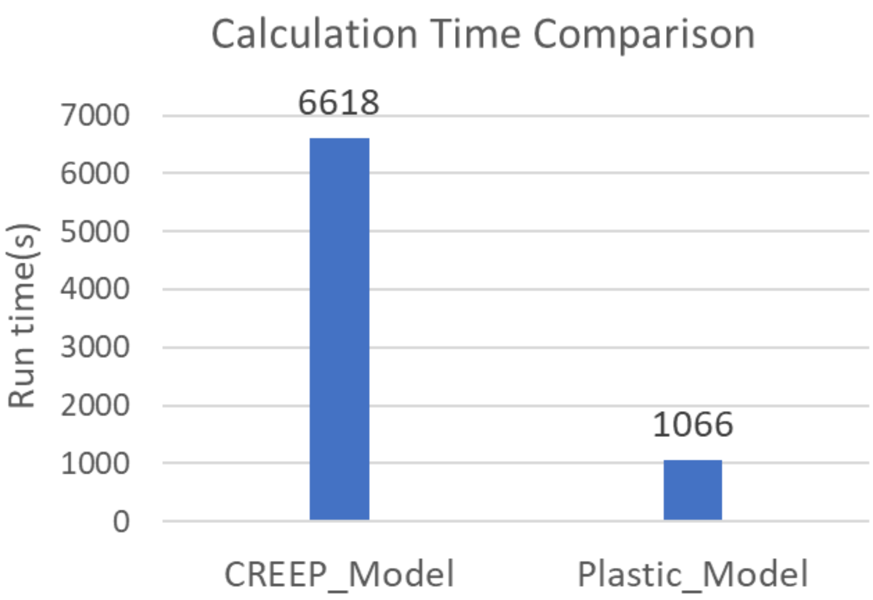

Figure 3.9 Comparison of calculation time for high-fidelity model and low-fidelity model.

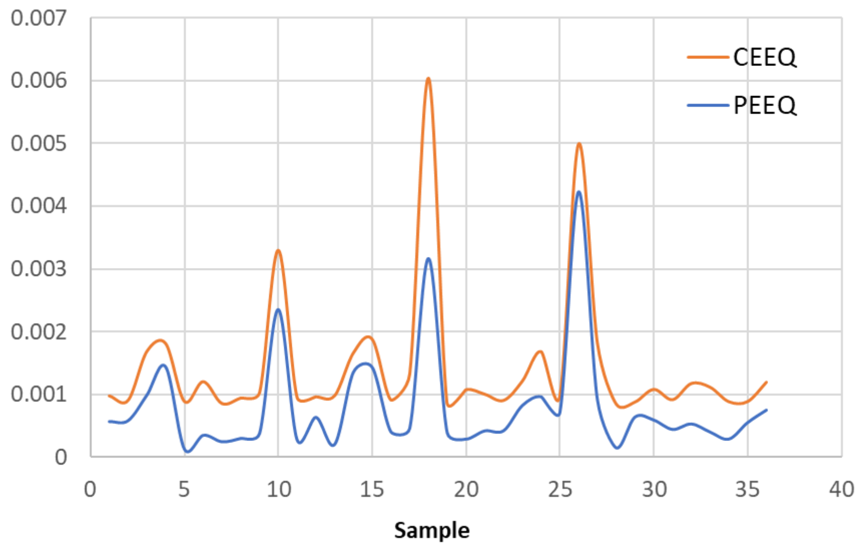

Figure 3.10 Comparison of solder strain of 35 samples for high-fidelity model and low-fidelity model.

3.4 Bayesian Optimization Algorithm Tuning

As introduced in the previous section, an AI-enabled design methodology uses the surrogate model to replace high-fidelity testing data or simulation results, and an optimization algorithm is used to search for the optimal solution. These design problems can be seen as such: we have a training dataset of T samples, (X_T, Y_T), and we want to perform a non-linear regression. Our goal is to learn this deterministic function, $f: X \rightarrow y$, or expressed as a probabilistic model that is $p(\theta|X_T, Y_T)$. The goal is to learn all possible model parameter combinations associated with the current model and dataset. Bayesian learning model, which uses the Gaussian process as a surrogate model and Bayesian Optimization as an optimization algorithm, has been shown to perform well in high-computational optimization scenarios due to its black-box optimization method. The schematic diagram can be seen below.

The Bayesian learning model collects existing data for training and gives the probability distribution of the unknown function under the current data. Then, the optimization algorithm selects the feature with unlabeled data that might be the optimal result and compares it with the criteria to determine whether the optimal solution is obtained.

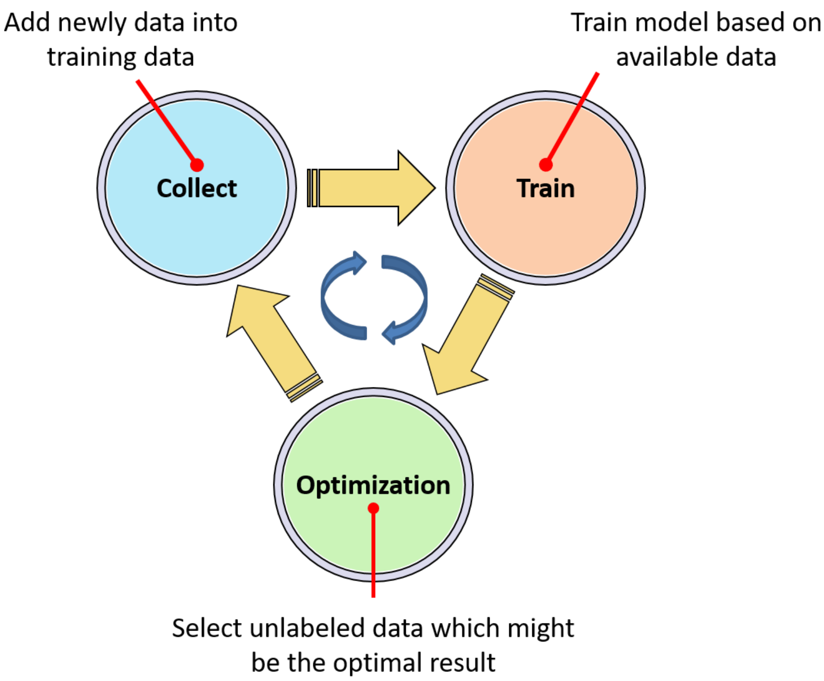

Figure 3.11 High-level summary of Bayesian learning optimization.

In this section, the Bayesian learning optimization method is used to design the EMC constitutive parameters with the lowest solder CEEQ. The specific model settings are shown in Table 3.1. The Gaussian process uses a joint Gaussian with the whole set of points to generate GP prior and calculate the posterior distribution of the unknown function, and the acquisition function using the posterior distribution to calculate and choose the following feature and label which might be the target output. Then, the feature will be used to build a parametric model in ABAQUS and run the simulation to get the high-fidelity label. The label will be evaluated via criteria to determine whether to be fed back into the loop to proceed to the next iteration or output the result.

Table 3.1 Models for pin location optimization

Optimization Model	Surrogate model	Optimization algorithm
Bayesian learning model	GP	BO

Acquisition functions are mathematical techniques that guide how the parameter space should be explored during Bayesian optimization. They use the predicted mean and variance generated by the Gaussian process model. For a set of such predictions on a set of candidate parameter sets, an acquisition function combines the means and variances into a criterion that will direct the search. The variance term generated by the Gaussian process model usually reflects the spatial aspects of the data. Candidate sets with high variance are not near any existing parameter values (i.e., those that have observed performance estimates). The predicted variance is very close to zero or near an existing result.

There is usually a trade-off between two strategies:

1. Exploitation focuses on results that are near the current best results by penalizing for higher variance values.

2. Exploration pushes the search towards unexplored regions.

The acquisition functions have quasi-tuning parameters, usually trade-offs between exploitation and exploration. This work selects the knowledge gradient (KG) and expected improvement (EI) as the MF and HF optimization acquisition functions, respectively. The definitions of these two functions are given as follows:

$$KG^n(x) = E\left[\max_{x' \in \chi} f^{n+1}(x') - \max_{x' \in \chi} f^n(x') \middle| S^n, x^n = x\right] \tag{3.4}$$

$$EI(x) = E\max\left(f(x) - f\left(x^+\right), 0\right)$$

$$= \left(f_{\min} - m(x)\right)\Phi\left(\frac{f_{\min} - m(x)}{s(x)}\right) + s(x)\Phi\left(\frac{f_{\min} - m(x)}{s(x)}\right) \tag{3.5}$$

where m(x) and s(x) are the mean and variance of the label at x, f is the function to be optimized with the estimated maximum at x+, ξ is a parameter controlling the degree of exploration and Φ(z), ϕ(z) denotes the cumulative distribution function and density function of a standard Gaussian distribution. EI takes both mean and variance into consideration and can better handle the exploration and exploitation trade-off in larger sample spaces.

Table 3.2 Design space for BO model.

Variable	Unit	Min	Max
E0	Mpa	20000	27000
E1-9	Mpa	100	5000
C1	/	15	100
C2	/	80	600
Tg	°C	105	200
α1	ppm/°C	5	15
α2	ppm/°C	15	80

After developing the FE model and Optimization model, iterative optimization can be performed to obtain the EMC constitutive parameters that minimize thermal fatigue damage to the package. Table 3.2 lists the range of the variables in the proposed framework. The range refers to the EMC samples in the previous section, which can cover most of the EMC.

Also, the physical constraint for the pony series should still be considered. Therefore, a constraint that E0 should be greater than the sum of the rest modulus Ei, which can be described as follows, is added to the optimization boundary [79].

$$\frac{\sum_{i=1}^{n} E_i}{E_o} < 1 \tag{3.6}$$

3.4.1 Model performance of the MFBO framework

The precision and efficiency of the MF model are compared to the HF model. As shown in Figure 3.12, the blue line with orange spot is the iteration result of HF model and the green line with yellow spot is the iteration result of MF model, the MF model exhibits a faster convergence speed in the initial stage, which is about 5hrs of the iteration. Compared with the HF model, the MF optimization model shows a faster convergence speed, and as the number of iterations increases, the two models will gradually converge to the same optimization result.

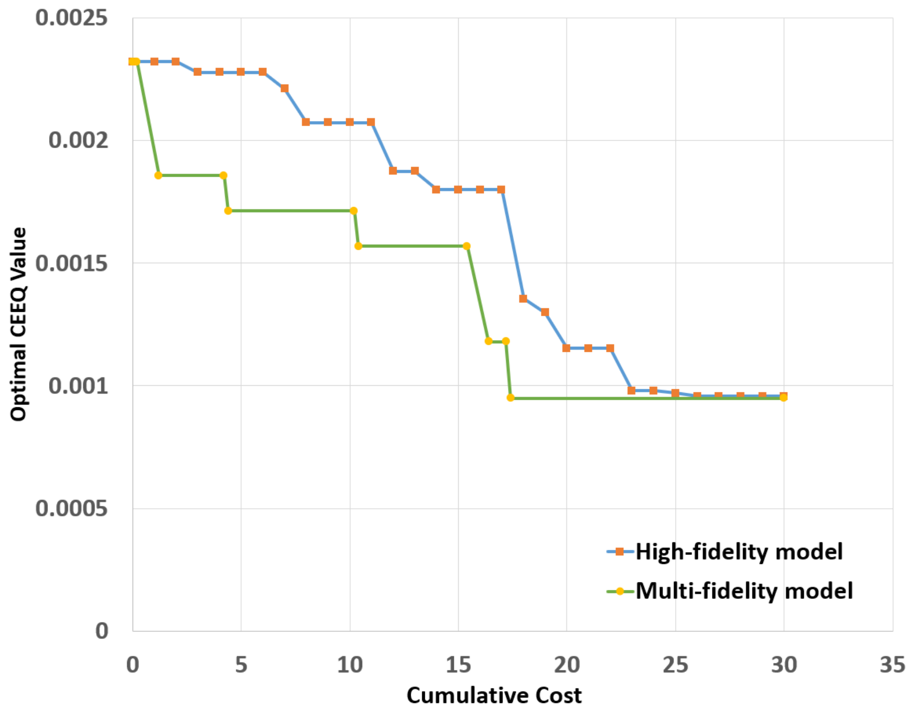

Figure 3.12 Comparison of the convergence for high-fidelity and multi-fidelity models on solder fatigue lifetime optimization.

Table 3.3 lists the cumulative cost and corresponding runtime of the two models. Using the MFBO approach, the runtime can be reduced from 42.9 hours to 32.6 hours, which is about 25% reduction. The results show that the use of a combination of elastic-plastic model and viscoelastic model to establish a MF database and conduct MF optimization can significantly improve the optimization efficiency while ensuring the consistency of the optimization results using only the HF viscoelastic model and provide ideas for the optimization of subsequent packaging models with more complex material and structural combinations.

Also, the optimal EMC constitutive parameters are listed in Table 3.4.

Table 3.3 Runtime of two optimization models.

Optimization model	Runtime (h)	Runtime Reduction
High-fidelity	42.9	\
Multi-fidelity	32.6	24.2%

Table 3.4 The optimal EMC property resulted from MFBO.

Variable	Unit	Optimized Value
E0	MPa	22598
E1	MPa	1321
E2	MPa	2289
E3	MPa	3203
E4	MPa	2990
E5	MPa	2782
E6	MPa	3809
E7	MPa	2857
E8	MPa	899
E9	MPa	1051
C1	/	23.6
C2	/	162.8
Tg	°C	106
α1	ppm/°C	13.1
α2	ppm/°C	32.6

Comparing the simulation result of optimal EMC and original EMC, the CEEQ distribution in the solder joint is shown in Figure 3.13 below. Due to the in-plane shear deformation caused by CTE mismatch, the risk area occurs at the solder joint–lead frame interface, as shown in the figure, while the solder CEEQ distribution for the optimal EMC shows a more uniform distribution.

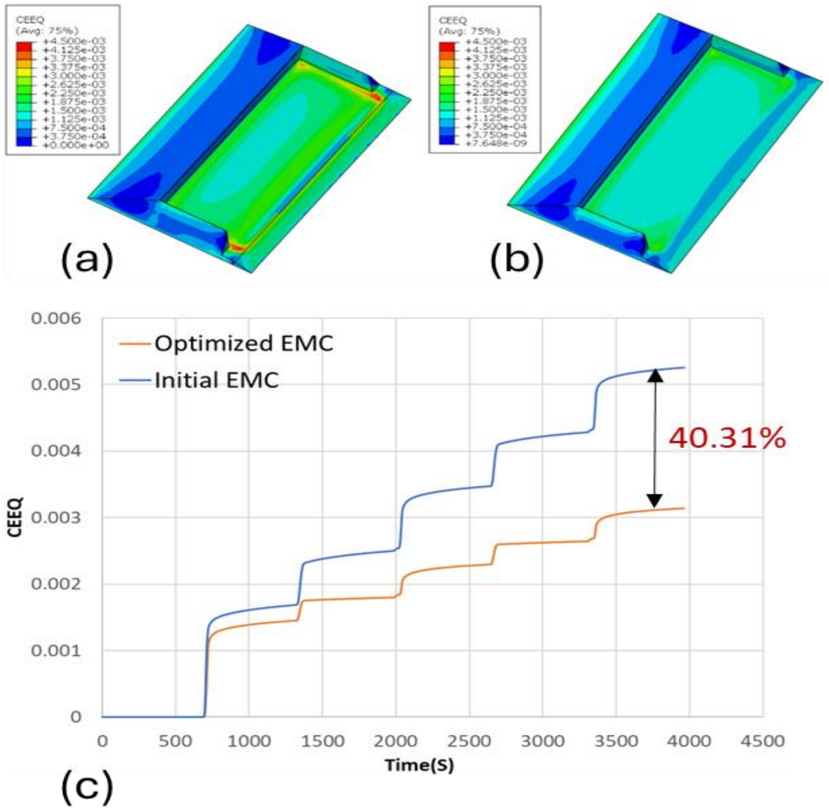

Figure 3.13 (a) Creep strain distribution in solder joint of Initial design, (b) Creep strain distribution in solder joint of Optimized design, (c) Comparison of Maximum Creep strain in solder joint for Initial design and optimized design.

The CEEQ-time curve is plotted by extracting the Max. CEEQ in the solder joint in Figure 3.13, and the curve demonstrates a 40% reduction in the Max. solder CEEQ for optimized EMC compared to the original EMC.

3.5 ANN Model for EMC Material Parameter Conversion

The previous sections have done the first and the second steps in the automatic EMC selection framework, and the Optimal EMC constitutive parameters are obtained, while this set of parameters is used for simulation and analysis and is not suitable for the selection on the supplier side. Therefore, seen in Figure 3.14, an ANN model is constructed to convert the EMC constitutive parameters into the modulus at different temperatures to meet the requirement of the supplier.

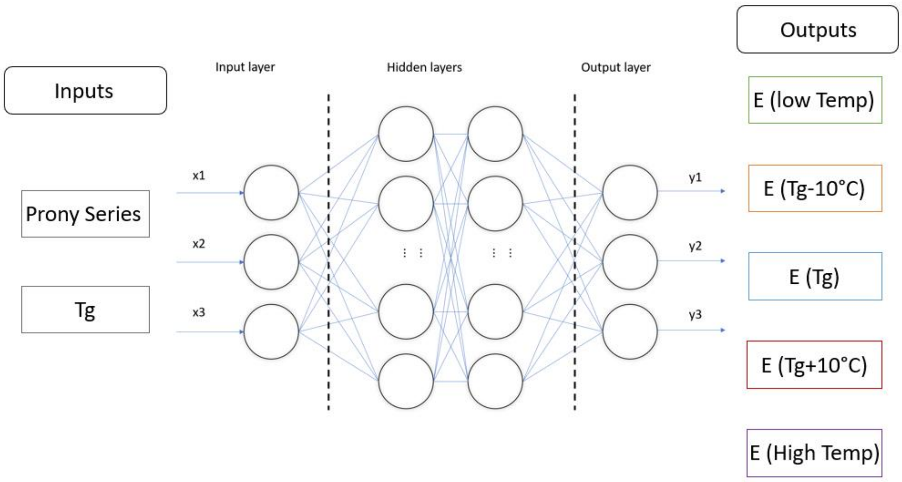

Figure 3.14 Proposed ANN model to transform the EMC constitutive parameters predicted from NPCNN to standard form.

Considering that the viscoelastic behavior of EMC is expressed in the form of Prony series in the previous MFBO model, these parameters are very convenient for simulation but are invalid parameters for the selection of EMC in industry. The Young's modulus of EMC at different temperatures, i.e. low temperature, tg-10 degrees, tg, tg+10 degrees, and high temperature, is used as the parameter for selecting the viscoelastic behavior of EMC in industry. Therefore, an ANN model is proposed in this section to uses the EMC constitutive parameters as input and the modulus at different temperature points as output to convert the parameters obtained by the previous optimization model to the parameters fit the industry requirement.

The ANN model is optimized by adjusting the learning rate, the number of hidden layers, and the number of nodes in each layer. The final network structure includes 2 hidden layers with 13 nodes in each layer, and the learning rate is 0.022. The training and prediction performance of the ANN model is presented in Figure 3.15. (a) shows the error of training and validation sets. (b)-(f) shows the prediction performances of modulus at different temperatures. The evaluation index for the five outputs is 3.43%, 2.13%, 5.44%, 5.25%, and 9.61%, respectively, for MAPE, and 0.9881, 0.9954, 0.9562, 0.9544, and 0.8783, respectively, for R-squared error. The prediction results demonstrate an acceptable accuracy, which reveals that this model can convert viscoelastic properties into reliable elastic parameters. The transformed elastic modulus at different temperatures is listed in Table 3.5.

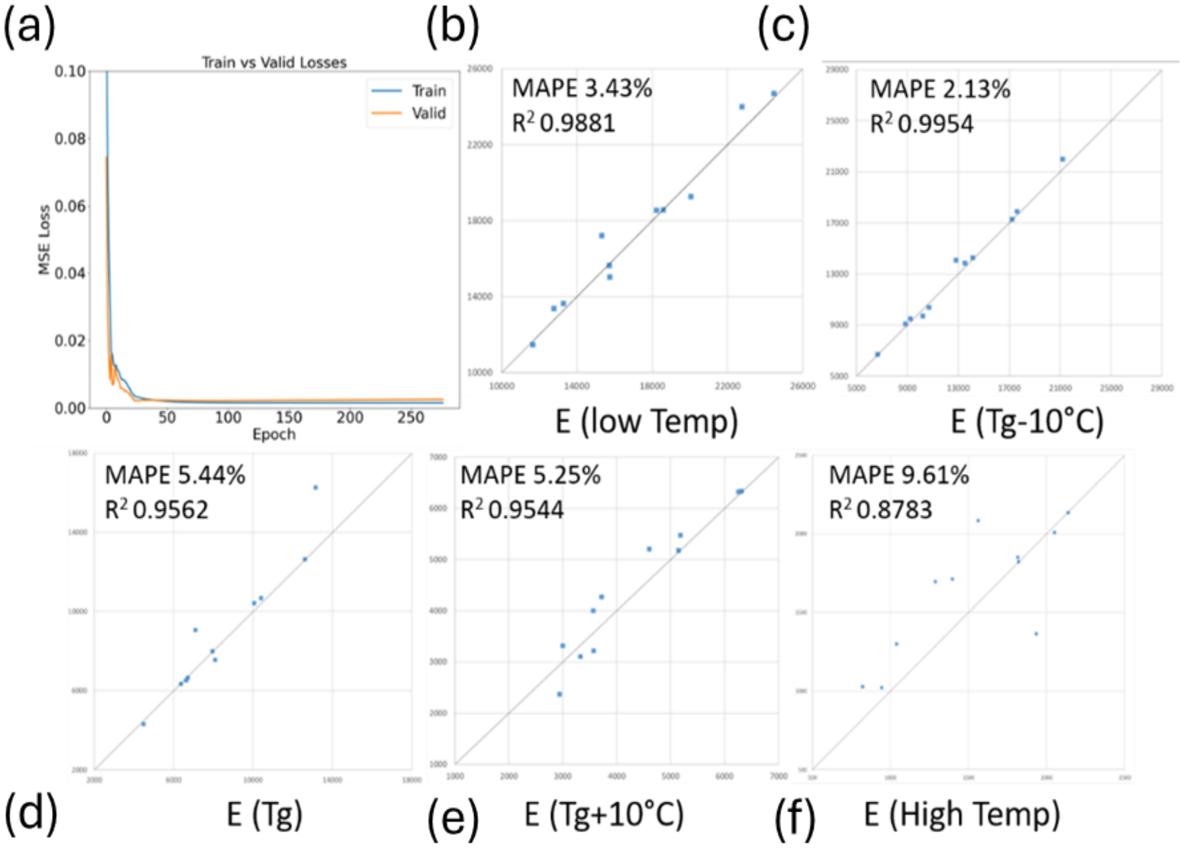

Figure 3.15 Illustration of the model loss, MAPE and R-squared error of the EMC parameter predicted from the proposed ANN model, (a) training loss and validation loss, (b) Modulus at low temperature, (c) Modulus at Tg-10°C, (d) Modulus at Tg, (e) Modulus at Tg+10°C, (f) Modulus at high temperature.

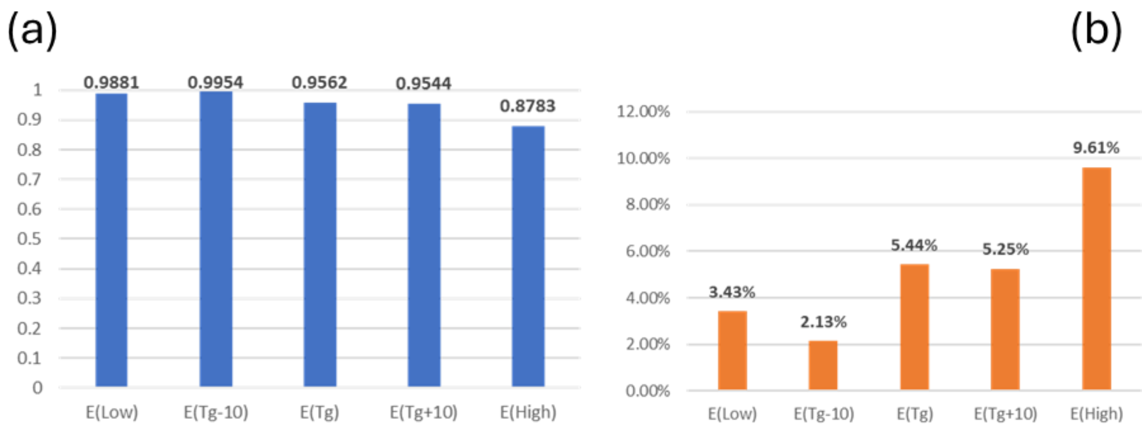

Figure 3.16 Summary of ANN model prediction performance for EMC parameters, (a) R-squared error, (b) MAPE.

Table 3.5 Result of automatic molding compound selection framework.

Variable	Unit	Optimal Value
E (25oC)	MPa	20744
E (Tg-10°C)	MPa	14619
E (Tg)	MPa	10946
E (Tg+10°C)	MPa	6240
E (250oC)	MPa	1704
C1	/	23.6
C2	/	162.8
Tg	°C	106
α1	ppm/°C	13.1
α2	ppm/°C	32.6

3.6 Model Validation

Experimental data is used to compare the performance of this automatic molding compound selection framework. Here, five types of EMC with different material properties are selected to build a DOE matrix, the fatigue lifetime under TCT process are measured as the objective for comparison. Table 3.6 lists the experimental results [92] with some crucial material properties and the corresponding fatigue lifetime. It can be seen from the table that EMC C, D, and E have much better fatigue performance than EMC A and B. These experimental results suggest that the acceptable material property range of EMC is within 19.7GPa to 22.4GPa for modulus, 11.3 ppm/°C to 13.0 ppm/°C for CTE, and 99°C to 104°C for Tg under TCT. These data are very close to the data listed in Table 3.6, generated from the automatic EMC selection framework, indicating that the proposed framework is valid.

Table 3.6 Experimental results with different EMCs.

EMC	E (MPa)	α1 (ppm/°C)	Tg (°C)	Failed units during TCT cycles		
				2k	3k	4k
A	26600	10.5	108	0	5	26
B	26700	10.9	98	0	5	28
C	22400	11.3	100	0	0	0
D	19700	12.3	104	0	0	0
E	21000	13.0	99	0	0	0

3.7 Chapter Summary

This chapter constructs an AI-enabled framework with a combination of FE simulation and machine learning model to automatically select the EMC with the highest performance under TCT. As evidenced by experimental data, this framework shows high prediction precision. Moreover, compared to traditional DOE methods, this framework's efficiency is remarkable, and it is very suitable for potential applications for package reliability assessment with a much shorter design loop. This scalable framework provides excellent potential opportunities in packaging automatic design.

The trained ANN model can transform the viscoelastic parameters into elastic parameters, which can fill the gap between the need for viscoelastic parameters for accurate FE simulation and the reality that only elastic parameters are provided by suppliers. Compared to the general BO method, the proposed MFBO method can effectively improve the optimization efficiency and reduce the computational cost. However, the runtime for the current MFBO is still relatively long, so further studies are needed to increase efficiency.

CHAPTER 4

MULTI-OBJECTIVE BAYESIAN OPTIMIZATION APPROACH FOR WARPAGE CONTROL

In the previous section, our work focused on one of the standard failure modes in package reliability tests: the solder crack under thermal fatigue loading. Considering that thermal cycling simulation is quite time-consuming and complex, a multi-fidelity approach is applied to the optimization framework to reduce the design cost. In this section, another standard failure mode frequently occurring in recent package design, warpage control, is considered, and a multi-objective approach is applied to help select the EMC effectively.

In recent years, based on the multi-functional requirements of electronic products, the Fan-Out Panel-Level Package (FOPLP), as an advanced packaging architecture with lower cost, thinner profile, and better electrical and thermal performance, has provided a potential solution for the semiconductor industry. However, FOPLP meets the warpage problem due to the shrinkage of the epoxy molding compound (EMC) and the mismatch of coefficients of thermal expansion (CTE) between EMC and the substrate during the assembly process, decreasing the yield rate and packaging reliability. Therefore, selecting a proper EMC to reduce thermally induced warpage is essential.

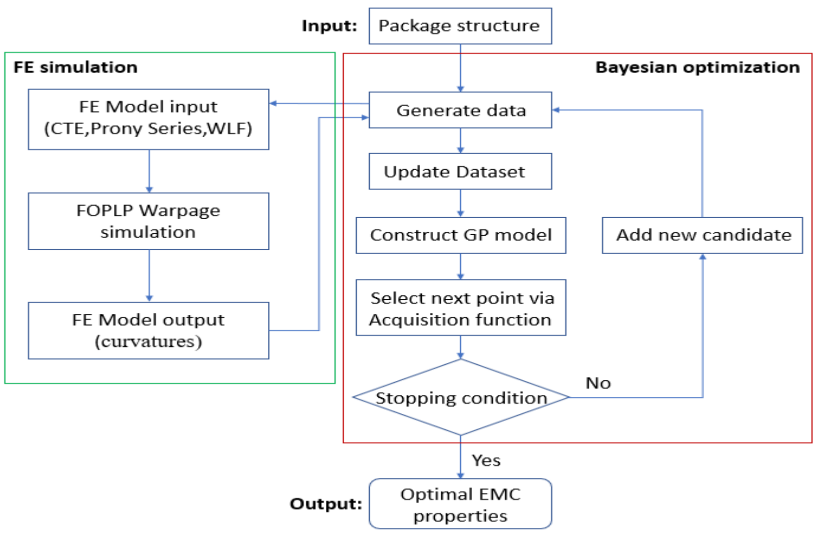

Figure 4.1 Workflow for the proposed MOBO framework for warpage control.

As shown in Figure 4.1, this section proposes an AI-based design framework to provide EMC material properties with minimal warpage deformation during assembly. A high-fidelity FE model of FOPLP with a 300 × 300 mm^2 carrier size is first built to characterize warpage deformation during assembly. The FE model is then regarded as a data source for the multi-objective Bayesian optimization model (MOBO). The material properties of EMC, including the Prony series, WLF shift function, and CTE, were employed as variables, and two curvatures at in-plane directions were employed and constructed as output for MOBO.

4.1 Introduction of Warpage Deformation in Packaging

FOPLP or FOWLP, as a large-scale IC packaging technology, owning advantages such as larger substrate size, high area utilization, high integration and flexibility and low costs, have been an ideal selection for applications requiring robust performance in compact forms, such as AI servers and modern communication devices. Figure 4.2 shows the typical structure of FOPLP and FOWLP, which the differences is whether the die is placed on round wafers or square panels, and for such packaging technology, many dies are usually placed on the substrate simultaneously and covered with EMC in the manufactory process. Due to the CTE mismatch between components, excessive warping, and structural failure can easily occur in the subsequent process.

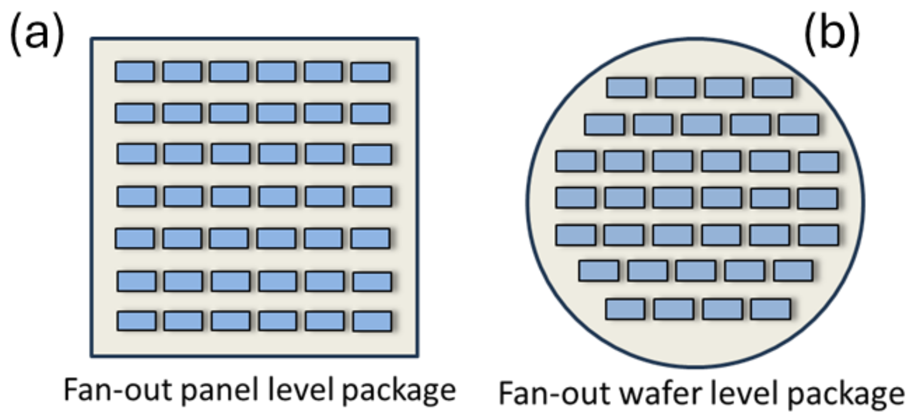

Figure 4.2 Large-scale IC packaging technology: (a) Fan-out panel level package, (b) Fan-out wafer level package.

Considering the in-plane dimension of large-scale packages is much larger than the out-of-plane dimension. The problem can be regarded as an out-of-plane deformation problem of multilayer shells, and the corresponding governing equation can be illustrated as follows [93].

Die stress along the top line:

$$\sigma_{Top} = \frac{\Delta\alpha\Delta T}{\lambda t_1}\left(1-3\frac{tD_1}{t_1 D}\right)\left(1-\frac{\cosh kx}{\cosh kl}\right) \tag{4.1}$$

Curvature:

$$\frac{1}{\rho(x)} = \frac{t\cdot\Delta\alpha\Delta T}{2\lambda D}\left(1-\frac{\cosh kx}{\cosh kl}\right) \tag{4.2}$$

Deflection:

$$\omega(x) = \frac{t\Delta\alpha\Delta T}{2\lambda D}\left[\frac{1}{2}x^2-\frac{\cosh kx-1}{k^2\cosh kl}\right] \tag{4.3}$$

Referring to the out-of-plane deformation governing equation of the multilayer structure, the key parameters that affect the warping deformation include:

1. Temperature
2. CTE difference
3. Thickness
4. Modulus
5. In-plane dimension

Among these critical factors, the EMC, as a viscoelastic material, plays a vital role in warpage control. Firstly, the instantaneous change of CTE when passing through the Tg point will cause a drastic change in the structural warping direction during the loading process, thus affecting the product yield. Secondly, the viscoelastic properties of EMC will significantly impact the deformation mode [94] of the packaging structure and affect the design method to minimize the warpage deformation. While such problems have not yet been fully understood, warpage orientation rotation cannot be explained using the conventional approach, giving a vast challenge to warpage control.

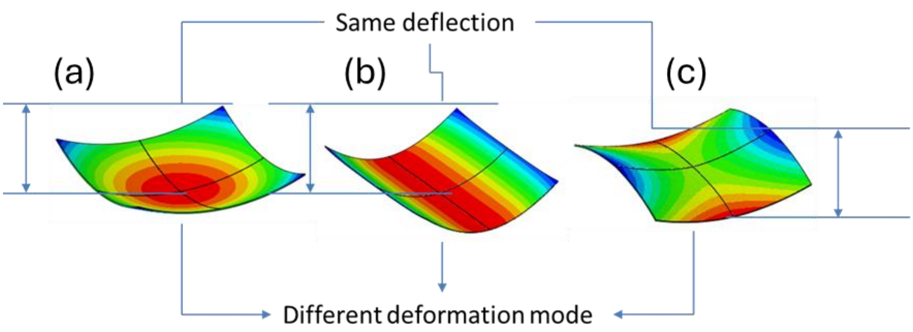

Figure 4.3 Warpage modes of large-scale IC packaging under thermal loading affected by EMC material property: (a) spherical shape, (b) cylindrical shape, (c) saddle shape.

Figure 4.3 shows the deformation mode that may occur in large-scale package warpage deformation. The three warping modes are different, namely spherical, cylindrical, and saddle-shaped, but the three warping deformations have the same deformation value, since the packaging process usually has many steps, unpredictable warpage deformation shape will bring great difficulties to the warpage release and matching of subsequent processes. Although the inner mechanism may not be fully understood, the FE simulation can perform different deformation modes under different EMC materials, and the machine learning algorithm can regard the unknown mechanism as a black box to help select the optimal EMC with minimum warpage deformation.

4.2 FE Model Development

Firstly, a detailed FOPLP FE model with a 300x300mm^2 carrier size is constructed and validated by experiment and then regarded as the benchmark for the proposed optimization model development. To reduce the simulation cost, a quarter model includes a die, clip, bump, copper trace, die pad, and molding compound developed for high efficiency. Figure 4.4 (a) shows the top view structure of FOPLP. In the quarter model, 1156 dies are located on the panel, and Figure 4.4 (b) shows the inner component of a single unit, the model contains die, clip, bump, lead frame and EMC, where the die and the clip are bonded by the bump, and the panel is connected via copper. Finally, the molding compound covers the structure.

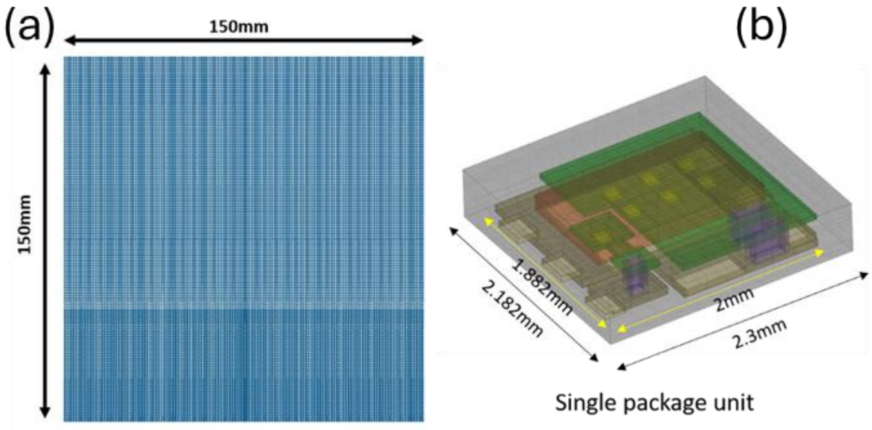

Figure 4.4 (a) Top view of the FOPLP package under consideration, (b) The dimensions and components for a single unit in FOPLP.

The typical material properties for components in FOPLP are listed in Table 4.1. The package structure will experience a high temperature to room temperature cooling stage during the assembly process and will cross the glass transition temperature. Therefore, EMC, as a time-dependent and temperature-dependent material, needs an accurate model to describe its mechanical properties. For the developed FE model, the clip, die pad, die, and bump are assumed to be elastic materials, and the EMC is considered a viscoelastic material.

Table 4.1 Material property of components in FOPLP.

Component	Elastic Modulus (GPa)	Poisson's Ratio	CTE (ppm/°C)
EMC	24.8	0.35	10.0(<125°C), 42.0(>125°C)
Die	169.0	0.23	2.6
Copper/Bump	105	0.35	17.7
Clip/Die pad	130.0	0.35	17.0

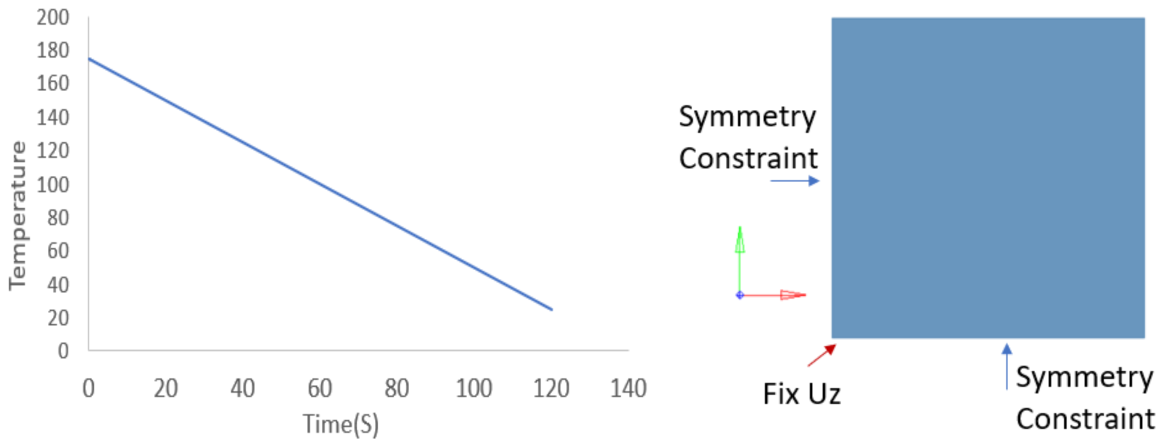

Figure 4.5 Temperature profile and boundary condition of FE model under the assembly process.

The typical temperature profile for the FOPLP assembly process is shown in Figure 4.5. The package is cooling down from 175°C, which is assumed to be strain-free, to room temperature (25 °C) in 120s, and as the quarter FE model is constructed, symmetry constraints are imposed on the surface of symmetry of the x and y axes, and out-of-plane displacement is constrained at the center of symmetry.

4.2.1 FE model validation

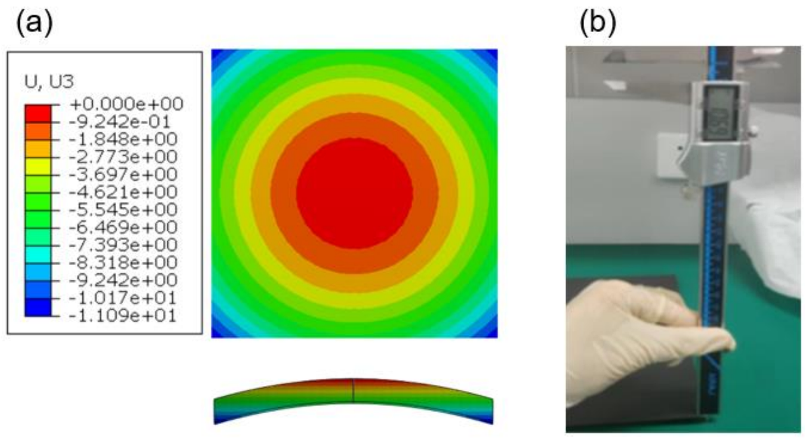

Figure 4.6 The maximum warpage deflection for FOPLP under assembly process: (a)Simulation result, (b)Experimental measurement.

Table 4.2 Maximum warpage deformation in FOPLP structure

Method	Warpage deformation(mm)	Difference
Experiment	10.69	\
FE simulation	11.09	+3.74%

The warpage simulation was constructed by Abaqus 2019, and to ensure the simulation can obtain a variety of possible warpage deformation modes, the Visco analysis procedure with explicit/implicit integration and an adaptive stabilization damping coefficient was set [95]. The warpage simulation result for the initial FOPLP design is shown in Figure 4.6 (a) and (b). Table 4.2 listed the warpage deformation measured from the simulation result and experiment. The warpage deformation from the simulation result is 11.09mm and that from the experiment is 10.69mm. The difference is about 3.74%, and this error is acceptable, so this model can be used as a benchmark to generate data for warping optimization.

4.3 Multi-Objective Bayesian Optimization Approach

As reviewed previously, the maximum deflection is a typical result of warpage deformation. However, this value has several limitations, making it inappropriate for optimization. Firstly, warpage deflection is size-dependent, due to the boundary effect, the value changes with package size. Secondly, previous studies have shown that the viscoelastic behavior of EMC will cause a different deformation mode in the package, including spherical mode, cylindrical mode, cylindrical diagonal mode, and saddle mode. On the one hand, different deformation modes will affect the position of the maximum displacement point, and secondly, different deformation modes may be dominated by different material properties. Therefore, only extracting the maximum displacement will cause information about the structural warpage deformation to be lost [96].

To avoid the boundary effect, show the package's deflection mode. This work extracts the curvature along two in-plane axes to replace the typical maximum warpage deformation. Then, the curvatures of the two axes are set as outputs for the next step, multi-objective optimization. In general, a FE model is first constructed and validated. Then, the developed FE model is considered as a data source for warpage management, where the EMC material properties, including CTE and viscoelastic behavior, are the input space, and the curvature for two in-plane axes is combined to generate the target function. A transforming method is used to convert the multi-objective optimization to mono-objective optimization [97-98]. The weighted sum of normalized objectives is used to generate the target function, and this can be expressed as follows:

$$Y_t = \sum_{i=1}^{n} W^i \times norm(y^i) \tag{4.4}$$

where Yt is the target function, n is the number of the variables, y is the objective function, and W is the corresponding weight. For maximization and minimization optimization, W is defined by the following formula:

$$W^i = \left(1 - \frac{y_{\max}^i - \delta y^i}{y_{\max}^i}\right)^{-1} \tag{4.5}$$

$$W^i = \left(1 - \frac{y_{\min}^i}{y_{\min}^i + \delta y^i}\right)^{-1} \tag{4.6}$$

where y_{max}^i and y_{min}^i represent the maximum result of each objective function obtained through single-objective optimization, respectively. $\delta y^i$is the controlling factor of weight.
The constructed FE model is then regarded as a data source to simulate specific EMC material properties guided by BO. For optimization, the design space is the material properties of EMC, a total of 15 variables, 9 of which are the parameters in the Prony series, 2 are the empirical parameters in the WLF equation, and 2 are the thermal expansion coefficients above and below the Tg point, and the last term is the Tg point. The variables and the corresponding range are listed in Table 4.3.

Table 4.3 Input space of FOPLP warpage optimization

Variable	Unit	Min	Max
E0	MPa	20000	27000
E1-9	MPa	1000	5000
C1	/	15	100
C2	/	80	600
Tg	°C	105	200
α1	ppm/°C	5	20
α2	ppm/°C	20	50

For the multi-objective optimization, the curvature for the x-axis and y-axis are simultaneously optimized by introducing the target function.

$$C_t\colon C_t = w^1 \times C_x + w^2 \times C_y \tag{4.7}$$

In this work, δCx is fixed to 1e-6, and a set of Pareto optimal points are calculated and determined by changing the δCy and maximizing Ct. Considering that the deformation in the X and Y directions of the FOPLP structure optimized in this work does not have a significant uneven weight distribution, the values of δCy are selected as 1e-7, 1e-6, 5e-6, 1e-5, 5e-5, 1e-4, 5e-4, and 1e-3, which are in a similar order of magnitude to δCx. Different δCy are respectively substituted into formula 4.7 for calculation and converted into the corresponding objective function, and the MOBO model is established to optimize and record the optimal results. The values of Cx and cy as δCy changes are recorded and shown in Figure 4.7.

The design boundary of Cy shows a sudden degradation when δCy is greater than 5e-5, which means the total objective function has reached the Pareto frontier and explore the optimal result. Thus, 1e-6 for δCx and 5e-5 for δCy are selected to develop the target function and for optimization.

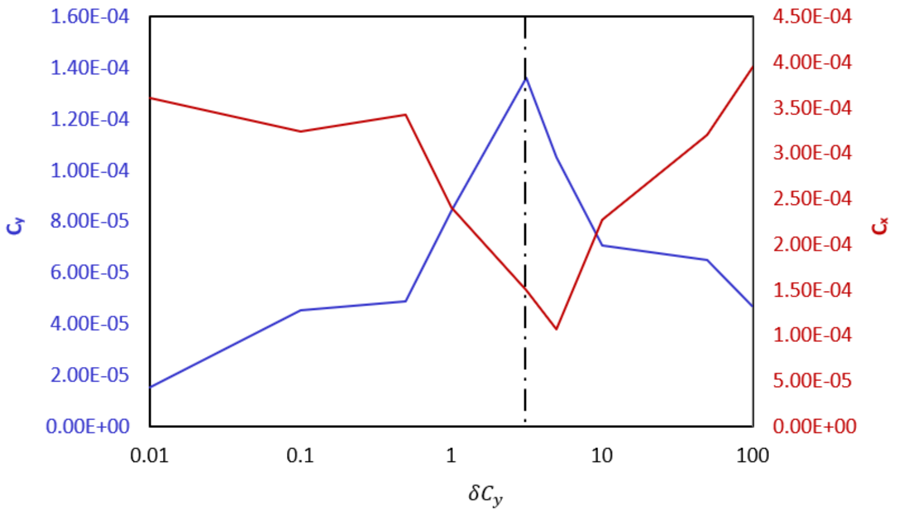

Figure 4.7 Pareto optimal points of Cx and Cy corresponding to δCy.

Figure 4.8 demonstrates the iteration process of warpage optimization. The objective function decreases rapidly in the first few iterations and enters a stable state after nearly 30 cycles, and there is no further optimization result until the 80th cycle. Therefore, it is assumed that the established MOBO model obtains the global optimal solution after 80 steps of iteration, and at this time, the objective function, Ct, is 2.1e-5.

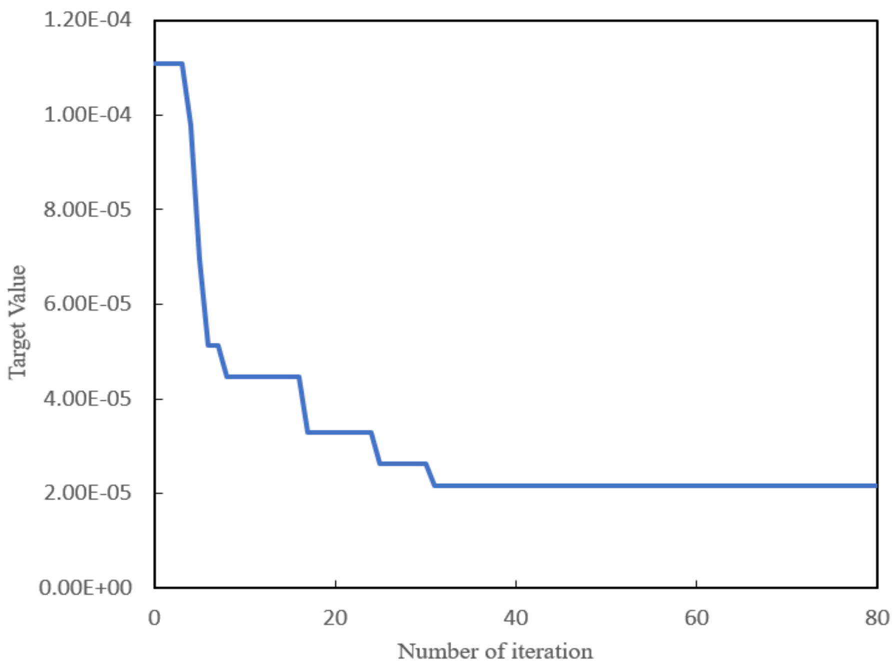

Figure 4.8 Convergence of curvature Ct for FOPLP under assembly process.

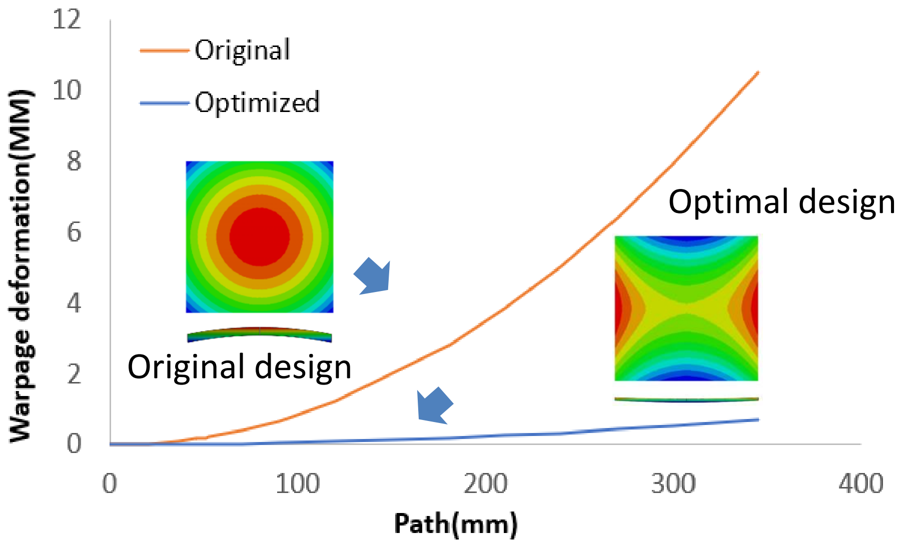

Figure 4.9 The maximum warpage deflection and warpage shape for initial and optimized EMC design of FOPLP.

The warpage index for the initial design and globally optimal design are listed in Table 4.4 and the deformation cloud map is shown in Figure 4.9. For the global maximum deformation, the objective function Ct changes from 1.108e-4 for initial design to 0.217e-4 for optimized design, and the corresponding deflection reduces from 11.09mm to 0.709mm, a drop of more than 93%. Meanwhile, seeing from Figure 4.9, the curve illustrates the maximum warpage deflection which the path is from the center of the package to the edge of the package, the deformation cloud map demonstrates that the deformation shape for the initial design is spherical shape while for the optimized design is saddle shape. This is because of the viscoelastic behavior of EMC, that is, the package will generate a sudden deformation which is due to the fast change of CTE when passing the Tg point.

Table 4.4 Target value in Optimization

FOPLP	Ct	Max deformation(mm)	Difference
Initial Design	1.108e-4	11.09	\
Optimal Result	0.217e-4	0.709	-93.6%

4.4 SHAP Analysis

Considering that the FOPLP panel's deformation shape has also changed from sphere mode to saddle mode, it is essential to explain the ML model and explore the critical factor of the deformation mode for further warpage control.

As the demand for interpretability in machine learning projects grows, the need to effectively communicate the complex outputs of model explanation techniques to non-technical stakeholders becomes more pressing. In this context, SHAP (Shapley Additive Explanations) [99] emerges as a powerful tool crucial in explaining how machine learning models make predictions.

The definition of Shapley value for each feature i is as follows.

$$\Phi_i(v) = \sum_{S \subseteq F \setminus \{i\}} \frac{|S|!\left(|F|-|S|-1\right)!}{|F|!}\left[v(S \cup i) - v(S)\right] \tag{4.8}$$

$$v_x(S) = \int f\left(x_1, \ldots x_F\right) d\mathrm{P}_{x \notin S} - E_X[f] \tag{4.9}$$

Where v is the target function and v(s) are the prediction for feature values in set S that are marginalized over features not included in set S minus the average prediction. Four fundamental properties of Shapley values help it be considered a definition of a fair payout and allow for an ML model explanation: Efficiency, Symmetry, Null Player, and Additivity. The definition of these properties is shown below respectively.

Efficiency,

$$\sum_{i=1}^{F} \Phi_i = f(x) - E_x[f] \tag{4.10}$$

Symmetry,

$$\forall S \subset F : f_{S \cup \{i\}}(x_{S \cup \{i\}}) = f_{S \cup \{j\}}(x_{S \cup \{j\}}) \Rightarrow \phi_i(x) = \phi_j(x) \tag{4.11}$$

Null Player,

$$\forall S \subset F: f_{S\cup\{i\}}(x_{S\cup\{i\}}) = f_S(x_s) \Rightarrow \phi_i(x) = 0 \quad (4.12)$$

Linearity:

$$\begin{aligned} &\phi_i^{f+g}(x) = \phi_i^f(x) + \phi_i^g(x) \\ &\forall a \in \mathbb{R}: \phi_i^{a \cdot f}(x) = a \cdot \phi_i^f(x) \end{aligned} \quad (4.13)$$

Here, efficiency means the sum of the Shapley values of all features equals the difference between the prediction for x and the average over the entire dataset; Symmetry means that the contributions of two feature values should be the same if they contribute equally to all possible coalitions; Null Player means a feature that does not change the predicted value regardless of which coalition of feature values it is added to should have a Shapley value of 0, and the linearity means If two models described by the prediction functions f and g are combined, the distributed prediction should correspond to the contributions derived from f and the contributions derived from g.

The linearity property of Shapley values is advantageous when dealing with ensemble models, such as random forests or the Xgboost model. In these cases, computing the Shapley Value for a feature involves averaging the single contributions of each tree, making the process more straightforward.

In this section, 378 FOPLP models with different EMC properties, different deformation, and different deformation shapes are collected as a dataset; an Xgboost model is constructed to train and explore the relationship between EMC material properties and the deformation mode. Then, the SHAP model is applied to explain the model and analyze the critical factors that influence the deformation shape.

Boosting is an ensemble learning technique that builds a robust classifier from several weak classifiers in a series. Boosting algorithms play a crucial role in dealing with bias-variance trade-offs, unlike bagging algorithms, which only control for high variance in a model, boosting controls both aspects (bias and variance) and is more effective. Figure 4.10 shows the flowchart of the Xgboost model.

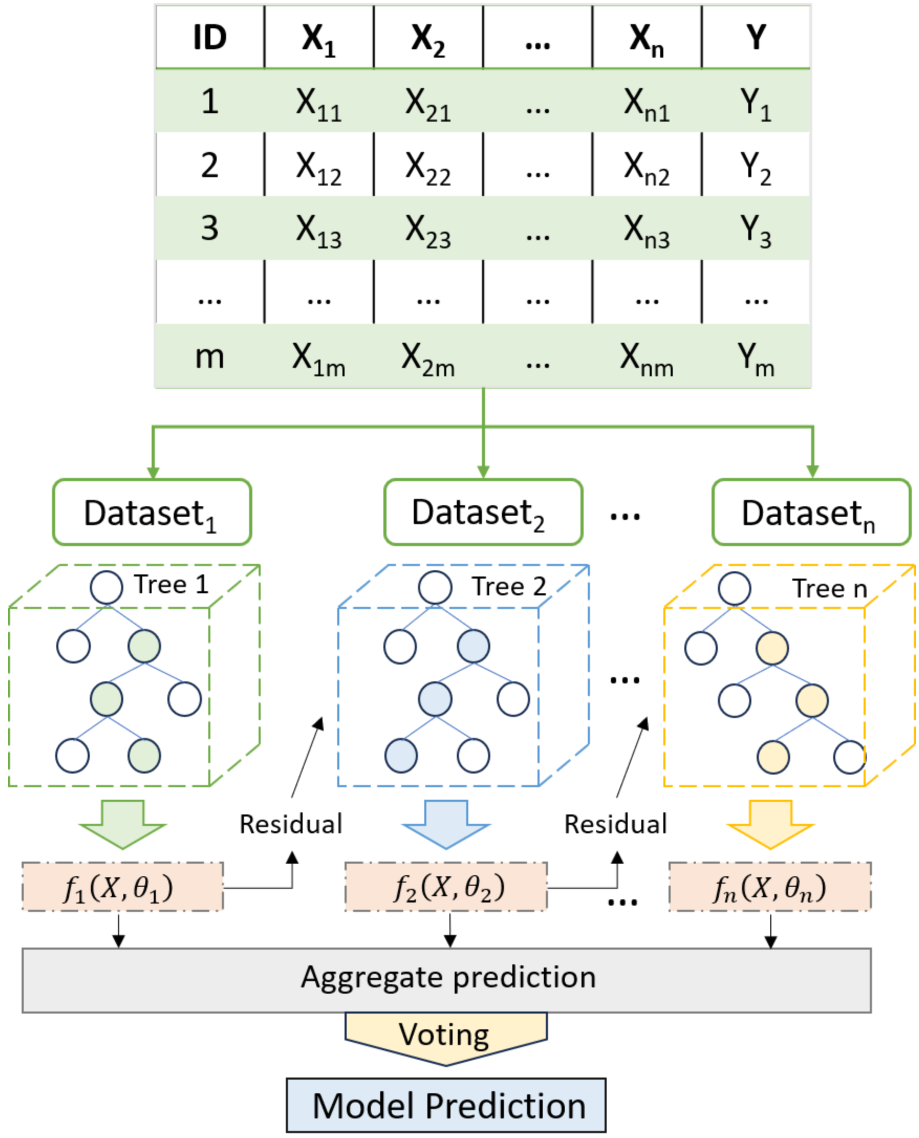

Figure 4.10 Illustration of the Xgboost model.

The training data is first separated into different datasets to construct classification and regression trees (CART). In CART, an actual score is associated with each leaf, giving a richer interpretation beyond classification. A single tree is not strong enough to be used in practice. So, an ensemble tree model that sums the prediction of multiple trees together is used.

Mathematically, the Xgboost model [100-101] can be written in the form.

$$\hat{y}_i = \sum_{k=1}^{K} f_k(x_i),\ f_k \in F \tag{4.14}$$

K is the number of trees, f_k is a function in the functional space F, and F is the set of all possible CARTs. The objective function can be written as follows.

$$obj(\theta) = \sum_{i}^{n} l(y_i, \hat{y}_i) + \sum_{k=1}^{K} \omega(f_k) \tag{4.15}$$

Here $\omega(f_k)$ is the complexity of the tree f_k.

To analyze how the EMC material properties affect the panel deformation mode, the curvature of the x-axis and the y-axis are extracted and divided as the model's objective function Z. For spherical deformation mode, the curvature in the x and y axes are both positive, and the value is equal, so the objective function Z should be around 1. For the cylindrical deformation mode, the curvature of the x and y axes should be one positive and the other close to zero, so the objective function Z should be close to zero or a more significant value. Last, for the saddle deformation mode, the curvature on the x and y axes should be a positive and a negative number with equal absolute values, so the objective function Z should be around -1, the definition of the objective function $obj(c)$ is shown as follows,

$$obj(c) = \frac{C_x}{C_y} \tag{4.16}$$

Where the C_x indicates the curvature along the x-axes in FOPLP package and the C_y indicates that in y axes. To ensure that the model has good prediction performance, the Xgboost model is optimized for hyperparameters. 5 hyperparameters are considered in parameter tuning, the max depth affects the model learning capability, the learning rate affects the convergence of the model, the n estimators affect the number of iterations of the model, and the subsample affects model generalization performance. All the hyperparameters are optimized by Bayesian optimization and the results for the trained Xgboost model are shown in Table 4.5.

Table 4.5 Hyperparameter tuning for Xgboost model.

Hyperparameter	Max depth	Learning rate	n estimators	Subsample	booster
Value	3	0.1	300	0.8	gbtree

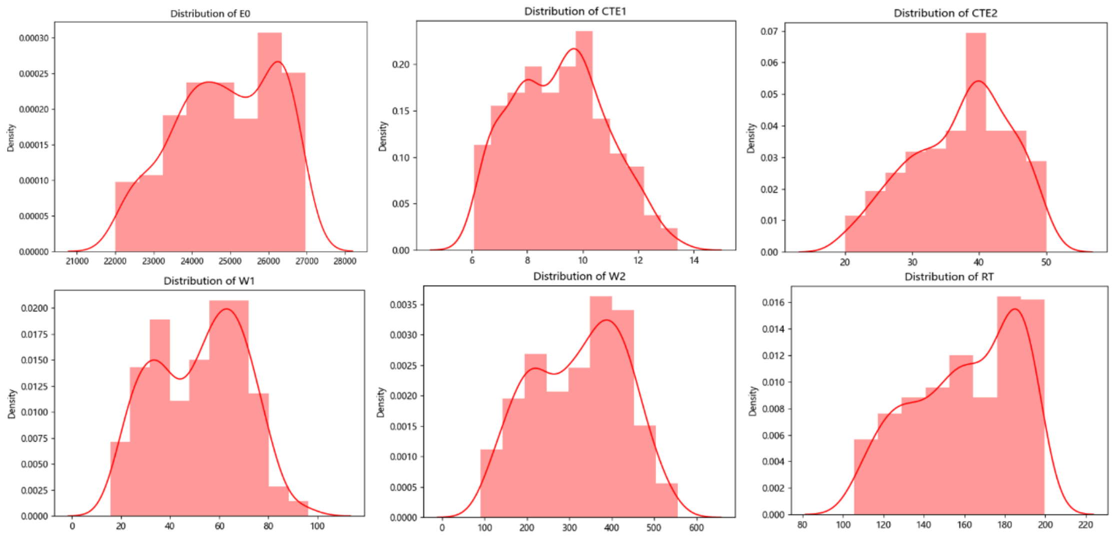

Figure 4.11 Data distribution of constitutive parameters of EMC material, the first row are E0, CTE1, CTE2 and the second row are W1, W2 and RT.

Table 4.6 Skewness and kurtosis value of features.

Feature	E0	CTE1	CTE2	W1	W2	RT
Skewness	-0.2815	0.1834	-0.3801	-0.0816	-0.1291	-0.3909
Kurtosis	-0.9548	-0.7249	-0.6873	-0.9781	-0.9693	-1.0306

Data distribution has a particular impact on the model's learning results. Normally distributed data is more conducive to model training, and skewness and kurtosis can be used to compare the differences in data relative to the normal distribution. Figure 4.11 shows the feature distribution for the proposed Xgboost model; most features demonstrate a normal distribution, and the skewness and kurtosis are shown in Table 4.6. The closer the result is to 0, the closer the distribution is to a normal distribution. The E0, CTE2, W1, W2, and RT show negative skewness values, indicating that the data is mainly concentrated in the middle and right, and some extreme values on the left may affect the training effect. Meanwhile, the CTE1 shows a positive value, meaning some extreme values might be in the right area. The kurtosis value is 0 to -1, indicating that less data is concentrated at the two ends and the middle. Hence, the data distribution is relatively flat, reflected in a platykurtic distribution.

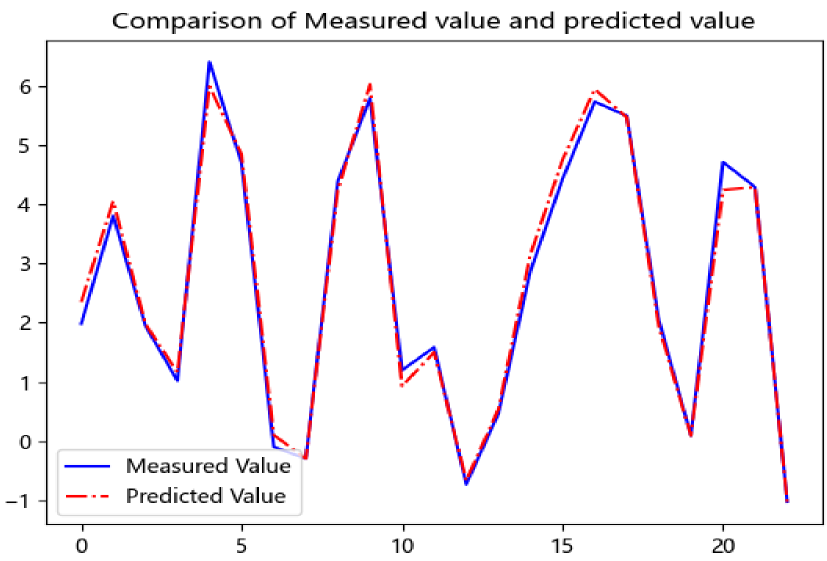

Figure 4.12 Performance of proposed Xgboost model for EMC constitutive parameters prediction in validation set.

Figure 4.12 shows the model performance of testing data. The model shows an acceptable result with the MSE and MAPE at 2.54 and 3.77%, respectively. Figure 4.13 shows the important analysis of the models; the x-axis is the mean SHAP value, and the y-axis lists the variable category. A higher SHAP value demonstrates a higher impact on the model for the specific variable.

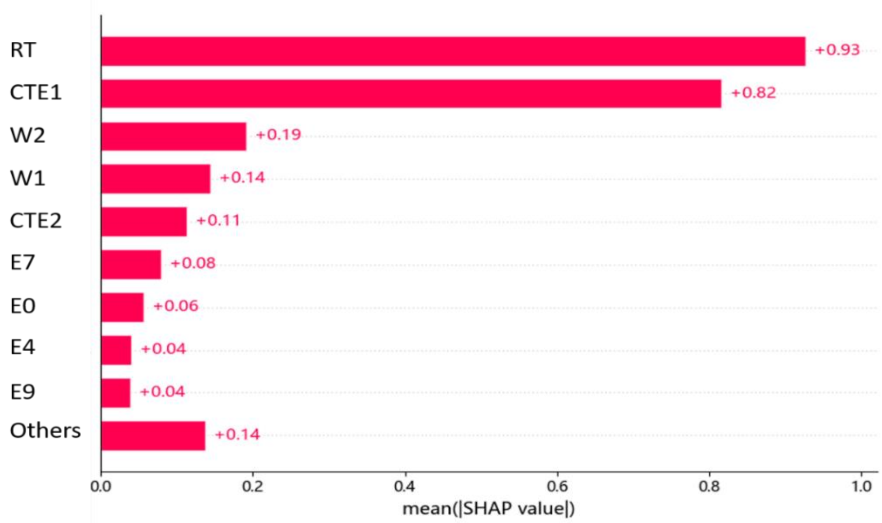

Figure 4.13 The important analysis of the EMC constitutive parameters for warpage control, where RT is the most important feature and CTE1 is the second one.

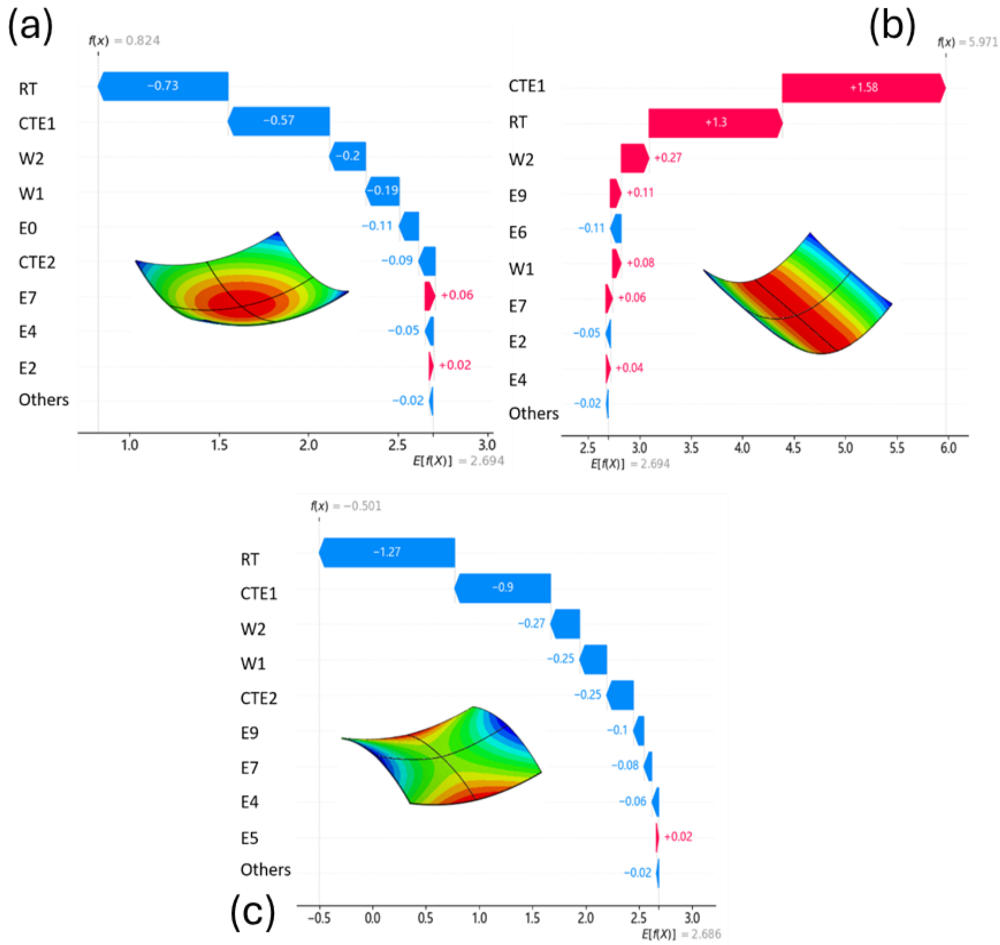

Figure 4.14 Contribution of each material constitutive parameter of EMC to warping deformation: (a) the spherical shape, (b) the cylindrical shape, and (c) the saddle shape.

The top five essential features are RT, CTE1, W2, W1, and CTE2; the RT and the CTE1 are the two most essential features, which show a significantly higher impact on the Mean SHAP value than other features.

Figure 4.14 shows the waterfall distribution of variables for different deformation shapes. The x-axis is the value of the objective function, and the expected value for the current training data is 2.694, which means that for the current structure, the cylindrical deformation mode is more accessible to generate. The y-axis lists the variables from most important to least important. Here, red means the positive effect, and blue reflects the adverse effect. The value expresses the variable's weight on the result. It can be seen from the result that for all three deformation shapes, CTE1 and RT are always the two most important factors. For saddle mode and spherical mode, RT is the most significant factor and has a similar contribution weight, which means the spherical mode and the saddle mode have a similar control constraint. For cylindrical mode, the most important factor is CTE1. Here, the RT controls the CTE dividing line, and the EMC demonstrates CTE1 and CTE2 when the temperature is lower and above RT, as the CTE mismatch generates the warpage. For a viscoelastic material, the modulus will drop fast with time when the temperature is above the RT, so most of the deformation would be contributed by CTE1. Hence, that might be the reason CTE1 is the most important factor for cylindrical mode while for spherical and saddle mode, a more complex correlation needs to be fulfilled to make a balanced deformation. Since RT is the most important feature affecting the change of panel shape to saddle and spherical modes, CTE1 and CTE2, which are highly correlated with RT, will significantly affect the final deformation mode and are likely to be mutually coupled, so a more in-depth analysis is needed. A more complex and informative display of SHAP values is a bee swarm plot that reveals the relative importance of features and their actual relationship to the predicted outcome.

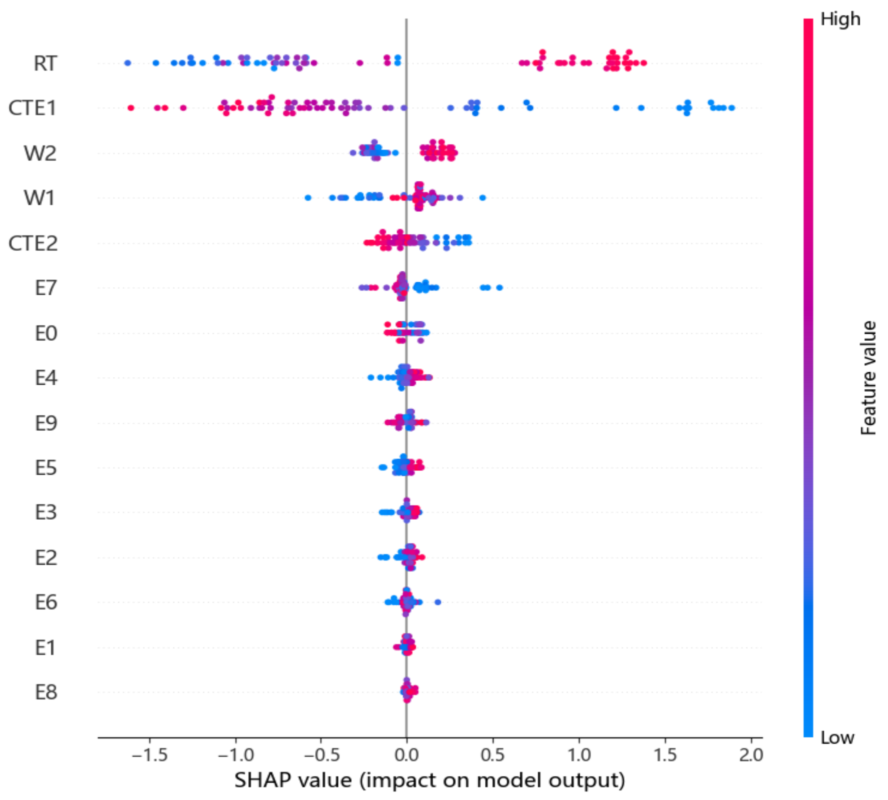

Figure 4.15 Trends in the influence of each EMC constitutive parameters on warpage deformation.

Figure 4.15 is the bee swarm plot for the proposed model. The input variables are ranked from top to bottom by their mean absolute SHAP values for the entire dataset on the left, and the color bar on the right corresponds to the raw values of the variables for each instance; if the value of a variable for a particular instance is relatively high, it appears as a red dot and on the other hand, the relatively low values appear as blue dots.

The first row shows the relationship between RT and objective function, the SHAP value is high when we select a high RT, and the SHAP value is decreased when a low RT is selected. Due to the loading profile of the cooling process being from 175 degrees to 25 degrees, when the RT is too high to exceed the initial temperature, there is no such thing as controlling warpage. Thus, the SHAP value is relatively high.

The second important feature, CTE1, shows an opposite trend: a higher CTE1 leads to a lower SHAP value, and a lower CTE1 leads to a higher SHAP value. The result may be due to the CTE matching. Looking at the CTE parameters of other components in the packaging structure, the CTE of LF is 17, and the CTE of Cu is 17.7. Therefore, the larger CTE1 can reduce the difference between these two components and thus reduce thermal strain to reduce warpage. This discovery sheds light on the model's behavior and enhances our understanding of its predictive power.

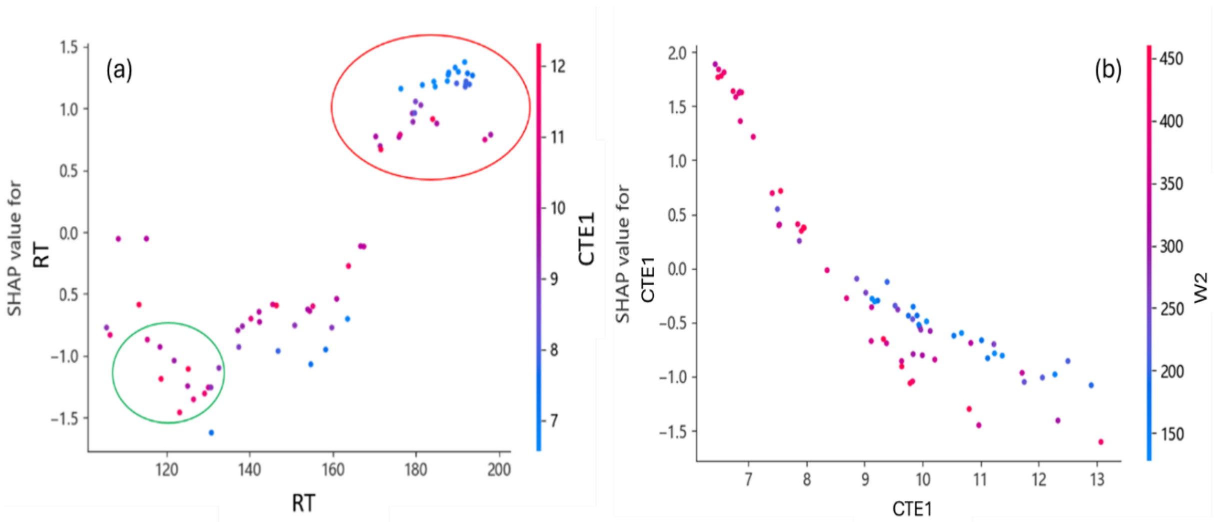

Figure 4.16 Coupling analysis of the three most important EMC constitutive parameters on warping deformation: (a) RT and CTE1, (b) CTE1 and W2.

The RT and the CTE1 are the essential features for the warpage control, and to fully explore the relationship of how these two features affect the warpage shape, the dependence analysis is implemented and shown in Figure 4.16. Here, (a) is the dependence plot for RT, the points are colored by CTE1 value, as indicated by the color bar, and this figure demonstrates the interaction effects of RT and CTE1 on warpage deformation shape. When RT is higher than 180 degrees, highlighted by the red circle in Figure 4.16 (a), it can be seen that no matter how CTE1 changes, the SHAP value is relatively high, knowing that the loading range is from 175 degrees to 25 degrees. When RT is beyond 175 degrees, the CTE2 does not affect deformation. Thus, the deformation shape is always cylindrical mode. When RT is about 120 degrees, highlighted with a green circle in Figure 4.16 (a), the SHAP value decreases as CTE1 increases, demonstrating that the deformation mode will change at a specific combination of RT and CTE1. Figure 4.16 (b) shows the dependence analysis between CTE1 and W2, and the relationship between these two features is a specific linear relationship. When CTE1 is less than 9ppm, the SHAP value decreases as CTE1 decreases, and there is no clear correlation between CTE1 and W2. When CTE1 is greater than 9ppm, there is an apparent negative correlation between W2 and SHAP value; as W2 decreases, the SHAP value increases.

Table 4.7 demonstrates the material properties of EMC before and after the optimization, comparing with the features listed as the essential features in SHAP analysis, namely, the RT and the CTE1. The value of RT for the initial spherical shape is 125 degrees, for the optimal saddle shape is 111.5 degrees, and the value of CTE1 for spherical mode and saddle mode is 10.0 and 12.3, respectively. The results match well with SHAP results; the saddle shape is more likely to generate when the EMC material properties own a lower RT and a higher CTE.

Table 4.7 Comparison of EMC parameters under different deformation shapes.

Design	Shape	RT (℃)	CTE1 (ppm/℃)	CTE2 (ppm/℃)	W1	W2
Initial	Spherical	125	10.0	42.0	24.98	126.6
Optimal	Saddle	111.5	12.3	35.1	25.71	170.9

4.5 Chapter Summary

In this chapter, an AI-based design framework is developed to control the FOPLP warpage deformation under the assembly process. The proposed framework combines FE simulation with BO, significantly improving FOPLP warpage management. Compared to the conventional DOE method, the AI-enabled optimization framework shows remarkable performance in design efficiency and design space and can be extended to related applications for package reliability assessment.

CHAPTER 5

AI-ENABLED CZM PARAMETERS CHARACTERIZATION UNDER MIX-MODE FAILURE

Delamination occurs frequently in structure and greatly impacts reliability; generally speaking, the interface shows a lower adhesive strength and has a higher risk of damage when an external load is applied. The delamination of polymer and metal is a standard failure mode in packaging reliability evaluation, and the cohesive zone model (CZM) as a numerical approach has been widely used in mechanics to study interface behavior. Precise determination of the traction-separation law parameters of CZM is crucial for analyzing adhesive layers and design structures. Traditional methods to evaluate such parameters need a series of testing and simulation, including designing fracture tests such as the double cantilever beam (DCB) test and end notch flexure (ENF), measuring the interfacial displacement field, and developing a numerical model to decouple the mix-mode fracture behavior. Most of these methods are complex and time-consuming and cannot obtain the CZM parameters directly. To simplify the steps and obtain parameters accurately and efficiently, a machine learning-based framework is developed in this study to determine the CZM parameters automatically.

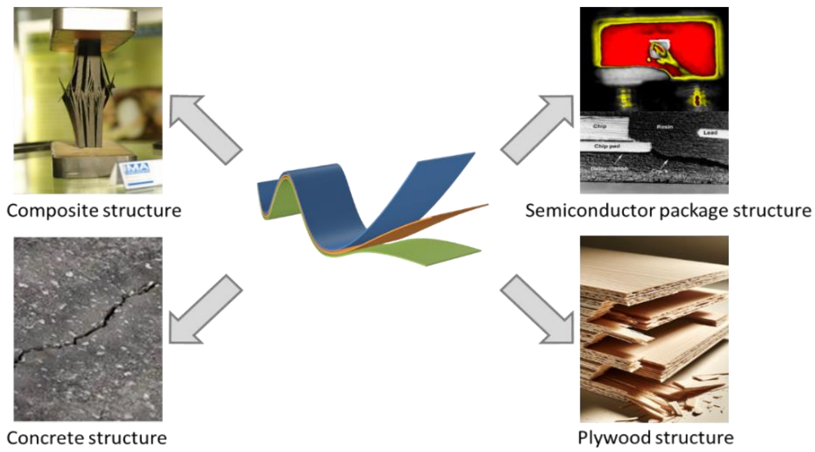

Figure 5.1 Typical scenarios of interface delamination failure, the first row are composite and semiconductor packaging, and the second row are concrete and plywood.

In this chapter, a three-dimensional (3D) and two-dimensional (2D) finite element (FE) model based on the button shear test is established, and the CZM parameters, loading force, interfacial energy, and loading height are extracted to generate a multiple database, the database contains 600 high-fidelity data from the 3D FE model and 3600 low-fidelity data from the 2D FE model. According to the database, a multi-fidelity neural network (MFNN) model is developed to identify the mix-mode CZM parameters from the database.

5.1 Failure Criteria for Interface Delamination

Interfacial delamination is a critical failure mode in composite materials and electronic packaging, significantly impacting reliability and performance. The assessment of interfacial delamination often employs various failure criteria to predict the onset and propagation of delamination under mechanical stress. Common approaches include linear elastic fracture mechanics (LEFM), which evaluates stress intensity factors, and cohesive zone modeling (CZM), which considers the nonlinear behavior of interfaces under load. These methods analyze the interaction between the adhesive strength and material properties, allowing for crack initiation and growth predictions. Recent advancements have introduced bilinear and energy-based failure criteria that enhance understanding of mixed-mode delamination, accommodating different loading conditions and material responses. Such criteria are essential for developing robust designs that mitigate the risks associated with interfacial delamination in practical applications.

5.1.1 Fracture failure criteria

Fracture failure criteria are essential in analyzing and predicting material failure under stress, particularly in the presence of cracks. These criteria provide a framework for understanding when and how materials will fail, enabling engineers to design safer and more reliable structures. The most widely used criteria include the stress intensity factor (K), which assesses the stress field near a crack tip, and the energy release rate (G), which evaluates the energy available for crack propagation. The failure occurs when these parameters exceed certain material-specific thresholds, such as fracture toughness (Kc) or critical energy release rates. Various models are employed to capture the complex interactions between material behavior and crack growth. By establishing these criteria, researchers can predict failure modes—ranging from brittle fracture to ductile tearing—thereby enhancing the durability and integrity of engineering components across diverse applications.

The stress intensity factor (K) [102] is central to these criteria, which quantifies the stress state near the tip of a crack subjected to external loads. The stress intensity factor is categorized into three modes: Mode I (opening mode), Mode II (sliding mode), and Mode III (tearing mode), each describing different loading conditions that can lead to crack growth, as Figure 5.2 shows. In the real problem, the fracture may demonstrate a mix-mode status, combining Mode I, II, and III. The mode mixite can be defined by the phase angle as illustrated below [103]:

$$\Psi = \tan^{-}\left(K_{II} / K_{I}\right) \tag{5.1}$$

Where K_I and K_{II} are the Mode I and Mode II stress intensity factors at the crack tip, and the corresponding energy release rate is shown as follows [104]:

$$G = \frac{1}{2}(1 - D_{\beta}^{2})(\frac{1}{\bar{E}_1} + \frac{1}{\bar{E}_2})(K_I^2 + K_{II}^2) \tag{5.2}$$

Here, D_β means the Dundurs mismatch parameter, and E_1 and E_2 are the elastic modulus for the bi-material structure under plain strain or plane stress condition, respectively.

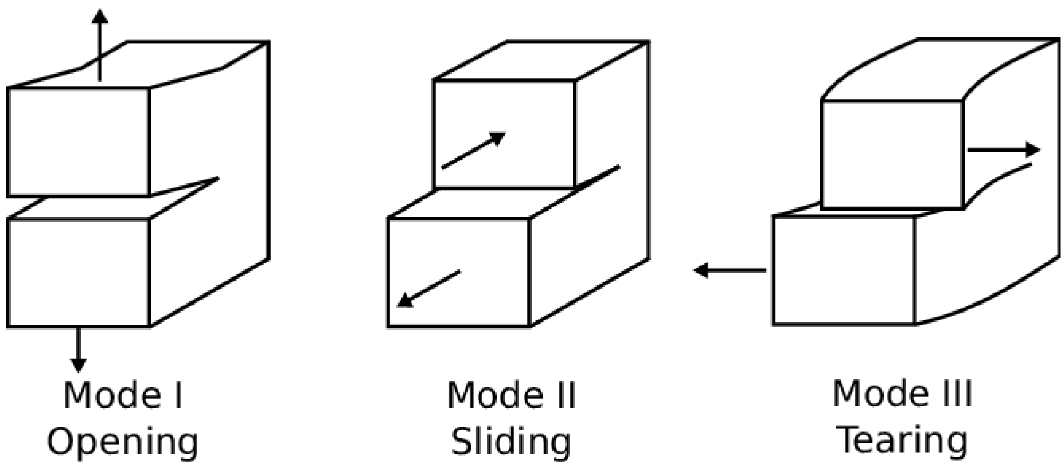

Figure 5.2 The basic crack modes.

5.1.2 Common methods for interface delamination analysis

To identify the crack failure behavior effectively and accurately, some standard methods have been proposed and widely used in research and engineering, that is, the Phase-field model [105], Cohesive Zone model (CZM) [106], Contour Integral methods [107], Virtual Crack Closure Technique (VCCT) [108], and Extended Finite Element method (XFEM) [109]. It is essential to consider their unique characteristics, advantages, and limitations in analyzing interface delamination.

1. Phase-Field Model

The phase-field model utilizes a continuous scalar order parameter to represent cracks, allowing for a diffuse transition between damaged and undamaged states. This method captures complex crack patterns and mixed-mode fractures without requiring explicit crack tracking. It is particularly advantageous for simulating intricate behaviors such as bifurcation and coalescence. However, it can be computationally intensive and may require careful calibration of parameters like fracture energy and length scale to ensure accurate results across various materials and loading conditions.

2. Cohesive Zone Model (CZM)

The cohesive zone method models the behavior of materials at the crack tip using a traction-separation law that describes how tractions change with separation distance. CZM is effective for simulating crack initiation and propagation under mixed-mode loading, clearly representing the fracture process zone. While it allows for accurate predictions of crack paths, it requires calibrating cohesive parameters and can be less effective in complex geometries or when cracks interact with other features in a material.

3. Contour Integral Methods

Contour integral methods, such as the J-integral approach, evaluate the energy release rate associated with crack growth through path-independent integrals. These methods are well-established for calculating stress intensity factors and are particularly useful in linear elastic fracture mechanics. However, they typically rely on predefined crack paths, which can limit their applicability in scenarios involving complex crack interactions or dynamic fracture processes.

4. Virtual Crack Closure Technique (VCCT)

The virtual crack closure technique calculates the energy release rate by analyzing the work done during the closure of a virtual crack segment. VCCT is straightforward and effective for steady-state crack growth in linear elastic materials. However, its limitations arise in dynamic scenarios or when dealing with complex interactions between multiple cracks or varying material properties.

5. Extended Finite Element Method (XFEM)

The extended finite element method enhances traditional finite element analysis by allowing discontinuities (such as cracks) to be represented within elements without a re-meshing technique. XFEM effectively captures complex crack geometries and can handle multiple interacting cracks simultaneously. This method is particularly useful for problems involving dynamic loading conditions but may require more sophisticated implementation techniques than other methods.

Table 5.1 Typical crack analyzing methods.

Method	Advantages	Limitations
Phase-Field Model	Captures complex patterns; no explicit tracking needed; handles mixed modes well	Computationally intensive; requires parameter calibration
Cohesive Zone Method	Clear definition of fracture process; good for mixed-mode loading	Requires parameter calibration; less effective in complex geometries
Contour Integral	Accurate stress intensity calculations; well-established	Relies on predefined paths; remeshing needed
VCCT	Simple and effective for steady-state growth	Limited in dynamic scenarios; less effective for complex interactions
XFEM	Handles multiple cracks without remeshing; captures complex geometries	More sophisticated implementation required

Table 5.1 shows the comparison of these common crack analyzing methods for interface delamination; the contour integral method is quite complex to use when simulating crack propagation and the VCCT method has many limitations in dynamic conditions. CZM, Phase-field model, and XFEM all demonstrate exemplary performance in simulating dynamic crack propagation, while for the interface delamination, the critical fracture energy of the interface is generally the lowest, so cracks usually propagate along the interface without random cracks. Hence, the high computational cost is a significant limitation of the phase field method and extended finite element method in the analysis of interface delamination. The CZM model can customize the crack propagation path and has a low computational cost. Therefore, we chose CZM as the analysis model for subsequent interface delamination.

5.2 Definition of Cohesive Zone Model

The CZM is widely used in engineering and analysis as a phenomenological model introduced by Barenblatt [110-111]. As one of the models for studying crack propagation, the cohesion model has high accuracy in analyzing interface crack propagation. Compared with other crack propagation analysis models, the CZM has the following advantages:

1. No need for preset cracks.
2. Low mesh sensitivity when simulating interface crack propagation.
3. Robust model customization.
4. Good convergence.

However, it also has some disadvantages. First, since the CZM is a model defined based on damage mechanics, it is difficult to obtain the parameters of the cohesion model under the mixed failure mode. Secondly, the parameters of the cohesion model are obtained based on the inversion method. Thus, ensuring the uniqueness of the inverted parameters isn't easy.

The schematic diagram for CZM is shown in Figure 5.3; The orange and green area in the diagram represents the pure shear and tensile behavior, and the blue area represents the mix-mode behavior. When the external loads are applied to the interface, the model will calculate the tensile stress and the shear stress among the interface; when the stress is beyond the peak stress, it is defined that the damage occurs, and it is expressed as the stiffness reduction. When the displacement reaches the max displacement, the point or the part is defined as failure.

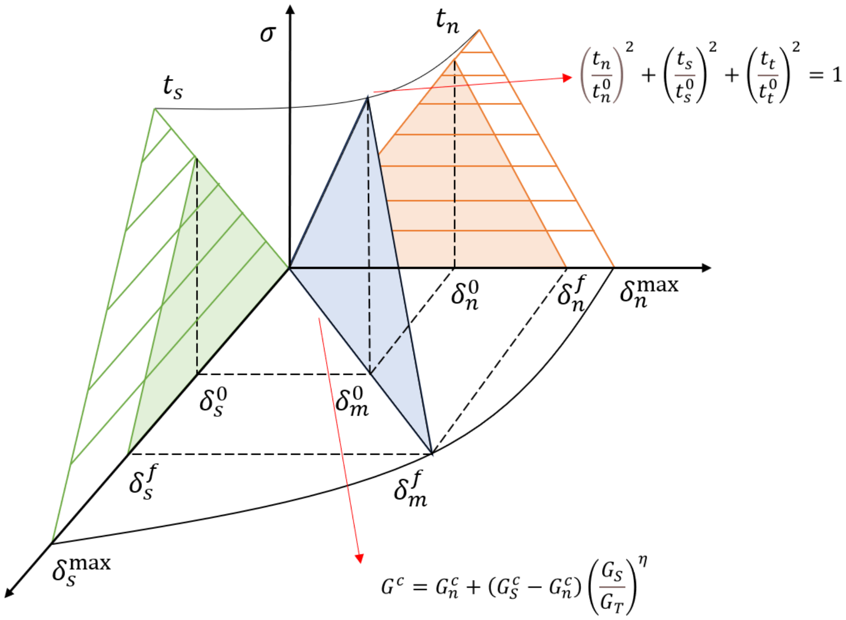

Figure 5.3 Definition of mix-mode cohesive zone model, the t is the traction stress and indicates the damage criterion, the G is the separation energy and indicates the failure criterion.

The Quads traction criterion and Benzeggagh-Kenane (BK) fracture criterion are commonly used for the mix-mode damage initiation and criterion. As the CZM obeys the traction–separate principle, the definition of the traction criterion and separate criterion are listed below.

$$\left\{\frac{\langle t_n \rangle}{t_n^0}\right\}^2 + \left\{\frac{t_s}{t_s^0}\right\}^2 + \left\{\frac{t_t}{t_t^0}\right\}^2 = 1 \tag{5.3}$$

$$G_n^C + (G_s^C - G_n^C)\left\{\frac{G_S}{G_T}\right\}^\eta = G^C \tag{5.4}$$

Where t_n^o and t_n are the peak and transient stress in the normal direction, respectively; t_s^0 and t_s are the peak and transient stress in the in-plane direction, respectively; t_t^0 and t_t are the peak and transient stress in the out-of-plane direction, respectively; G^c is the critical fracture energy; η is the material-related dimensional parameters; G_n^c and G_n are the critical and transient fracture energy in normal direction, respectively; G_s^c and G_s are the critical and transient fracture energy in in-plane direction, respectively; G_t^c and G_t are the critical and transient fracture energy in out-of-plane direction, respectively; The CZM can be identified precisely by obtaining these parameters, and considering that the in-plane shear behavior is equal to the out-of-plane shear behavior, the mix-mode CZM parameters can be reduced to 5 variables, that is $t_n^0, t_s^0, G_n^c, G_s^c, \eta$.

5.3 Methodology for CZM Parameter Prediction

As reviewed in the previous section, the CZM parameters are identified through inversion techniques, which bring some difficulties to the application of CZM. Firstly, standard testing methods such as DCB and ENF are needed to calculate the interface critical energy release rate. Secondly, a mixed-mode testing method should be designed to generate force and fracture energy under different conditions. Last, suitable inversion techniques need to be selected to fulfill the requirement. Such a measuring method has some limitations; the first one is that not all materials can be prepared with standard-sized samples. Second, the design of mixed-mode tests is usually complex and challenging to ensure repeatability. Third, the inversion method is usually limited and only works within a specific range.

To solve these problems, an ML-based CZM parameter prediction framework is established to identify the EMC-lead frame interface behavior.

The workflow of the proposed framework is shown in Figure 5.4 above. The first step selects the Button shear test as the benchmark for the simulation model development and validation. The button shear test is easy to prepare and has good repeatability. Button shear tests can obtain different mixed-mode fracture failure scenarios by adjusting different shear heights, as the schematic diagram shows below.

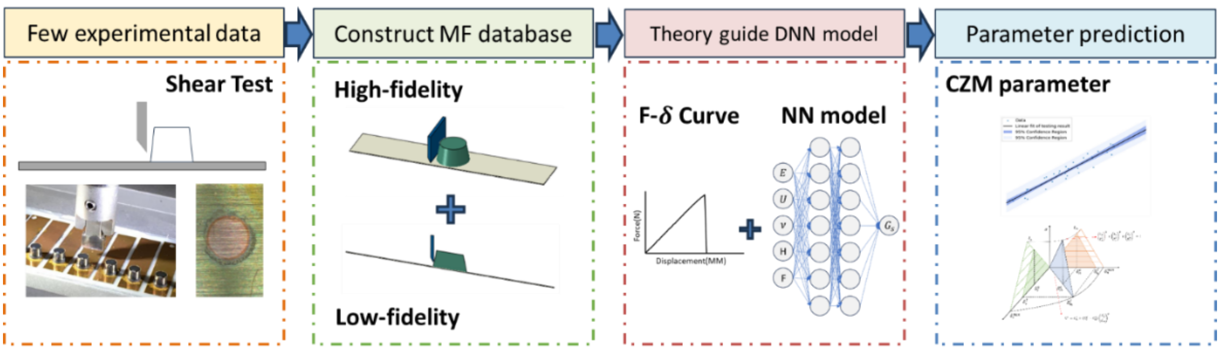

Figure 5.4 The proposed ML-based CZM parameter prediction framework.

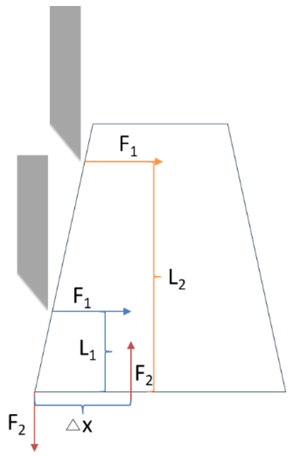

Figure 5.5 Mechanism of mixed-mode fracture using button shear tests, F is the force and l is the arm.

The shear force at different shear heights is the same; with the shear height changes, the bending moments change, too. Assume that a small length at the fixed edge is the moment arm; the higher the shear height, the higher the moment, and the higher the reaction force at the fixed edge, namely F2. Therefore, a mixed failure mode is shown in Figure 5.5. When the shear height is lower than a specific value, the test results tend to be a shear failure. As the height gradually increases, the failure mode will change to tensile failure.

The second step is to build an FE model based on the button shear test; in this work, a multi-fidelity FE simulation is calculated to generate a multiple database, the high-fidelity model is developed in 3D elements, and the low-fidelity model is developed in 2D elements. Considering the button shear test can only generate the force-displacement curve. Therefore, the third step is constructing an NN model to learn the CZM parameters through the force-displacement curve.

5.4 Button Shear FE Model Development

The established geometric model and dimensions are shown in Figure 5.6 below. The simulation model uses a C3D8R full integration element to improve the calculation accuracy. The button material is set to EMC, the substrate is copper, and a zero-thickness CZM element is inserted to simulate the delamination.

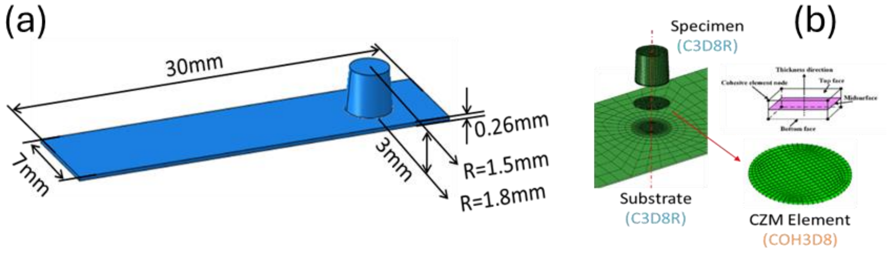

Figure 5.6 (a) the shape and dimensions of the button shear FE model, (b) the element type used in the FE model.

Similarly, to ensure the reliability of the simulation results, the mesh sensitivity analysis of the established finite element model was carried out. Seen from Figure 5.7, the mesh size was set from 0.05mm to 0.2mm, with an interval of 0.025mm. The established finite element model was subjected to button shear simulation at different heights, and the histogram of the results is shown in Figure 5.8 below. The histogram on the left shows the relationship between force and mesh size. Figure 5.8 (a) shows the comparison of peak force at different shear height, When the mesh size is 0.05, 0.075 and 0.1, the simulation results show that the models have some differences only at 60% shear height, and at other heights, the results of the three models are very close. When the mesh size continues to expand to 0.125, the results begin to fluctuate greatly. As the mesh size continues to increase, the instability of the simulation results will increase further, which indicates the error of the results is too large to be credible. Figure 5.8 (b) shows the relationship between fracture energy, shear height and mesh size. The fluctuation of fracture energy at different heights for models with different mesh sizes is smaller than that of force results, but the error will slowly increase when the mesh size is greater than 0.125. Considering the balance between efficiency and accuracy, a model with a mesh size of 0.1 mm was selected as the benchmark model for subsequent work.

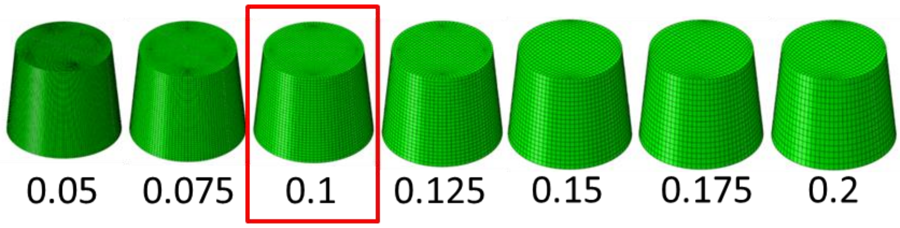

Figure 5.7 Different mesh size FE models prepared for mesh sensitivity analysis.

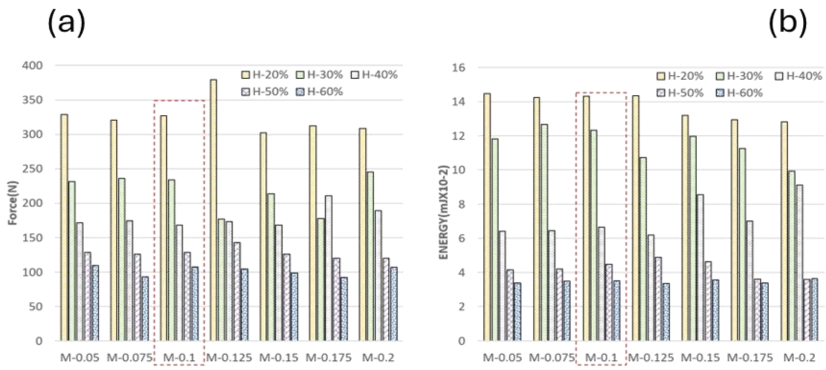

Figure 5.8 Comparison of peak force and fracture energy with mesh size at different shear heights: (a) Force, (b) Energy.

A dynamic simulation is also necessary to obtain the force-displacement curve and extract the peak force and interface fracture energy. Thus, a quasi-static simulation based on the explicit step is established. Figure 5.9 shows the simulation result for the whole button shear process with the crack initiation, propagation, and final delamination.

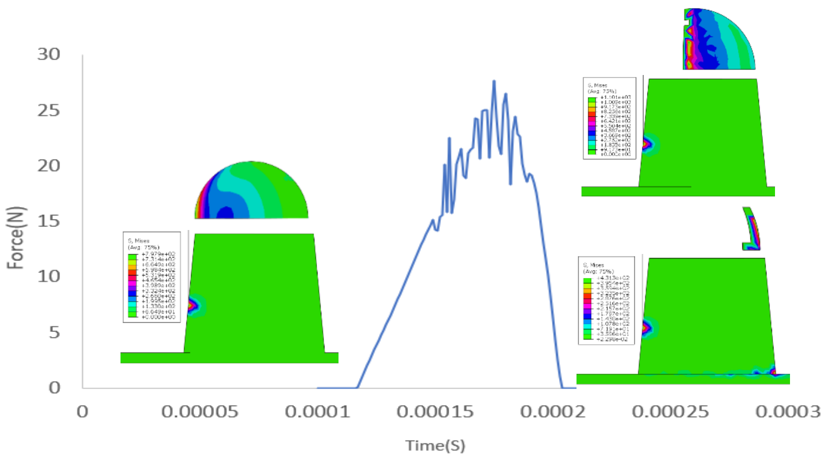

Figure 5.9 Illustration of the failure modes and interface behavior during the button shear test.

The energy evaluation is done as shown in Figure 5.10 to ensure the dynamic simulation result can be considered a quasi-static process. Here, the ALLWK, ALLFD, ALLIE, ALLKE, ALLSE, and ALLVD represent total energy, Frictional energy, Internal energy, kinetic energy, strain energy, and viscous energy, respectively. The total energy equals the external energy, and the value should be constant when the shear process is done.

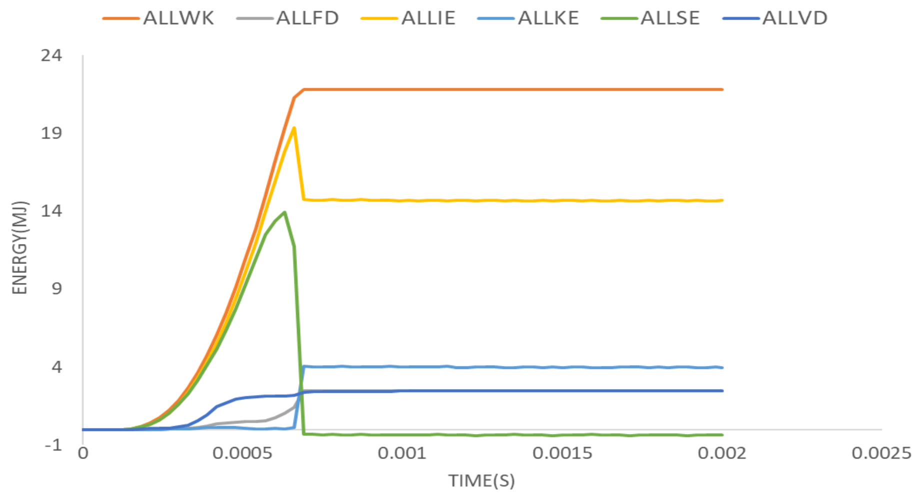

Figure 5.10 Illustration of the different energies over time during button shear test.

Here, the strain energy represents elastic deformation, so the value decreases to zero when the shear process is done, and the internal energy, kinetic energy, is two balanced energies with one increase and another decrease. To ensure the quasi-static process, the kinetic energy to internal energy ratio should be lower enough to add little external energy to the system. Figure 5.11 shows the ratio of internal and kinetic energy; the ratio is lower than 5% from 0 to 0.0006s. Observing the simulation result cloud map shows that the model is in the shear deformation stage between 0 and 0.0006s. The bonding layer has experienced four stages: receiving loading, unit damage, crack initiation, and crack propagation.

By observing the simulation result cloud map, the model is in the shear deformation stage between 0 and 0.0006s, and the bonding layer has experienced four stages: receiving loading, unit damage, crack initiation, and crack expansion. When the simulation time exceeds 0.0006s, the bonding layer is delaminated from the substrate, and there is no longer a constraint relationship between the button and the substrate. At the same time, the strain energy accumulated in the button is released, and the internal energy is converted into kinetic energy, so the entire button flies out. Considering that the change in kinetic energy after subsequent delamination has nothing to do with the bonding layer, we only focus on the change in the ratio of internal energy to kinetic energy between 0 and 0.0006 s. The ratio of kinetic energy to internal energy during this period has never been higher than 5%, indicating that the established dynamic simulation model does not add sufficient energy to affect the accuracy of the results.

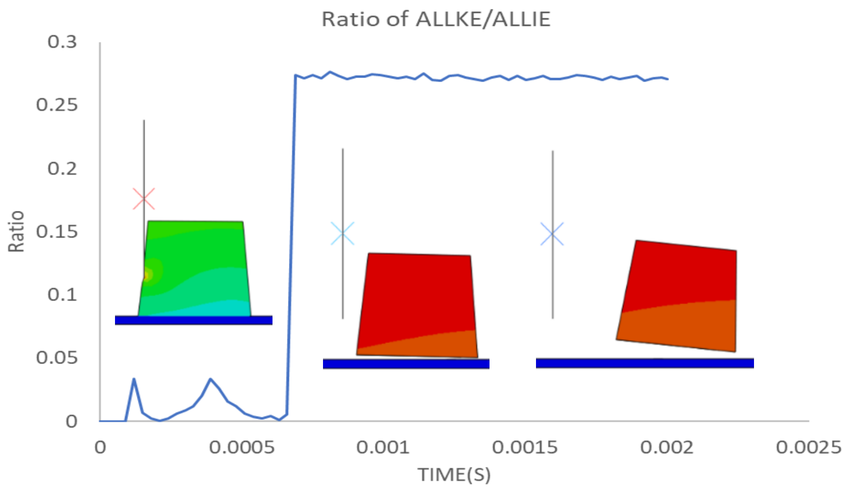

Figure 5.11 Illustration of the internal and kinetic energy ratio over time during button shear test, the ratio was less than 5% before the interface delamination.

Therefore, the simulation process can be regarded as quasi-static loading. To construct the multi-fidelity database for further model training, a 2D FE model is developed. The 2D model uses a plane strain element, and the mesh size is set the same as the 3D FE model. In this work, 600 data from the 3D model and 3600 from the 2D model are collected to construct multiple databases for model training.

Figure 5.12 shows the correlation between parameters; on the x-axis is the peak force in the 2D model, the interface fracture energy in the 2D model, and the same parameters in the 3D model. On the y-axis is also the peak force and energy in the 2D and 3D modes. The figures show the distribution and correlation of the simulation results of the 2D and 3D models when the shearing height is 20%, 30%, 40%, and 50%. The lower left corner of each figure is a scatter plot of data distribution, which reflects the data distribution, including the distribution range and distribution trend. The middle part is the distribution histogram, which reflects the relationship of data distribution. The upper right corner is the Pearson correlation coefficient, representing the significant data correlation relationship. The Pearson correlation coefficient is a statistical measure that quantifies the strength and direction of the linear relationship between two continuous variables. One of the most used and well-known correlation coefficients in statistics, and the definition of the Pearson correlation coefficient is as follows:

$$\rho_{X,Y} = \frac{\operatorname{cov}(X,Y)}{\sigma_X \sigma_Y} = \frac{E\left[(X-\mu_X)(Y-\mu_Y)\right]}{\sigma_X \sigma_Y} \tag{5.5}$$

Here, σ_X and σ_Y are the standard deviation for the two variables and $cov(X,Y)$ are the covariance. The Pearson correlation coefficient of energy and force for 2D model and 3D model at 20%, 30%, 40% and 50% shear height are 0.9, 0.99, 0.93, 0.98, 0.91, 0.96, 0.90, 0.97, respectively. All the coefficients are higher than 0.9, which demonstrate that there is a strong positive linear relationship between 2D model result and 3D model result and prove that a multi-fidelity database based on the combination of 3D high-fidelity data and 2D low-fidelity data is feasible.

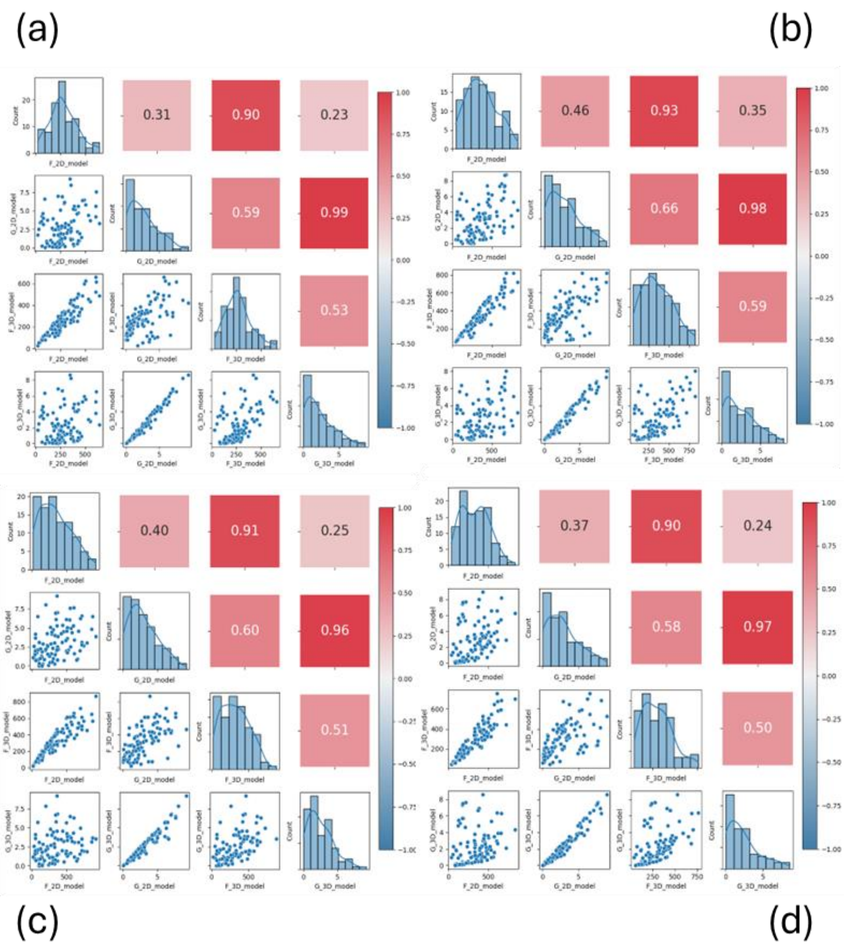

Figure 5.12 Pearson correlation coefficient distribution between 3D model and 2D model to demonstrate the degree of linear correlation (a) 20% shear height, (b) 30% shear height, (c) 40% shear height, (d) 50% shear height.

5.5 Theory-Guide NN Model for CZM Parameter Identification

Considering that in the simulation view, the mix-mode CZM parameters are first set up, and the force-displacement curve is the simulation result. For the prediction view, it is an inverse problem, which needs to predict the CZM parameters based on a load-displacement curve. This work extracts the peak load and the interface fracture energy from the load-displacement curve. These two parameters, material properties and shear height, are used as input variables to train the proposed model. The critical tensile stress, critical shear stress, tensile fracture energy, shear fracture energy, and mix-mode coefficient are extracted as target functions.

Figure 5.13 shows the correlation between those core variables. The peak force F, interface energy U, shear fracture energy GII, shear stress S, and mix-mode coefficient BK are listed in the chart. It can be seen that the shear stress has a specific relevance with peak force, shear energy shows a high relevance with interface fracture energy, and the mix-mode coefficient BK shows a low correlation with other parameters.

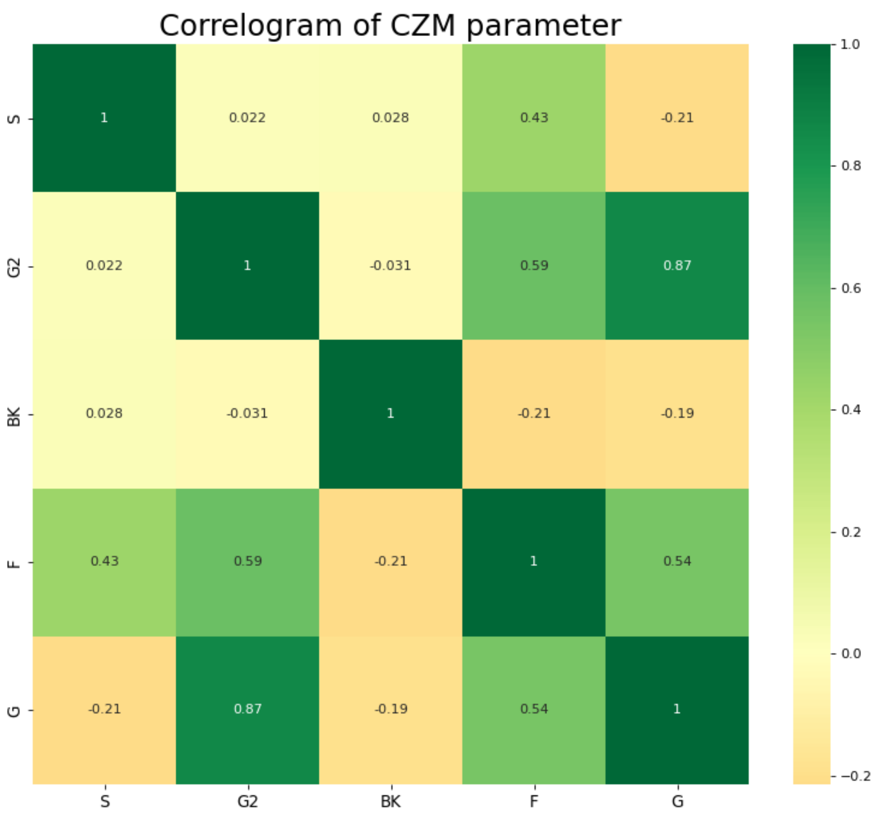

Figure 5.13 Pearson correlation coefficient distribution between each CZM parameter.

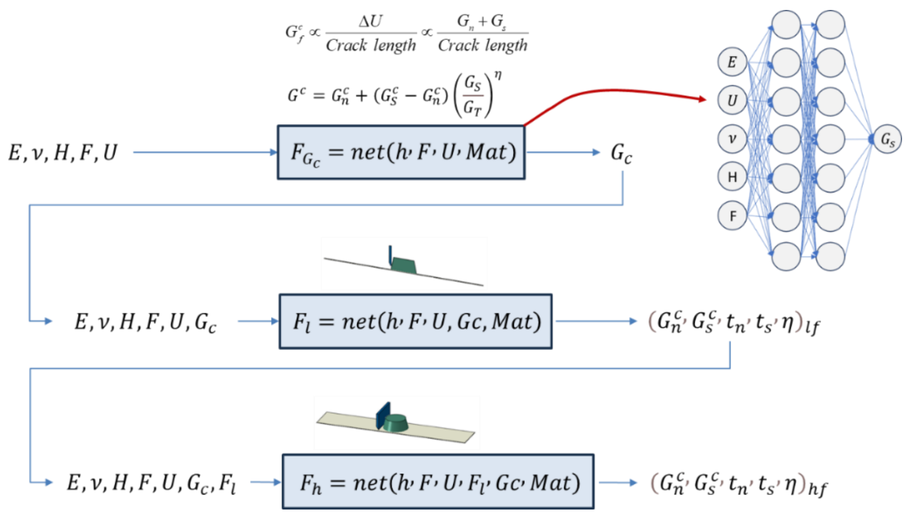

Figure 5.14 The framework of the theory-guide MFNN model, three sub-NN models are constructed to predict the total fracture energy, the LF CZM parameters and the HF CZM parameters respectively.

Based on the data analysis above, a theory guide MFNN model is proposed to predict the CZM parameters via force, interface energy, shear height, and material properties. The workflow of the proposed model is shown in Figure 5.14. The input variables are elastic modulus, Poisson's ratio of the button, shear height, force, and interface fracture energy. Knowing that the total fracture energy in CZM has such a relationship with U, GI, GII, and η, as shown below:

$$G_f^c \propto \frac{\Delta U}{Crack\ length} \propto \frac{G_n + G_s}{Crack\ length} \tag{5.6}$$

$$G^c = G_n^c + (G_S^c - G_n^c)\left(\frac{G_S}{G_T}\right)^\eta \tag{5.7}$$

$$G_f^c = G_I^C + G_{II}^C \tag{5.8}$$

The first NN model, FGc, predicts the intermediate variables Gc and helps the model provide prior knowledge to the subsequent prediction. The trained model will predict the Gc as another input variable to train the Fl, which is the low-fidelity model, and this model will output the low-fidelity CZM parameters, and such output parameters will then be regarded as the input parameters to train the high-fidelity model and obtain the actual CZM parameters.

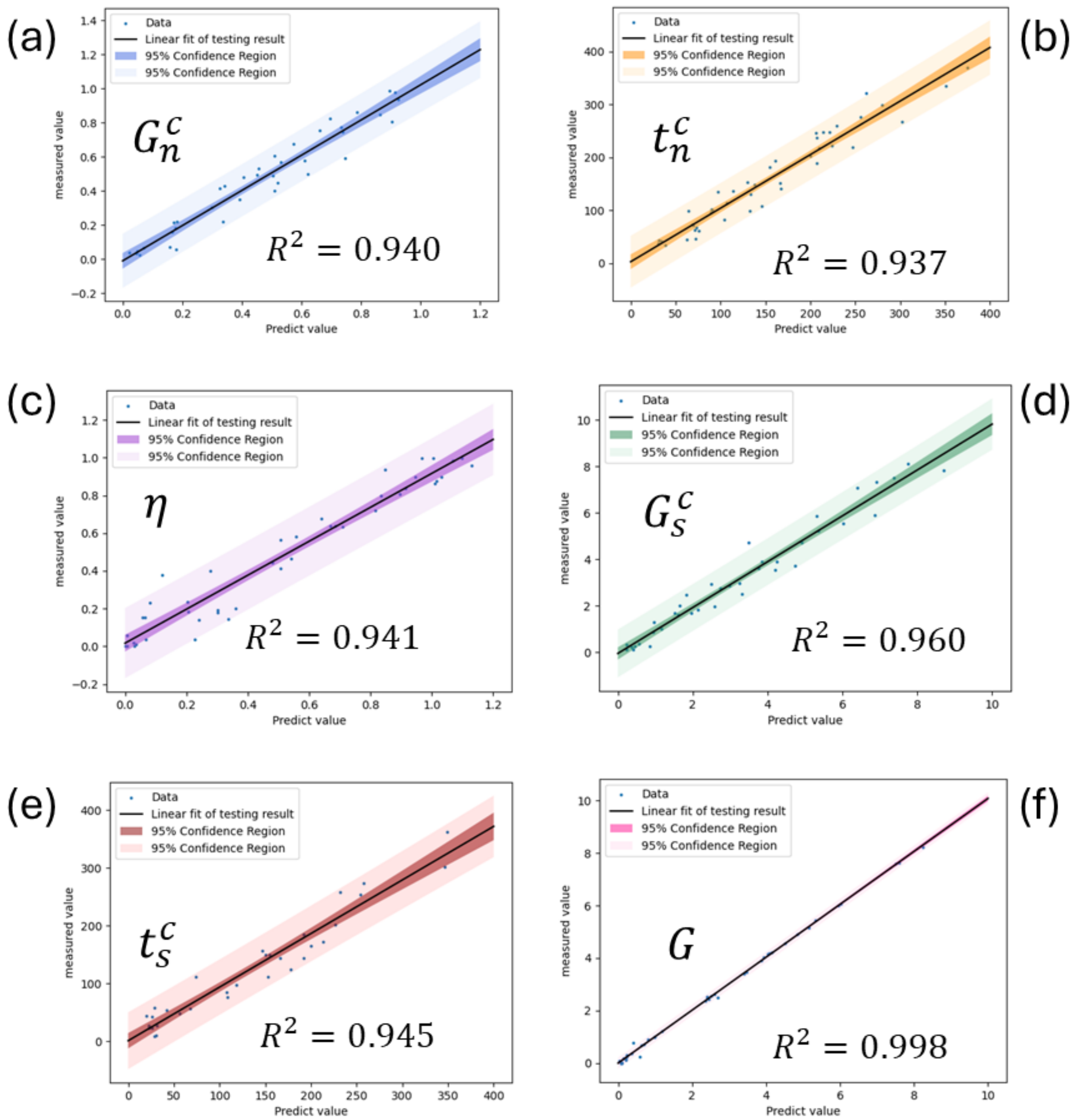

Figure 5.15 Illustration of the R-squared error and data distribution of CZM parameters obtained by theory-guide MFNN model for validation sets, Shaded areas represent 95% confidence bounds, (a) G_n^c, (b) t_n^c, (c) η, (d) G_s^c, (e) t_s^c, (f) G.

Figure 5.15 shows the model performance of the theory-guided MFNN model. The model shows an acceptable prediction result. The R^2 values for all the CZM parameters are around 0.95. Considering that the error may accumulate in such nested structures, the R^2 for parameter G is 0.998, which reduces the impact of error accumulation on subsequent models to a certain extent.

5.6 Model Validation and Comparison

Two types of samples are prepared to validate the accuracy and efficiency of the proposed model. The sample size is illustrated in Figure 5.16; the shape of the first sample is the same as the one for machine learning model development and training. The sample is shaped like a truncated cone, with the radius of the top and bottom surfaces being 3 and 3.5 mm respectively, and the height being 3 mm. Another sample is also in the shape of a truncated cone, while the radius of the top and bottom surfaces is 9 and 11.3 mm, respectively, and the height is 6mm. The aim of preparing these two validation tests is to confirm the identified CZM parameters are unique and can be applied in different structures.

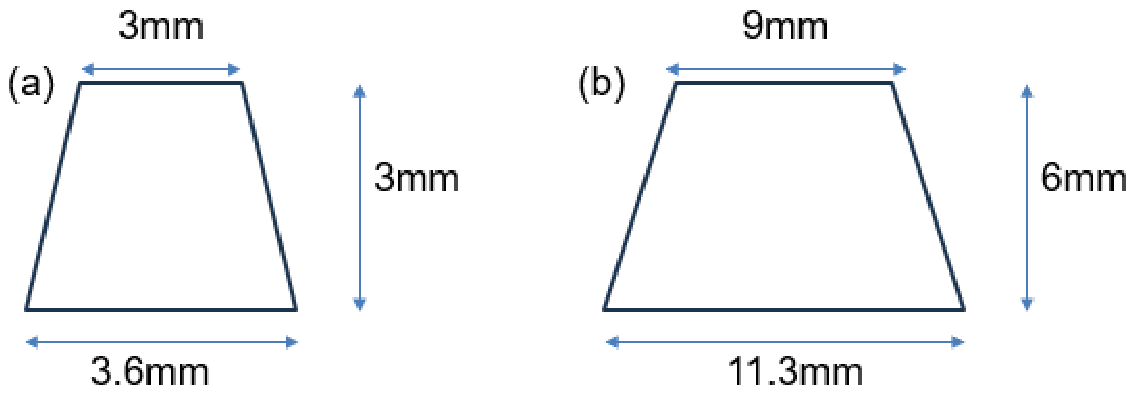

Figure 5.16 Two types of samples for theory-guide MFNN model validation: (a) small-size samples and (b) large-size samples.

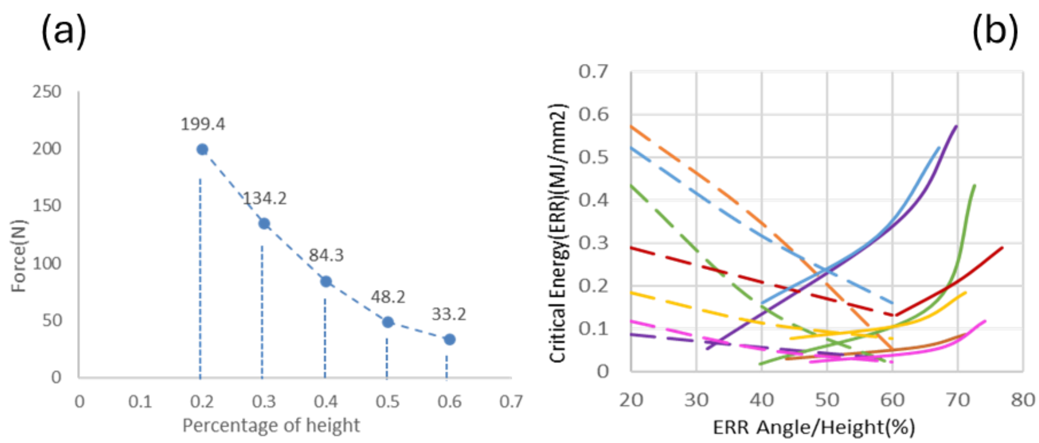

Figure 5.17 (a) Illustration of the forces at different shear heights obtained from the experiments for model validation, (b) the fracture energy over phase angle and shear height.

Five samples with the same dimension as in the FE model were prepared, and button shear tests were performed at heights of 20%, 30%, 40%, 50%, and 60%, respectively. The testing results are shown in Figure 5.17(a), and the results show a linear trend; that is, as the shear height decreases, the critical force gradually increases. The force-displacement curve, shear height, and material properties obtained from the test were input into the trained model, and the CZM parameters predicted by the model were averaged and brought into the finite element model to calculate the simulation results and compare them with the test results.

Considering the accuracy of the test instrument, for the button shear test, usually, only the peak force is valid, while the displacement result is not accurate. In this case, the parameter G representing fracture energy cannot be effectively obtained directly through experimental methods, to solve this problem, several simulations at different shear heights are implemented, and the result is shown in Figure 5.17(b), where the y-axis is the critical energy, and the x-axis can be regarded as shear height for the solid line and as phase angle for the dashed line. The simulation result demonstrates that the critical fracture energy has a monotonic relationship with the shear height and the fracture phase angle. As the shear height increases, the fracture phase angle decreases, and the critical fracture energy decreases. Similar trends are met at shear heights of 20%-60%. Therefore, based on this finding, we established the following optimization model to help obtain CZM parameters when the critical fracture energy is unknown.

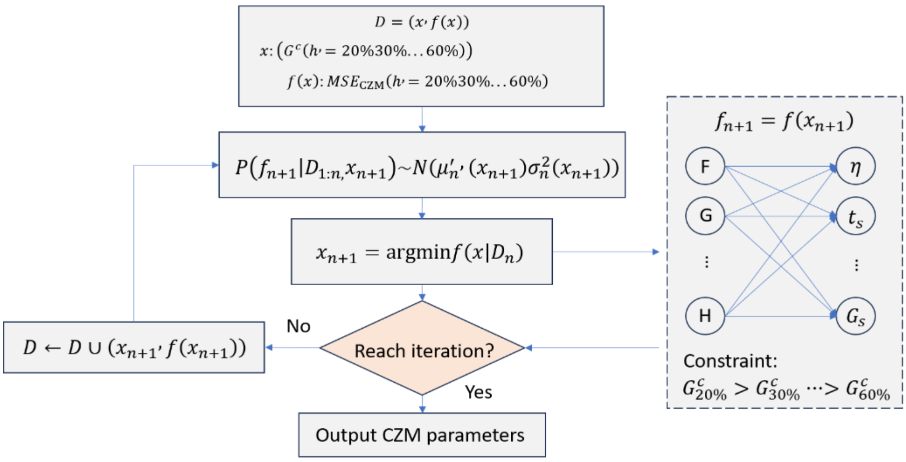

Figure 5.18 BO framework to predict CZM parameters with unknown energy, the proposed theory-guide MFNN model is part of the optimization for fracture energy identification.

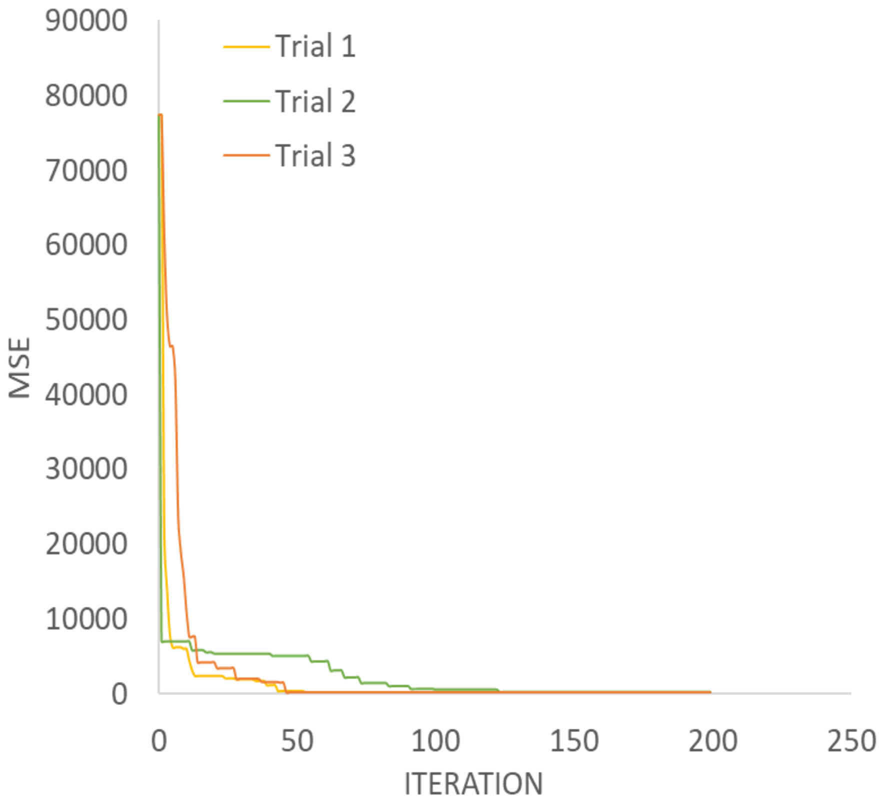

Figure 5.19 Illustration of convergence for BO framework on CZM parameters prediction under three trials.

Figure 5.18 shows the optimization workflow for the CZM parameters characterization when fracture energy is unknown. Here, the input space D is consistent with x and f(x), the x is the critical fracture energy at different heights, and the f(x) is the MSE of CZM parameters under different heights. The model will first deliver a set of critical fracture energy at different heights into datasets, and the fracture energy will combine with the corresponding measured peak loads into the theory-guide MFNN model proposed in the previous section to make a CZM parameters prediction. Each shear height will obtain a set of CZM parameters, and the MSE of all CZM parameters is regarded as the f(x) to check if the value fulfills our requirement. If yes, the model outputs the calculated results, and if not, these data sets will be sent to the database and start another loop until the MSE meets our requirements. The optimization result is shown in Figure 5.19. Three trials are done to calculate the average value of CZM parameters, and after about 150 iterations, the model obtains the optimal result.

Table 5.2 The optimization results of CZM parameters.

CZM parameter	t_n	t_s	G_n	G_s	η
Predict value	19.1MPa	382.7MPa	0.608J	244.1J	2.51

Table 5.2 shows the CZM parameters predicted from proposed model, and to validate the model performance and if the predicted CZM parameters are unique, a comparison of simulation results using predicted CZM parameters and testing results based on two different sample mentioned in previous section are implemented. Figure 5.20 shows the shear height – force curve of button shear process. 5 data points are obtained from the experimental result as the shear height is 20%, 30%, 40%, 50%, and 60%, the predicted CZM parameters are then delivered to the FE model to calculate the peak force. Here, the orange line is the results of FE simulation, and the blue line is the results of test, although there are some differences in absolute values, the two results show the same trend, that is, the peak force decreases with the decrease of shear height. The specific results are shown in Table 5.3, the MSE and MAPE for the experimental and simulation results are 114.9 and 13.9%, respectively, demonstrating an acceptable result in prediction performance.

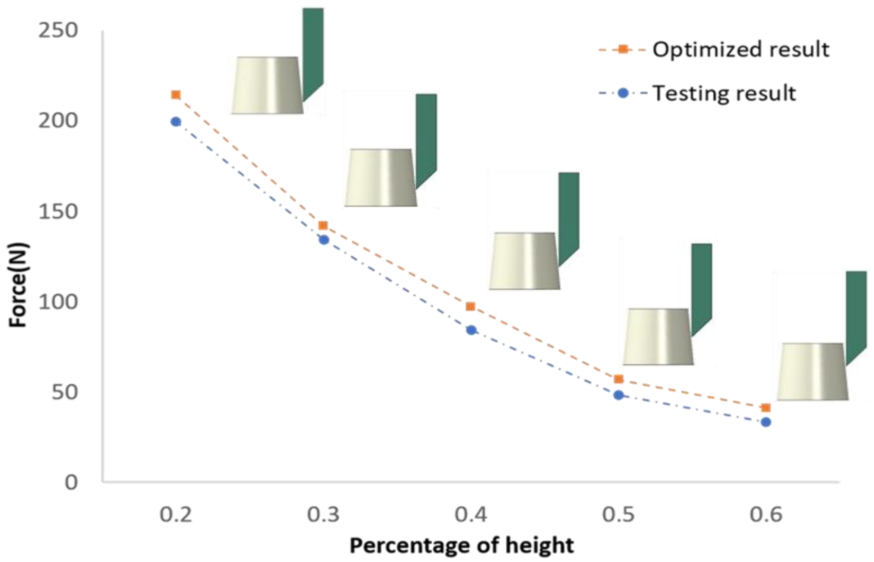

Figure 5.20 Comparison of shear force between BO results and experimental results for small-sized samples.

Table 5.3 Comparison of optimization results and experimental results for small-sized samples.

Shear height (%)	20	30	40	50	60	MSE	MAPE
Measured value	199.4	134.2	84.3	48.2	33.2	\	\
Predicted value	214.2	141.7	97.2	56.7	41.1	114.9	13.9%

To ensure the obtained CZM parameters are unique, as mentioned in previous section, a large-sized sample is conducted to implement the same button shear process through FE simulation and test by using the CZM parameters obtained from proposed model. As Figure 5.21 shows, the orange line is the simulation result, and the blue line is the testing result. Same as the results of the previous small-size model, although there are some differences in absolute values, the two models have the same trend, that is, the peak force decreases as the shear height decreases. The value of the force is shown in table 5.4, the MSE and the MAPE for these samples are 10.84 and 3.85%, respectively, demonstrating that the CZM parameters obtained from the proposed model are unique.

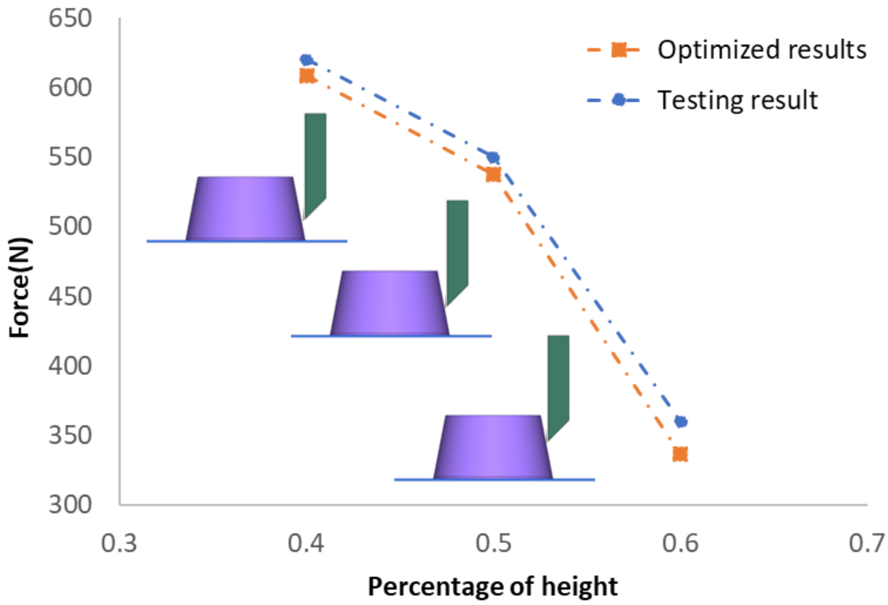

Figure 5.21 Comparison of shear force between BO results and experimental results for large-sized samples.

Table 5.4 Comparison of optimization results and experimental results for large-sized samples.

Shear height (%)	40	50	60	MSE	MAPE
Measured value	620	550	360	\	\
Predicted value	609.4	538.7	337	10.84	3.45%

5.7 Chapter Summary

This chapter develops a theory-guided MFNN model to learn the CZM parameters under mix-mode fracture mode. The proposed model shows an acceptable performance, and the prediction value can be validated by testing results.

The proposed prediction model can avoid the parameter sensitivity problem of traditional inversion methods and can be applied to a broader range of CZM parameter identification. Meanwhile, the predicted CZM parameters are also unique after experimental verification.

CHAPTER 6

SUMMARY AND FUTURE WORK

6.1 Summary

In this work, to meet the design requirements of the advanced packaging industry in the new era, an AI-enabled method for the microelectronic packaging industry is proposed and applied to optimize and select epoxy molding compound (EMC) in IC packaging design. In the first part of the work, an automatic EMC material selection framework is proposed for IC packaging design, and the framework contains two sub-models. The first prediction model is developed to identify EMC constitutive parameters through the elastic modulus at several temperature points, and the second optimization model with a multi-fidelity and multi-objective approach is constructed and applied in thermal fatigue management and warpage control to help select appropriate EMC.

The second part develops a theory-guided multi-fidelity neural network model for identifying the cohesive zone model parameters under mix-mode failure to identify and analyze the EMC-lead frame interface behavior. The proposed model shows acceptable performance and has been validated by testing results.

In summary, this work proposed an AI-enabled framework focused on EMC optimization and selection; the proposed framework can characterize the EMC constitutive parameter and EMC-lead frame interface parameter and effectively and accurately select the suitable EMC under certain conditions.

6.2 Future Work

The following research areas are suggested for the future work:

1. Three-dimensional integrated circuits (3-D IC) technology [112-113] has emerged in the past few decades, driven partly by the techno-economic difficulties of dimensional scaling and the increasing interconnect delay in microelectronics. In a 3-D IC, vertical integration of multiple transistor planes, either through monolithic integration or through bonding of

various strata, offers reduced interconnect delay and enhanced design flexibility. However, vertical integration in a 3-D IC also poses severe thermal management challenges. Higher heat generation in a concentrated area fundamentally results in a more significant temperature rise, as shown in Figure 6.1, which may cause performance reduction and, in an extreme case, physical damage to the circuit. Therefore, accurate thermal modeling, thermal–electrical co-design and co-optimization, and practical thermal management approaches for heat dissipation are needed to address the thermal challenge in 3-D ICs.

2. The AI technique has shown a powerful ability to explore the root causes affecting material and mechanical performance. Hence, more work needs to be done to find which components are the critical factors and why they affect the correlation performance.
3. Current work focuses more on the single problems in IC packaging design, and a more general model needs to be established. Hence, the Gen AI model needs to be constructed; the model consists of three main components, as Figure 6.2 shows: the LLMs provide a question-answering interface, the transfer learning model aims to improve generalizability, and the reinforcement learning model is to train and make decision.

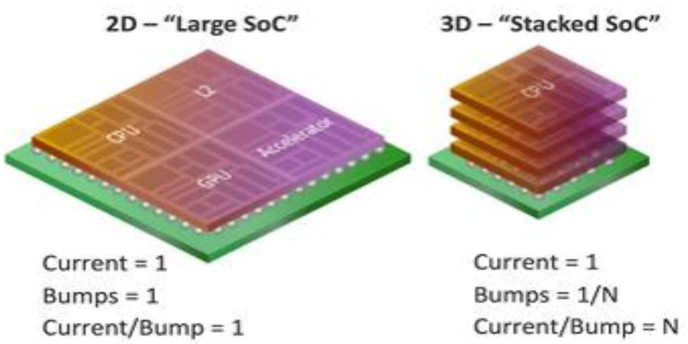

Figure 6.1 Power density distribution in 2D and 3D Structures.

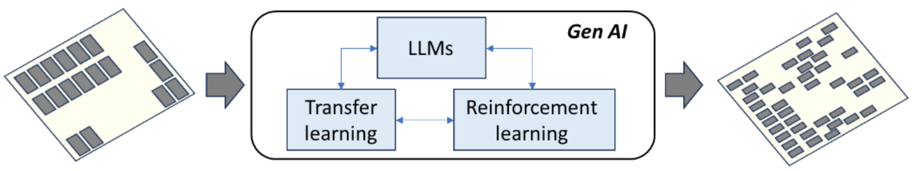

Figure 6.2 The workflow of the Gen AI design framework, where LLMs are used as interfaces, transfer learning improves model generalization, and reinforcement learning makes decisions.

REFERENCES

[1] N. Thomas. "The end of Moore’s law: A new beginning for information technology." Computing in science & engineering 19.2 (2017): 41-50.

[2] https://www.linkedin.com/pulse/guide-large-language-models-how-work-yugank-aman-g1wif.

[3] https://semiengineering.com/advanced-packagings-next-wave/

[4] H. Mohd, et al. "A review: Fundamentals of computational fluid dynamics (CFD)." AIP conference proceedings. Vol. 2030. No. 1. AIP Publishing, 2018.

[5] S. Singiresu. The finite element method in engineering. Elsevier, 2010.

[6] V. Jagota, P. Aman, and K. Khushmeet. "Finite element method: an overview." Walailak Journal of Science and Technology (WJST) 10.1 (2013): 1-8.

[7] R. Sonam, et al. "A review on Design of Experiments (DOE)." Int. J. Pharm. Chem. Sci 3.1 (2014): 216-24.

[8] S. Weissman, N. Anderson. Design of experiments (DoE) and process optimization. A review of recent publications[J]. Organic Process Research & Development, 2015, 19(11): 1605-1633.

[9] K. Chiang and C. Yuan. "An overview of solder bump shape prediction algorithms with validations." IEEE transactions on advanced packaging 24.2 (2001): 158-162.

[10] X. Gao, et al. "Dimension optimization of through silicon via (TSV) through simulation and design of experiment (DOE)." 2012 13th International Conference on Electronic Packaging Technology & High Density Packaging. IEEE, 2012.

[11] Q. Quan, and H. Carlton. "A design-of-experiment (DOE) to optimize a SiP design for connectivity applications." 2016 IEEE 18th Electronics Packaging Technology Conference (EPTC). IEEE, 2016.

[12] W. Sun, et al. "Warpage simulation and DOE analysis with application in package-on-package development." EuroSimE 2008-International Conference on Thermal, Mechanical and Multi-Physics Simulation and Experiments in Microelectronics and Micro-Systems. IEEE, 2008.

[13] L. Xu, et al. "A simulation-based multi-objective design optimization of electronic packages under thermal cycling and bending." Microelectronics Reliability 44.12 (2004): 1977-1983.

[14] M. Wu and J. Lan. "Simulation and experimental study of the warpage of fan-out wafer-level

packaging: The effect of the manufacturing process and optimal design." IEEE Transactions on Components, Packaging and Manufacturing Technology 9.7 (2018): 1396-1405.

[15] Raz, Ali K., et al. "An overview of systems engineering challenges for designing AI-enabled aerospace systems." AIAA Scitech 2021 Forum. 2021.

[16] Z. Zhao, et al. "Challenges and opportunities of AI-enabled monitoring, diagnosis & prognosis: A review." Chinese Journal of Mechanical Engineering 34.1 (2021): 56.

[17] P. Yao, H. Chen, J. Yang, and J. Wu, "Machine Learning Enabled Optimization of Pick-up Process for Thin Die," in 2021 22nd International Conference on Electronic Packaging Technology (ICEPT), 2021, pp. 1-5.

[18] W. Dai, Z. Wang, and K. Xue, "Multi-objectives design optimization based on multi-objectives Gaussian processes for System-in-Package," in 2021 IEEE 71st Electronic Components and Technology Conference (ECTC), 2021, pp. 2151-2157.

[19] Y. Chen, R. Sun, Y. Gao, and J. Leopold, "A nested-ANN prediction model for surface roughness considering the effects of cutting forces and tool vibrations," Measurement, vol. 98, pp. 25-34, 2017.

[20] V. Samavatian, M. Fotuhi-Firuzabad, M. Samavatian, P. Dehghanian, and F. Blaabjerg, "Correlation-driven machine learning for accelerated reliability assessment of solder joints in electronics," Scientific reports, vol. 10, pp. 1-14, 2020.

[21] P. Yao, H. Chen, H. Fan, J. Wu and J. Yang, "AI-enabled Automatic Molding Compound Selection for A Power Device with High Solder Joint Reliability," 2022 23rd International Conference on Electronic Packaging Technology (ICEPT), Dalian, China, 2022, pp. 1-6, doi: 10.1109/ICEPT56209.2022.9873400.

[22] P. Yao, H. Chen, J. Yang and J. Wu, "Machine Learning Enabled Optimization of Pick-up Process for Thin Die," 2021 22nd International Conference on Electronic Packaging Technology (ICEPT), Xiamen, China, 2021, pp. 1-5, doi: 10.1109/ICEPT52650.2021.9568235.

[23] P. Yao et al., "Physics-based Nested-ANN Approach for Fan-Out Wafer-Level Package Reliability Prediction," 2022 IEEE 72nd Electronic Components and Technology Conference (ECTC), San Diego, CA, USA, 2022, pp. 1827-1833, doi: 10.1109/ECTC51906.2022.00287.

[24] H. Fan, C. Li, N. b. Azman, P. Yao and H. Chen, "Robust Package Study for A Power Package

by Simulation," 2021 22nd International Conference on Thermal, Mechanical and Multi-Physics Simulation and Experiments in Microelectronics and Microsystems (EuroSimE), St. Julian, Malta, 2021, pp. 1-4, doi: 10.1109/EuroSimE52062.2021.9410850.

[25] Y. Zhang, P. Yao, Y. Han, J. Yang, H. Chen, J. Wu, et al., "Microfluidic electroless deposition for uniform stacking chip interconnection: simulation framework and experimental validation," Chemical Engineering Journal, p. 134684, 2022.

[26] C. Selvanayagam, et al. "Machine learning approach to improve accuracy of warpage simulations." 2019 IEEE 69th Electronic Components and Technology Conference (ECTC). IEEE, 2019.

[27] T. Ahmed, et al. "Warpage prediction of Injection-molded PVC part using ensemble machine learning algorithm." Materials Today: Proceedings 50 (2022): 565-569.

[28] A. Prisacaru, et al. "Degradation prediction of electronic packages using machine learning." 2019 20th International Conference on Thermal, Mechanical and Multi-Physics Simulation and Experiments in Microelectronics and Microsystems (EuroSimE). IEEE, 2019.

[29] J. Lai, et al. "RLC circuit forecast in Analog IC packaging and testing by machine learning techniques." Micromachines 13.8 (2022): 1305.

[30] S. Park, et al. "Application of machine learning for optimization of 3-D integrated circuits and systems." IEEE transactions on very large scale integration (VLSI) systems 25.6 (2017): 1856-1865.

[31] M. Mao, et al. "Machine learning for board-level drop response of BGA packaging structure." Microelectronics Reliability 134 (2022): 114553.

[32] M. Su, et al. "Identification of the interfacial cohesive law parameters of FRP strips externally bonded to concrete using machine learning techniques." Engineering Fracture Mechanics 247 (2021): 107643.

[33] Y. Freed, S. Marco, and Z. Navid. "Implementation of a probabilistic machine learning strategy for failure predictions of adhesively bonded joints using cohesive zone modeling." International Journal of Adhesion and Adhesives 118 (2022): 103226.

[34] C. Wei, et al. "Data driven modeling of interfacial traction–separation relations using a thermodynamically consistent neural network." Computer Methods in Applied Mechanics and

Engineering 404 (2023): 115826.

[35] Z. He, G. Liu, X. Zhao, et al. Overview of Gaussian process regression[J]. Control and Decision, 2013, 28(8): 1121–1129.

[36] C. Rasmussen. Gaussian processes in machine learning[C]//Summer school on machine learning. Springer, Berlin, Heidelberg, 2003: 63-71.

[37] K. Liu, B. Liu, C. Xu. Intelligent analysis model of slope nonlinear displacement time series based on Genetic-Gaussian process regression algorithm of combined kernel function[J]. Chinese Journal of Rock Mechanics and Engineering, 2009, 10: 2128-2134.

[38] A. Abdessalem, N. Dervilis, D. Wagg, et al. Automatic kernel selection for Gaussian processes regression with approximate Bayesian computation and sequential Monte Carlo [J]. Frontiers in Built Environment, 2017, 3: 52.

[39] H. James, F. Nicolo, and D. Neil. Gaussian processes for big data.arXiv preprint arXiv:1309.6835, 2013.

[40] H. James, F. Nicolo, A. Ricardo, D. Nicolas, S. Alan, Z. Max, and D. Neil. GPy: A Gaussian process framework in Python, 2014b.

[41] J. Vanhatalo, J. Riihimäki, J. Hartikainen, et al. GPstuff: Bayesian modeling with Gaussian processes[J]. Journal of Machine Learning Research, 2013, 14(Apr): 1175-1179.

[42] H. Nickisch, C. Rasmussen. Gaussian mixture modeling with Gaussian process latent variable models[C]. Joint Pattern Recognition Symposium. Springer, Berlin, Heidelberg, 2010: 272-282.

[43] E. Snelson, Z. Ghahramani. Sparse Gaussian processes using pseudo-inputs[J]. Advances in neural information processing systems, 2006, 18: 1257.

[44] J. Quinonero, C. Rasmussen. A unifying view of sparse approximate Gaussian process regression[J]. The Journal of Machine Learning Research, 2005, 6: 1939-1959.

[45] Z. Liu, H. Xu. Kernel parameter selection for support vector machine classification[J]. Journal of Algorithms & Computational Technology, 2014, 8(2): 163-177.

[46] C. Selvanayagam, P. Duong, R. Mandal, et al. Machine learning approach to improve the accuracy of warpage simulations[C]//2019 IEEE 69th Electronic Components and Technology Conference (ECTC). IEEE, 2019: 834-841.

[47] V. Samavatian, M. Fotuhi-Firuzabad, M. Samavatian, et al. Correlation-driven machine learning for accelerated reliability assessment of solder joints in electronics[J]. Scientific Reports, 2020, 10(1): 1-14.

[48] Y. Huang, C. Pan, S. Lin, et al. The machine-learning approach in detection and classification for defects in TSV-based 3-D IC[J]. IEEE Transactions on Components, Packaging and Manufacturing Technology, 2018, 8(4): 699-706.

[49] J. Springenberg, A. Klein, S. Falkner, and F. Hutter, “Bayesian optimization with robust bayesian neural networks,” in Advances in Neural Information Processing Systems 29, 2016, pp. 4134–4142.

[50] F. Hutter, H. Hoos, K. Leyton-Brown. Sequential model-based optimization for general algorithm configuration[C]. International conference on learning and intelligent optimization. Springer, Berlin, Heidelberg, 2011: 507-523.

[51] J. Bergstra , R. Bardenet, Y. Bengio, et al. Algorithms for hyper-parameter optimization[J]. Advances in neural information processing systems, 2011, 24.

[52] R. Neal. Bayesian learning for neural networks[M]. Springer Science & Business Media, 2012.

[53] S. Li, W. Xing, R. Kirby, et al. Multi-fidelity Bayesian optimization via deep neural networks[J]. Advances in Neural Information Processing Systems, 2020, 33.

[54] H. J. Kushner, "A new method of locating the maximum point of an arbitrary multipeak curve in the presence of noise", *J. Basic Eng.*, vol. 86, no. 1, pp. 97-106, Mar. 1964.

[55] J. Mockus, V Tiesis and A. Zilinskas, "The application of Bayesian methods for seeking the extremum" in Toward Global Optimization 2, Amsterdam, The Netherlands:North Holland, 1978.

[56] N. Srinivas, A. Krause, S. Kakade and M. W. Seeger, "Gaussian process optimization in the bandit setting: No regret and experimental design", *Proc. Int. Conf. Mach. Learn.*, pp. 1015-1022, 2010.

[57] Z. Wang, M. Zoghi, F. Hutter, D. Matheson and N. De Freitas, "Bayesian optimization in high dimensions via random embeddings", *Proc. Int. Joint Conf. Artif. Intell.*, pp. 1778-1784, Jul. 2013.

[58] C. Oh, E. Gavves and M. Welling, "BOCK: Bayesian optimization with cylindrical kernels", *Proc. 35th Int. Conf. Mach. Learn.*, pp. 3868-3877, 2018.

[59] A. Nayebi, A. Munteanu and M. Poloczek, "A framework for Bayesian optimization in embedded subspaces", *Proc. Int. Conf. Mach. Learn.*, pp. 4752-4761, 2019.

[60] J. Kirschner, M. Mutny, N. Hiller, R. Ischebeck and A. Krause, "Adaptive and safe Bayesian optimization in high dimensions via one-dimensional subspaces", *Proc. Int. Conf. Mach. Learn.*, pp. 3429-3438, 2019.

[61] D. Hernández-Lobato, J. Hernandez-Lobato, A. Shah and R. Adams, "Predictive entropy search for multi-objective Bayesian optimization", *Proc. Int. Conf. Mach. Learn.*, pp. 1492-1501, 2016.

[62] P. Feliot, J. Bect and E. Vazquez, "A Bayesian approach to constrained single-and multi-objective optimization", *J. Global Optim.*, vol. 67, no. 1, pp. 97-133, 2017.

[63] K. V. Price, "Differential evolution vs. the functions of the 2nd ICEO", *Proc. IEEE Int. Conf. Evol. Comput.*, pp. 153-157, 1997-Apr.

[64] K. V. Price and R. Storn, "Differential evolution: A simple evolution strategy for fast optimization", *Dr. Dobb's J.*, vol. 22, no. 4, pp. 18-24, 1997.

[65] R. Storn and K. V. Price, *Differential evolution: A simple and efficient adaptive scheme for global optimization over continuous spaces*, 1995.

[66] R. Storn and K. V. Price, "Minimizing the real functions of the ICEC 1996 contest by differential evolution", *Proc. IEEE Int. Conf. Evol. Comput.*, pp. 842-844, 1996.

[67] R. Storn, "On the usage of differential evolution for function optimization", *Proc. North Am. Fuzzy Inform. Process. Soc.*, pp. 519-523, 1996.

[68] K. Price, R. Storn and J. Lampinen, Differential EvolutionA Practical Approach to Global Optimization, Germany, Berlin:Springer, 2005.

[69] S. Das, A. Abraham, U. K. Chakraborty and A. Konar, "Differential evolution using a neighborhood based mutation operator", *IEEE Trans. Evol. Comput.*, vol. 13, pp. 526-553, Jun. 2009.

[70] S. Rahnamayan, H. R. Tizhoosh and M. M. A. Salama, "Opposition-based differential evolution", *IEEE Trans. Evol. Comput.*, vol. 12, pp. 64-79, Feb. 2008.

[71] J. Vesterstrm and R. A. Thomson, "Comparative study of differential evolution particle swarm optimization and evolutionary algorithms on numerical benchmark problems", *Proc. IEEE Congr. Evol. Comput.*, pp. 1980-1987, 2004.

[72] J. Brest, S. Greiner, B. Bokovi, M. Mernik and V. umer, "Self-adapting control parameters in differential evolution: A comparative study on numerical benchmark problems", *IEEE Trans. Evol. Comput.*, vol. 10, pp. 646-657, Dec. 2006.

[73] A. K. Qin, V. L. Huang and P. N. Suganthan, "Differential evolution algorithm with strategy adaptation for global numerical optimization", *IEEE Trans. Evol. Comput.*, vol. 13, pp. 398-417, Apr. 2009.

[74] J. Zhang and A. C. Sanderson, "JADE: Adaptive differential evolution with optional external archive", *IEEE Trans. Evol. Comput.*, vol. 13, pp. 945-958, Oct. 2009.

[75] K. Zielinski and R. Laur, "Constrained single-objective optimization using differential evolution", *Proc. IEEE Congr. Evol. Comput.*, pp. 927-934, 2006-Jul.

[76] J. Ronkkonen, S. Kukkonen and K. V. Price, "Real parameter optimization with differential evolution", *Proc. IEEE CEC*, vol. 1, pp. 506-513, 2005.

[77] A. M. Sutton, M. Lunacek and L. D. Whitley, "Differential evolution and non-separability: Using selective pressure to focus search", *Proc. 9th Annu. Conf. GECCO*, pp. 1428-1435, 2007-Jul.

[78] K. V. Price, "An Introduction to Differential Evolution," in New Ideas in Optimization, D. Corne, M. Dorigo, and F. Glover, Eds. London: McGraw-Hill, 1999, pp. 79-108.

[79] Y. Godovsky, et al. "Epoxy molding compounds as encapsulation materials for microelectronic devices." Speciality Polymers/Polymer Physics (1989): 1-48.

[80] Phansalkar, S. Prashant, C. Kim, and B. Han. "Effect of critical properties of epoxy molding compound on warpage prediction: A critical review." Microelectronics Reliability 130 (2022): 114480.

[81] C. Gan, et al. "Evolution of epoxy molding compounds and future carbon materials for thermal and mechanical stress management in memory device packaging: a critical review." Journal of Materials Science: Materials in Electronics 34.30 (2023): 2011.

[82] M. Sadeghinia, J. Kaspar, and J. Leo. "Characterization of the viscoelastic properties of an epoxy molding compound during cure." Microelectronics Reliability 52.8 (2012): 1711-1718.

[83] Y. Lin, et al. "Modeling of viscoelastic behavior of an epoxy molding compound during and after curing." IEEE Transactions on Components, Packaging and Manufacturing

Technology 1.11 (2011): 1755-1760.

[84] M. Baumgaertel and H. Winter, "Determination of discrete relaxation and retardation time spectra from dynamic mechanical data," Rheologica Acta, vol. 28, pp. 511-519, 1989.

[85] M. L. Williams, R. F. Landel, and J. D. Ferry, "The temperature dependence of relaxation mechanisms in amorphous polymers and other glass-forming liquids," Journal of the American Chemical society, vol. 77, pp. 3701-3707, 1955.

[86] W. Lee, L. T. Nguyen, and S. Guna. "Solder joint fatigue models: review and applicability to chip scale packages." Microelectronics reliability 40.2 (2000): 231-244.

[87] J. Depiver, M. Sabuj, and H. Dani. "Solder joint failures under thermo-mechanical loading conditions–A review." Advances in Materials and Processing Technologies 7.1 (2021): 1-26.

[88] Li, Lei, et al. "Thermal Fatigue Failure of Micro-Solder Joints in Electronic Packaging Devices: A Review." Materials 17.10 (2024): 2365.

[89] Depiver, A. Joshua, M. Sabuj, and H. Dani. "Solder joint failures under thermo-mechanical loading conditions–A review." Advances in Materials and Processing Technologies 7.1 (2021): 1-26.

[90] A. Syed. Accumulated creep strain and energy density based thermal fatigue life prediction models for SnAgCu solder joints[C]//2004 Proceedings. 54th electronic components and technology conference (IEEE Cat. No. 04CH37546). IEEE, 2004, 1: 737-746.

[91] J. Song, Y. Chen, Y. Yue. A general framework for multi-fidelity bayesian optimization with gaussian processes[C]//The 22nd International Conference on Artificial Intelligence and Statistics. PMLR, 2019: 3158-3167.

[92] Fan H, Boettcher T, Li C, et al. Solder Crack Improvement for A Power Package During TCT And PTC By Simulation[C]//2020 21st International Conference on Thermal, Mechanical and Multi-Physics Simulation and Experiments in Microelectronics and Microsystems (EuroSimE). IEEE, 2020: 1-4.

[93] Tzou, H. (2019). Multi-layered Shell Actuators. In: Piezoelectric Shells. Solid Mechanics and Its Applications, vol 247. Springer, Dordrecht.

[94] Kim, Cheolgyu, et al. "Warpage analysis of electroplated Cu films on fiber-reinforced polymer packaging substrates." polymers 7.6 (2015): 985-1004.

[95] A. Mallik, R. Stout and J. Ackaert, "Finite-Element Simulation of Different Kinds of Wafer Warpages: Spherical, Cylindrical, and Saddle," in IEEE Transactions on Components, Packaging and Manufacturing Technology, vol. 4, no. 2, pp. 240-247, Feb. 2014, doi: 10.1109/TCPMT.2013.2293873.

[96] Kim, Cheolgyu, et al. "Mechanism of warpage orientation rotation due to viscoelastic polymer substrates during thermal processing." Microelectronics Reliability 73 (2017): 136-145.

[97] R. T. Marler and J. S. Arora, "Function-transformation methods for multi-objective optimization," Eng. Optim., vol. 37, no. 6, pp. 551–570, Sep. 2005.

[98] B. Kim and M. Shin, "Bayesian Optimization of MOSFET Devices Using Effective Stopping Condition," in IEEE Access, vol. 9, pp. 108480-108494, 2021, doi: 10.1109/ACCESS.2021.3101812.

[99] V. Broeck, Guy, et al. "On the tractability of SHAP explanations." Journal of Artificial Intelligence Research 74 (2022): 851-886.

[100] T. Chen, and G. Carlos. "Xgboost: A scalable tree boosting system." Proceedings of the 22nd acm sigkdd international conference on knowledge discovery and data mining. 2016.

[101] Chen, T. "Xgboost: extreme gradient boosting." R package version 0.4-2 1.4 (2015).

[102] Shih, C. F. "Cracks on bimaterial interfaces: elasticity and plasticity aspects." Materials Science and Engineering: A 143.1-2 (1991): 77-90.

[103] J. Hutchinson, and Z. Suo. "Mixed mode cracking in layered materials." Advances in applied mechanics 29 (1991): 63-191.

[104] A. G. Evans, et al. "The fracture energy of bimaterial interfaces." Metallurgical Transactions A 21 (1990): 2419-2429.

[105] J. Wu, et al. "Phase-field modeling of fracture." Advances in applied mechanics 53 (2020): 1-183.

[106] K. Park, and H. Glaucio. "Cohesive zone models: a critical review of traction-separation relationships across fracture surfaces." Applied Mechanics Reviews 64.6 (2011): 060802.

[107] B. Moran, and C. F. Shih. "A general treatment of crack tip contour integrals." International journal of fracture 35 (1987): 295-310.

[108] R. Krueger. "Virtual crack closure technique: History, approach, and applications." Appl. Mech.

Rev. 57.2 (2004): 109-143.

[109] S. Mohammadi. Extended finite element method: for fracture analysis of structures. John Wiley & Sons, 2008.

[110] Barenblatt GI. The formation of equilibrium cracks during brittle fracture. General ideas and hypotheses. Axially-symmetric cracks. J Appl Math Mech 1959;23(3): 622–36. Jan 1.

[111] Barenblatt GI. The mathematical theory of equilibrium cracks in brittle fracture. Adv. Appl. Mech. 1962;7:55–129. Jan 1[Elsevier]. [49] Needleman A. A continuum model for void nucleation b

[112] Lancaster, Austin, and K. Manish. "Integrated circuit packaging review with an emphasis on 3D packaging." Integration 60 (2018): 204-212.

[113] Lancaster, Austin, and K. Manish. "Integrated circuit packaging review with an emphasis on 3D packaging." Integration 60 (2018): 204-212.